Date	Version	User	Changes	Remarks
26/02/2016	3.00	RahmC	Initial object model for Package 3 based on latest P2 model: - Version 3 of association (class 15) - Version 1 of security setup (class 64) - Version 1 of M-Bus client (class 72) - M-Bus disconnect control activity calendar and special days - 8 M-Bus value registers per channel - Disconnect control activity calendar and special days - Load Mgmt control activity calendar and special days - Power quality profile + status, average current - Local consumer display configuration object - Extended communication log - Enhanced number of entries in profile 2 - GSM diagnostics	Function activation not added yet Update of Events pending Extension sheets need to be updated
24/01/2017	3.01	RahmC	Data types of M-Bus value registers extended to also cover signed values as per EN13757	
24/04/2017	3.02	RahmC	Optional action request added to add/remove_IPv6_address methods of IPv6 setup object Data types of event counters changed from enum to double_long_unsigned	
10/11/2017	3.03	RahmC	IDIS device type changed to 104 and 105 (IDIS package 3 1ph and 3ph meter) Optional set access added to attribute 3 of tcp udp setup to allow switching between IPv4 and IPv6 Extension sheets updated IDIS function control objects added including new types Scaler of voltage thresholds (sag, swell, magnitude, cut) aligned , i.e. choice between 0 or -1 (the same as for instantaneous and average values) Events updated Extension -G more generalised to cover all mobile networks	Open items: - Extend M-Bus client / setup for wireless devices
13/11/2017	3.04	RahmC	Function activation script tables extended to also cover combined activation and deactivation of all profiles	
06/12/2017	3.05	C. Rahm	Function activation extended with activation and deactivation of the optical port Function activation script table extended with scripts to activate and deactive the optical port Push setup version 1 introduced Events aligned again to spec	

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IDIS
Interoperability Specification

Package 3
Smart Metering Objects

Edition 1.0 , xx-xx-2017

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Syntax

Syntax of ''Attribute Type'' and ''Default Value'' columns of Object Model file

The { } construction denotes an iteration of 0..n elements
The | construction denotes a variant
The [ ] construction denotes an optional element
The red elements are terminal symbols

Syntax for the ''Attribute Type'' column

TypeSpecifier ::= StandardType | PredefinedType
StandardType ::= SimpleType | ArrayType | StructType | OctetStringType | ChoiceType
PredefinedType ::= types defined by the application, for example ass_list_type, and that can be defined with standard types or other predefined types.
unsigned | long_unsigned | long64 | long64_unsigned | enum | float32 | float64 | date_time | time | date
ArrayType ::= array [SizeConstraint] [(TypeSpecifier)]
StructType ::= struct{TypeSpecifier {, TypeSpecifier }}
OctetStringType ::= octet_string [SizeConstraint]
ChoiceType ::= choice(TypeSpecifier {, TypeSpecifier})
SizeConstraint ::= [Boundary [..Boundary]]
Boundary ::= IntegerValue | *

Syntax for the Default Value column

Value ::= IntegerValue | RangeValue | StructuredValue | StringValue | VariantValue| OBISValue | *
IntegerValue ::= [-] ?sequence of digits? | 0x ?Sequence of hex-digits?
RangeValue ::= [Boundary.. Boundary]
Boundary ::= IntegerValue | *
StructuredValue ::= {Value {, Value} [,...]}
StringValue ::= "\?Sequence of characters? | ?even sequence of hex-digits"
VariantValue ::= choice(Value {,Value})
OBISValue ::= OBISField - OBISField: OBISField. OBISField. OBISField. OBISField
OBISField ::= IntegerValue | RangeValue | *

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Extensions which control the supported objects

O=always optional

M=always mandatory

M-x mandatory for extension x

The '-x' defines the condition which applies to M, i.e. mandatory in case the condition is true.

-D=Extension D

-M=Extension M

-L=Extension L

-G=mobile communication channel is used (e.g. GPRS, UMTS, LTE, etc.)

-E=Ethernet communication channel is used

-G3=G3 PLC communication channel is used

-I4=IPv4 protocol is used

-I6=IPv6 protocol is used

-C=CIP supported

NA=not applicable

Colour Codes

Group Assignment
Instance
Attribute
Method

Predefined types

type_name=type_string
The identifier before the '=' is replaced by the string after the '='
No spaces are allowed before and after the '='

Types for OM Package 1

asslist_type=array(struct(long_unsigned, octet_string))
object_list_type=array(object_list_element)
object_list_element=struct(long_unsigned, unsigned, octet_string, access_right)
access_right=struct(attribute_access_descriptor, method_access_descriptor)
attribute_access_descriptor=array(attribute_access_item)
method_access_descriptor=array(method_access_item)
attribute_access_item=struct(integer, enum, access_selector)
access_selector=choice(array(integer),null)
method_access_item=struct(integer, enum)
associated_partners_type=struct(integer,long_unsigned)
application_context_name=choice(struct(unsigned, unsigned, long_unsigned, unsigned, unsigned, unsigned, unsigned)
xDLMS_context_type=struct(bit_string, long_unsigned, long_unsigned, unsigned, integer, octet_string)
mechanism_name_structure=application_context_name
scal_unit_type=struct(integer, enum)
scripts=array(script9)
script9=struct(long_unsigned, array(action_specification))
action_specification=struct(enum, long_unsigned, octet_string[6], integer, any)
script22=struct(octet_string[6], long_unsigned)
value_definition=struct(long_unsigned, octet_string, integer)
threshold=any
emergency_profile_type=struct(long_unsigned, octet_string, double_long_unsigned)
action_set=struct(action_item, action_item)
action_item=struct(octet_string, long_unsigned)
frequencies_type=struct(double_long_unsigned, double_long_unsigned)
initiator_descriptor=struct(octet_string, long_unsigned, unsigned)
desynchronization_listing_type=struct(double_long_unsigned, double_long_unsigned, double_long_unsigned,
capture_object_definition=struct(long_unsigned, octet_string, integer, long_unsigned)

Additional types for OM Package 2

send_destination_and_method_type=struct(enum, octet_string, enum)
window_element=struct(octet_string, octet_string)
caller_id=(octet_string)
list_of_allowed_callers_element=struct(octet_string, enum)
sender_id=(octet_string)
senders_and_actions_element=struct(octet_string, script22)
callback_data=struct(boolean, unsigned, octet_string)
LCP_options_element=struct(unsigned, unsigned, choice(unsigned, boolean, long_unsigned, double_long_unsigned, ca
IPCP_options_element=struct(unsigned, unsigned, choice(boolean, long_unsigned, double_long_unsigned, array(doub
PPP_authentication_type=choice(NULL, struct(octet_string, octet_string), struct(octet_string, unsigned), struct(boolear
quality_of_service_type=struct(qos_element, qos_element)
qos_element=struct(unsigned, unsigned, unsigned, unsigned, unsigned)
nr_rings_type=struct(unsigned, unsigned)
IPv6_address=octet_string[0..48]
IPv6_address_entry=(enum, IPv6_address)
register_assignment_item=struct(long_unsigned,octet_string[6])
execution_time_date=struct(octet_string,octet_string)
reply_status=struct(unsigned,unsigned)
neighbour_discovery_setup_element=struct(unsigned,long_unsigned,double_long_unsigned)

Additional types for OM Package 3

function_status_type=struct(octet_string, boolean)
functional_block=struct(octet_string, array(function_specification))
function_specification=struct(long_unsigned,octet_string)
object_definition=struct(long_unsigned, octet_string, integer, long_unsigned,restriction_element)
restriction_element=struct(enum,choice(NULL,struct(octet_string,octet_string),struct(octet_string,octet_string)))

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#	Object / Attribute Name	Supported Objects		Attribute Type	IC		SN	OBIS code / Default Value	Access rights [Get, Set, Action], optional if in '( )'			Comments
	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
	Abstract objects - Association & Security											
	SAP Assignment	**M**	**M**		**17**	**0**	**0xFC00**	**0-0:41.0.0.255**				
1	logical_name			octet_string[6]				"0000290000FF"	Get	--	Get	
2	SAP_assignment_list			asslist_type			0x08	{{1, *}}	Get	--	Get	One logical device, Server SAP 1
1	connect_logical_device						0x20		--	--	--	
	Current association	**M**	**M**		**15**	**3**		**0-0:40.0.0.255**				
1	logical_name			octet_string[6]				"0000280000FF"	Get	--	Get	
2	object_list			object_list_type			0x08		Get	--	Get	
3	associated_partners_id			associated_partners_type			0x10		Get	--	Get	
4	application_context_name			application_context_name			0x18		Get	--	Get	
5	xDLMS_context_info			xDLMS_context_type			0x20		Get	--	Get	
6	authentication_mechanism_name			mechanism_name_structure			0x28		Get	--	Get, (Set)	Optionally Set can be used to restrict the next association to a specific mechanism_name e.g. to mechnaism_name(5) after commisioning.
7	secret			octet_string			0x30		--	--	(Set)	In case of using LLS one can use attr 7 of Current Assoc. to change the LLS PW For HLS using GMAC secret is handled via Security setup
8	association_status			enum			0x38		Get	--	Get	
9	security_setup_reference			octet_string[6]			0x40	"00002B0000FF"	Get	--	Get	
10	user_list			array			0x48		Get	--	Get	
11	current_user			struct(unsigned,visible_string)			0x50		Get	--	Get	
1	reply_to_HLS_authentication						0x60		--	--	Action	
2	change_HLS_secret						0x68		--	--	(Action)	For changing the HLS secret in case of using MD5 or SHA-1
3	add_object						0x70		--	--	--	
4	remove_object						0x78		--	--	--	
5	add_user						0x80		--	--	--	
6	remove_user						0x88		--	--	--	
	Security setup	**M**	**M**		**64**	**1**		**0-0:43.0.0.255**				
1	logical_name			octet_string[6]				"00002B0000FF"	--	--	Get	
2	security_policy			enum			0x08		--	--	Get,Set	
3	security_suite			enum			0x10	0	--	--	Get,Set	0 = AES-GCM-128/AES-128
4	client_system_title			octet_string[8]			0x18		--	--	Get	
5	server_system_title			octet_string[8]			0x20		--	--	Get	
6	certificates			octet_string[8]			0x28		--	--	Get	
1	security_activate			enum			0x30		--	--	Action	
2	global_key_transfer			key_data			0x38		--	--	Action	
3	key_agreement			key_data			0x40		--	--	Action	
4	generate_key_pair			key_data			0x48		--	--	Action	
5	generate_certificate_request			key_data			0x50		--	--	Action	
6	import_certificate			key_data			0x58		--	--	Action	
7	export_certificate			key_data			0x60		--	--	Action	
8	remove_certificate			key_data			0x68		--	--	Action	
	Security - Receive frame counter - broadcast key	**M**	**M**		**1**	**0**		**0-0:43.1.1.255**				
1	logical_name			octet_string[6]				"00002B0101FF"	Get	--	Get	
2	value			double_long_unsigned			0x08		Get	--	Get	
	Security - Receive frame counter - unicast key	**M**	**M**		**1**	**0**		**0-0:43.1.0.255**				
1	logical_name			octet_string[6]				"00002B0100FF"	Get	--	Get	
2	value			double_long_unsigned			0x08		Get	--	Get	
	Abstract objects - ID's & Control information											
	COSEM logical device name	**M**	**M**		**1**	**0**	**0xFD00**	**0-0:42.0.0.255**				
1	logical_name			octet_string[6]				"00002A0000FF"	Get	--	Get	
2	value			octet_string[16]			0x08		Get	--	Get	unique identification of the logical device
	Device ID 1, manufacturing number	**M**	**M**		**1**	**0**		**0-0:96.1.0.255**				
1	logical_name			octet_string[6]				"0000600100FF"	--	--	Get	
2	value			octet_string[0..16]			0x08		--	--	Get	E-meter serial number (Serial number of the device, handled by the manufacturer); ASCII coded
	Device ID 2	**M**	**M**		**1**	**0**		**0-0:96.1.1.255**				
1	logical_name			octet_string[6]				"0000600101FF"	--	--	Get	
2	value			octet_string[0..48]			0x08		--	--	Get,Set	E-meter equipment identifier
	Device ID 3	**M**	**M**		**1**	**0**		**0-0:96.1.2.255**				
1	logical_name			octet_string[6]				"0000600102FF"	--	--	Get	
2	value			octet_string[0..48]			0x08		--	--	Get,Set	function location
	Device ID 4	**M**	**M**		**1**	**0**		**0-0:96.1.3.255**				
1	logical_name			octet_string[6]				"0000600103FF"	--	--	Get	
2	value			octet_string[0..48]			0x08		--	--	Get,Set	location information
	Device ID 5	**M**	**M**		**1**	**0**		**0-0:96.1.4.255**				
1	logical_name			octet_string[6]				"0000600104FF"	--	--	Get	
2	value			octet_string[0..48]			0x08		--	--	Get,Set	no special meaning defined
	Device ID 6, IDIS certification number	**M**	**M**		**1**	**0**		**0-0:96.1.5.255**				
1	logical_name			octet_string[6]				"0000600105FF"	--	--	Get	
2	value			octet_string[0..16]			0x08		--	--	Get	IDIS certification number

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
	Currently active energy tariff	**M**	**M**		**1**	**0**		**0-0:96.14.0.255**				
1	logical_name			octet_string[6]				"0000600E00FF"	--	--	Get	
2	value			octet_string[0..8]			0x08		--	--	Get	currently active tariff = active_mask of register activation object (mask_name of the 0-0:14.0.1.255) Empty octet string represents no tariff is active.
	IDIS function control	**M**	**M**		**122**	**0**		**0-0:44.1.0.255**				
1	logical_name			octet_string[6]				"00002C0100FF"	--	--	Get	
2	activation_status			array					--	--	Get	
3	function_list			array[3](functional_block)				{ {"\LPCAP_1",{7,1-0:99.1.0.255}}, {"\LPCAP_2",{7,1-0:99.2.0.255}}, {"\IF_LO_2W",{23,0-0:22.0.0.255}} }	--	--	Get	
1	set_function_status							array	--	--	Action	
	IDIS function control scheduler	**M**	**M**		**22**	**0**		**0-0:15.0.7.255**				
1	logical_name			octet_string[6]				"00000F0007FF"	--	--	Get	
2	executed_script			script22			0x08	{0-0:10.0.111.255, [1..8]}	--	--	Get, Set	
3	type			enum			0x10	1	--	--	Get, (Set)	1 is mandatory (others are optional. If only type 1 is supported Set is optional)
4	execution_time			array[0..1](execution_time_date)			0x18		--	--	Get, Set	dedicated timepoint for enabling and disabling functions; the execution can be disabled with writing empty array or {{"FFFFFFFF", "FFFFFFFFF"}}
	IDIS function control script table	**M**	**M**		**9**	**0**		**0-0:10.0.111.255**				
1	logical_name			octet_string[6]				"00000A006FFF"	--	--	Get	
2	scripts			array[8](script9)			0x08	{ {1,{{2,122,0-0:44.1.0.255,1,{{"\LPCAP_1",1},{"\LPCAP_2",1}}}}}, //enable LP1+2 {2,{{2,122,0-0:44.1.0.255,1,{{"\LPCAP_1",0},{"\LPCAP_2",0}}}}}, //disable LP1+2 {3,{{2,122,0-0:44.1.0.255,1,{{"\LPCAP_1",1}}}}}, //enable LP1 {4,{{2,122,0-0:44.1.0.255,1,{{"\LPCAP_1",0}}}}}, //disable LP1 {5,{{2,122,0-0:44.1.0.255,1,{{"\LPCAP_2",1}}}}}, //enable LP2 {6,{{2,122,0-0:44.1.0.255,1,{{"\LPCAP_2",0}}}}}, //disable LP2 {7,{{2,122,0-0:44.1.0.255,1,{{"\IF_LO_2W",1}}}}}, //enable 2 way coms on local interface {8,{{2,122,0-0:44.1.0.255,1,{{"\IF_LO_2W",0}}}}} //disable 2 way coms on local interface }	--	--	Get	scripts to enable and disable the capturing of load profile 1 and 2 (opt-in, opt-out)
1	execute			long_unsigned			0x20		--	--	--	Executes the script specified in parameter data. If data matches one of the script_identifiers in the script table, then the corresponding action_specification is executed. No direct method call possible via remote communication
	Abstract objects - Time related issues											
	Clock	**M**	**M**		**8**	**0**		**0-0:1.0.0.255**				
1	logical_name			octet_string[6]				"0000010000FF"			Get	
2	time			octet_string[12]			0x08		--	Set	Get, Set	Current date and time as local time, deviation can be handeled or set to 0x8000, status must be handled. With Set deviation=0x8000 and status=0xFF
3	time_zone			long			0x10	-60	--	Set	Get, Set	
4	status			unsigned			0x18		--		Get	
5	daylights_savings_begin			octet_string[12]			0x20	"FFFF03FE07020000008000FF"	--	Set	Get, Set	last sunday in March at 02:00
6	daylights_savings_end			octet_string[12]			0x28	"FFFF0AFE07030000008000FF"	--	Set	Get, Set	last sunday in October at 03:00
7	daylights_savings_deviation			integer			0x30	60	--	Set	Get, Set	
8	daylights_savings_enabled			boolean			0x38	1	--	Set	Get, Set	
9	clock_base			enum			0x40	1	--		Get	internal crystal (by default)
	Local Time	**O**	**O**		**1**	**0**		**1-0:0.9.1.255**				For display and readout purposes, not used for remote communication
1	logical_name			octet_string[6]				"0100000901FF"	--	--	Get	
2	value			octet_string[4]			0x08		--	--	Get	
	Local Date	**O**	**O**		**1**	**0**		**1-0:0.9.2.255**				For display and readout purposes, not used for remote communication
1	logical_name			octet_string[6]				"0100000902FF"	--	--	Get	
2	value			octet_string[5]			0x08		--	--	Get	
	Clock Time Shift Limit	**M**	**M**		**3**	**0**		**1-0:0.9.11.255**				
1	logical_name			octet_string[6]				"010000090BFF"	--	--	Get	
2	value			unsigned			0x08	60	--	Set	Get, Set	Maximum allowed time shift without registration of a time shift event
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--		
	NTP setup	**O**	**O**		**100**	**0**		**0-0:25.10.0.255**				
1	logical_name			octet_string[6]				"0000190A00FF"	--	--	Get	
2	activated			boolean			0x08	0	--	Set	Get, Set	
3	server_address			octet_string[0..*]			0x10		--	--	Get, Set	
4	server_port			long_unsigned			0x18		--	--	Get, Set	
5	authentication_method			enum			0x20	1	--	--	Get, Set	Only "no secret" or "shared secrets" supported.
6	authentication_keys			array			0x28		--	--	Set	
7	client_key			octet_string[0..*]			0x30		--	--	Set	
1	synchronize_time			integer			0x38		--	Action	Action	
2	add_authentication_key						0x40		--	--	--	
3	delete _authentication_key						0x48		--	--	--	
	Activity Calendar	**M**	**M**		**20**	**0**		**0-0:13.0.0.255**				

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
1	logical_name			octet_string[6]			"00000D0000FF"	--	--	Get	
2	calendar_name_active			octet_string[0..8]		0x08		--	--	Get	The calendar_name_active has a size of 8 characters at least, i.e. [0..8] is mandatory but can be longer
3	season_profile_active			array[0..*]		0x10		--	--	Get	
4	week_profile_table_active			array[0..*]		0x18		--	--	Get	
5	day_profile_table_active			array[0..*]		0x20		--	--	Get	
6	calendar_name_passive			octet_string[0..8]		0x28		--	Set	Get, Set	see calendar_name_active
7	season_profile_passive			array[0..*]		0x30		--	Set	Get, Set	
8	week_profile_table_passive			array[0..*]		0x38		--	Set	Get, Set	
9	day_profile_table_passive			array[0..*]		0x40		--	Set	Get, Set	
10	activate_passive_calendar_time			octet_string[12]		0x48		--	Set	Get, Set	
1	activate_passive_calendar			integer		0x50		--	(Action)	(Action)	
	Special Days Table	**M**	**M**		**11**	**0**	**0-0:11.0.0.255**				
1	logical_name			octet_string[6]			"00000B0000FF"	--	--	Get	
2	entries			array[0..30]		0x08		--	Set	Get, Set	
	Register activation - Energy	**M**	**M**		**6**	**0**	**0-0:14.0.1.255**				
1	logical_name			octet_string[6]			"00000E0001FF"	--	--	Get	
2	register_assignment			array[32..*](register_assignment_item)		0x08		--	(Set)	Get, (Set)	Array size depends on the amount of rated registers supported; 32 mandatory, others optional
3	mask_list			array[0..*]		0x10		--	Set	Get, Set	
4	active_mask			octet_string[0..8]		0x18		--	--	Get	
	Register activation - Maximum Demand	**M**	**M**		**6**	**0**	**0-0:14.0.2.255**				
1	logical_name			octet_string[6]			"00000E0002FF"	--	--	Get	
2	register_assignment			array[16..*](register_assignment_item)		0x08		--	(Set)	Get, (Set)	Array size depends on the amount of rated registers supported; 16 mandatory, others optional
3	mask_list			array[0..*]		0x10		--	Set	Get, Set	
4	active_mask			octet_string[0..8]		0x18		--	--	Get	
	Tariffication script table	**M**	**M**		**9**	**0**	**0-0:10.0.100.255**				
1	logical_name			octet_string[6]			"00000A0064FF"	--	--	Get	

2	scripts			scripts		0x08	Choice(
							// First variant
							{
							{1,{
							{1,6,0-0:14.0.1.255,4,*},
							{1,6,0-0:14.0.2.255,4,*},
							{1,70,0-1:96.3.10.255,3,[1..2]}
							}
							},
							
							{2,{
							{1,6,0-0:14.0.1.255,4,*},
							{1,6,0-0:14.0.2.255,4,*},
							{1,70,0-1:96.3.10.255,3,[1..2]}
							}
							},
							
							{3,{
							{1,6,0-0:14.0.1.255,4,*},
							{1,6,0-0:14.0.2.255,4,*},
							{1,70,0-1:96.3.10.255,3,[1..2]}
							}
							},
							
							{4,{
							{1,6,0-0:14.0.1.255,4,*},
							{1,6,0-0:14.0.2.255,4,*},
							{1,70,0-1:96.3.10.255,3,[1..2]}
							}
							}
							,...
							},
								--	--	Get, Set	Set to be able to add also the tariff activation script for load management.

1	execute			long_unsigned		0x20		--	--	(Action)	Execute typically not needed since tariffication always controlled only internaly, never over remote communication.
	Abstract objects - Billing period reset										
	Predefined Scripts - MDI reset / end of billing period	**M**	**M**		**9**	**0**	**0-0:10.0.1.255**				
1	logical_name			octet_string[6]			"00000A0001FF"	--	--	Get	
2	scripts			scripts		0x08		--	--	(Get)	The attribute's access rights must be declared in CTI file. Allowed possibilities - no access or Get access
1	execute			long_unsigned		0x20	1	--	--	Action	The activation of this scripts is performed by calling the execute() method to the script identifier 1 of the corresponding script object
	End of billing period 1 scheduler	**M**	**M**		**22**	**0**	**0-0:15.0.0.255**				Single action scheduler for Stored Billing Values Profile
1	logical_name			octet_string[6]			"00000F0000FF"	--	--	Get	
2	executed_script			script22		0x08	{0-0:10.0.1.255,1}	--	--	Get	"MDI reset / end of billing period"
3	type			enum		0x10	1	--	--	Get, (Set)	1 is mandatory (Other are optional. If only type 1 is supported Set is optional)
4	execution_time			array[0..1](execution_time_date)		0x18		--	--	Get, Set	at midnight (or any other time) of the first day of every month; the execution can be disabled with writing empty array or {{"FFFFFFFF", "FFFFFFFF"}}
	Data of billing period 1	**M**	**M**		**7**	**1**	**0-0:98.1.0.255**				Stored Billing Values Profile
1	logical_name			octet_string[6]			"0000620100FF"	--	--	Get	
2	buffer			array		0x08		--	--	Get	selective access per range and entry

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
3	capture_objects			array[1..*](capture_object_definition)			0x10 {{8,0-0:1.0.0.255,2,0}, {3,1-0:1.8.1.255,2,0}, {3,1-0:1.8.2.255,2,0}, {3,1-0:2.8.1.255,2,0}, {3,1-0:2.8.2.255,2,0}, ...}	--	--	Get, Set	clock;+A rate 1;+A rate 2;–A rate 1;–A rate 2; etc.
4	capture_period			double_long_unsigned			0x18 0	--	--	Get	
5	sort_method			enum			0x20 1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28 {*,*-*:*.*.*.*,*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30	--	--	Get	
8	profile_entries			double_long_unsigned			0x38 [13..*]	--	--	Get	≥ 13 months
1	reset			integer			0x58	--	--	Action	
	Abstract objects - Disconnector, Load mgmt, Supervision										
	Disconnect control scheduler	**M-D**	**M-D**		**22**	**0**	**0-0:15.0.1.255**				
1	logical_name			octet_string[6]			"00000F0001FF"	--	--	Get	
2	executed_script			script22			0x08 {0-0:10.0.106.255, [1..4]}	--	--	Get, Set	Connect or disconnect script of disconnector script table (0-0:10.0.106.255)
3	type			enum			0x10 1	--	--	Get, (Set)	1 is mandatory (others are optional. If only type 1 is supported Set is optional)
4	execution_time			array[0..1](execution_time_date)			0x18	--	--	Get, Set	dedicated timepoint for connection or disconnection; the execution can be disabled with writing empty array or {{"FFFFFFFF", "FFFFFFFFF"}}
	Disconnector script table	**M-D**	**M-D**		**9**	**0**	**0-0:10.0.106.255**				
1	logical_name			octet_string[6]			"00000A006AFF"	--	--	Get	

2	scripts			array[4](script9)			0x08 {
							{1,{{1,70,0-0:96.3.10.255,3,2}}}, //set to ready_for_reconnection
							{2,{{1,70,0-0:96.3.10.255,3,1}}}, //set to connected
							{3,{{2,70,0-0:96.3.10.255,1,0}}}, //execute remote disconnect
							{4,{{2,70,0-0:96.3.10.255,2,0}}}, //execute remote connect
							}
								--	--	Get	connect and disconnect scripts, 1 and 2 model local disconnection and local reconnection, 3 and 4 model remote disconnection and connection

1	execute			long_unsigned			0x20	--	--	--	Executes the script specified in parameter data. If data matches one of the script_identifiers in the script table, then the corresponding action_specification is executed. No direct method call possible via remote communication
	Disconnect Control Activity Calendar	**M-D**	**M-D**		**20**	**0**	**0-0:13.0.1.255**				
1	logical_name			octet_string[6]			"00000D0001FF"	--	--	Get	
2	calendar_name_active			octet_string[0..8]			0x08	--	--	Get	
3	season_profile_active			array[0..*]			0x10	--	--	Get	
4	week_profile_table_active			array[0..*]			0x18	--	--	Get	
5	day_profile_table_active			array[0..*]			0x20	--	--	Get	
6	calendar_name_passive			octet_string[0..8]			0x28	--	Set	Get, Set	
7	season_profile_passive			array[0..*]			0x30	--	Set	Get, Set	
8	week_profile_table_passive			array[0..*]			0x38	--	Set	Get, Set	
9	day_profile_table_passive			array[0..*]			0x40	--	Set	Get, Set	
10	activate_passive_calendar_time			octet_string[12]			0x48	--	Set	Get, Set	
1	activate_passive_calendar			integer			0x50	--	(Action)	(Action)	
	Disconnect Control Special Days Table	**M-D**	**M-D**		**11**	**0**	**0-0:11.0.1.255**				
1	logical_name			octet_string[6]			"00000B0001FF"	--	--	Get	
2	entries			array[0..30]			0x08	--	Set	Get, Set	
	Disconnect control	**M-D**	**M-D**		**70**	**0**	**0-0:96.3.10.255**				
1	logical_name			octet_string[6]			"000060030AFF"	--	--	Get	
2	output_state			boolean			0x08	--	--	Get	
3	control_state			enum			0x10	--	--	Get	
4	control_mode			enum			0x18	--	--	Get, Set	if only one conrol mode is available set is not allowed.
1	remote_disconnect			integer			0x20	--	--	Action	
2	remote_connect			integer			0x28	--	--	Action	
	Event Object - Disconnector Control log	**M-D**	**M-D**		**1**	**0**	**0-0:96.11.2.255**				
1	logical_name			octet_string[6]			"0000600B02FF"	--	--	Get	
2	value			enum			0x08	--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255; at first power up or if no events were yet generated the value should return 255
	Disconnector Control Log	**M-D**	**M-D**		**7**	**1**	**0-0:99.98.2.255**				
1	logical_name			octet_string[6]			"0000636202FF"	--	--	Get	
2	buffer			array			0x08	--	--	Get	selective access per range
3	capture_objects			array[2..3](capture_object_definition)			0x10 {{8, 0-0:1.0.0.255, 2, 0}, {1, 0-0:96.11.2.255, 2, 0}, {71, 0-0:17.0.0.255, 3, 0}}	--	--	Get, (Set)	clock;control event code;limiter threshold
4	capture_period			double_long_unsigned			0x18 0	--	--	Get	asynchronously
5	sort_method			enum			0x20 1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28 {*,*-*:*.*.*.*,*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30	--	--	Get	
8	profile_entries			double_long_unsigned			0x38 [10..*]	--	--	Get	
1	reset			integer			0x58	--	--	Action	
	Load Mgmt script table	**M-L**	**M-L**		**9**	**0**	**0-0:10.0.103.255**				
1	logical_name			octet_string[6]			"00000A0067FF"	--	--	Get	

2	scripts			array[4..*](script9)			0x08 {
							{1,{{1,70,0-1:96.3.10.255,3,2}}},
							{2,{{1,70,0-1:96.3.10.255,3,1}}},
							{3,{{2,70,0-1:96.3.10.255,1,0}}},
							{4,{{2,70,0-1:96.3.10.255,2,0}}}, ...
							}
								--	--	Get	connect and disconnect methods of the load mgmt objects (relays), see also disconnector script table for further details

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value		Public (16)	Preestablished (102)	Management (1)	
1	execute			long_unsigned			0x20		--	--	--	Executes the script specified in parameter data. If data matches one of the script_identifiers in the script table, then the corresponding action_specification is executed. No direct method call possible via remote communication
	Load Mgmt Activity Calendar	**M-L**	**M-L**		**20**	**0**		**0-0:13.0.2.255**				
1	logical_name			octet_string[6]				"00000D0002FF"	--	--	Get	
2	calendar_name_active			octet_string[0..8]			0x08		--	--	Get	
3	season_profile_active			array[0..*]			0x10		--	--	Get	
4	week_profile_table_active			array[0..*]			0x18		--	--	Get	
5	day_profile_table_active			array[0..*]			0x20		--	--	Get	
6	calendar_name_passive			octet_string[0..8]			0x28		--	Set	Get, Set	
7	season_profile_passive			array[0..*]			0x30		--	Set	Get, Set	
8	week_profile_table_passive			array[0..*]			0x38		--	Set	Get, Set	
9	day_profile_table_passive			array[0..*]			0x40		--	Set	Get, Set	
10	activate_passive_calendar_time			octet_string[12]			0x48		--	Set	Get, Set	
1	activate_passive_calendar			integer			0x50		--	(Action)	(Action)	
	Load Mgmt Special Days Table	**M-L**	**M-L**		**11**	**0**		**0-0:11.0.2.255**				
1	logical_name			octet_string[6]				"00000B0002FF"	--	--	Get	
2	entries			array[0..30]			0x08		--	Set	Get, Set	
	Load Mgmt - Relay control 1	**M-L**	**M-L**		**70**	**0**		**0-1:96.3.10.255**				Channel (filed B) to be incremented if additional instances needed
1	logical_name			octet_string[6]				"000160030AFF"	--	--	Get	
2	output_state			boolean			0x08		--	--	Get	
3	control_state			enum			0x10		--	--	Get	
4	control_mode			enum			0x18	6	--	--	Get, (Set)	If only one control mode is supported then control mode shall be 6. If more than one mode is supported then optional Set should be allowed. For relay only mode 4 or 6 should be used.
1	remote_disconnect			integer			0x20		--	--	Action	
2	remote_connect			integer			0x28		--	--	Action	
	Load Mgmt - Relay control 2	**O**	**O**		**70**	**0**		**0-2:96.3.10.255**				Channel (filed B) to be incremented if additional instances needed
1	logical_name			octet_string[6]				"000260030AFF"	--	--	Get	
2	output_state			boolean			0x08		--	--	Get	
3	control_state			enum			0x10		--	--	Get	
4	control_mode			enum			0x18	6	--	--	Get, (Set)	If only one control mode is supported then control mode shall be 6. If more than one mode is supported then optional Set should be allowed. For relay only mode 4 or 6 should be used.
1	remote_disconnect			integer			0x20		--	--	Action	
2	remote_connect			integer			0x28		--	--	Action	
	Limiter	**M-D**	**M-D**		**71**	**0**		**0-0:17.0.0.255**				
1	logical_name			octet_string[6]				"0000110000FF"	--	--	Get	
2	monitored_value			value_definition			0x08	choice({3, 1-0:90.7.0.255, 2}, {5,1-0:1.24.0.255, 2}, {5,1-0:15.24.0.255, 2})	--	--	Get, Set	instantaneous current or sliding demand
3	threshold_active			threshold			0x10		--	--	Get	
4	threshold_normal			threshold			0x18		--	--	Get, Set	
5	treshold_emergency			threshold			0x20		--	--	Get, Set	
6	min_over_threshold_duration			double_long_unsigned			0x28	180	--	--	Get, Set	
7	min_under_threshold_duration			double_long_unsigned			0x30	180	--	--	Get, Set	
8	emergency_profile			emergency_profile_type			0x38		--	--	Get, Set	
9	emergency_profile_group_id_list			array			0x40		--	--	Get, Set	
10	emergency_profile_active			boolean			0x48		--	--	Get	
11	actions			action_set			0x50	choice({ {0-0:10.0.106.255,1} , {0-0:10.0.106.255,2}}, { {0-0:10.0.106.255,1} , {0-0:10.0.106.255,0}})	--	--	Get, (Set)	
	Supervision monitor 1 - Fuse supervision L1	**M-D**	**M-D**		**21**	**0**		**1-0:31.4.0.255**				
1	logical_name			octet_string[6]				"01001F0400FF"	--	--	Get	
2	thresholds			array[1]			0x08		--	--	Get, Set	The threshold is of the same type as the monitored attribute of the referenced object
3	monitored_value			value_definition			0x10	{5, 1-0:31.4.0.255, 2}	--	--	Get, Set	Instantaneous value current per phase or average current for supervision
4	actions			array[1](action_set)			0x18	choice({{{0-0:10.0.106.255,1} , {0-0:10.0.106.255,2}}}, {{{0-0:10.0.106.255,1} , {0-0:10.0.106.255,0}}})	--	--	Get, Set	
	Supervision monitor 2 - Fuse supervision L2	**NA**	**M-D**		**21**	**0**		**1-0:51.4.0.255**				
1	logical_name			octet_string[6]				"0100330400FF"	--	--	Get	
2	thresholds			array[1]			0x08		--	--	Get, Set	The threshold is of the same type as the monitored attribute of the referenced object
3	monitored_value			value_definition			0x10	{5, 1-0:51.4.0.255, 2}	--	--	Get, Set	Instantaneous value current per phase or average current for supervision
4	actions			array[1](action_set)			0x18	choice({{{0-0:10.0.106.255,1} , {0-0:10.0.106.255,2}}}, {{{0-0:10.0.106.255,1} , {0-0:10.0.106.255,0}}})	--	--	Get, Set	
	Supervision monitor 3 - Fuse supervision L3	**NA**	**M-D**		**21**	**0**		**1-0:71.4.0.255**				
1	logical_name			octet_string[6]				"0100470400FF"	--	--	Get	
2	thresholds			array[1]			0x08		--	--	Get, Set	The threshold is of the same type as the monitored attribute of the referenced object
3	monitored_value			value_definition			0x10	{5, 1-0:71.4.0.255, 2}	--	--	Get, Set	Instantaneous value current per phase or average current for supervision
4	actions			array[1](action_set)			0x18	choice({{{0-0:10.0.106.255,1} , {0-0:10.0.106.255,2}}}, {{{0-0:10.0.106.255,1} , {0-0:10.0.106.255,0}}})	--	--	Get, Set	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
	Abstract objects - Errors & Alarms, Event logs											
	Error Register	**M**	**M**		**1**	**0**		**0-0:97.97.0.255**				
1	logical_name			octet_string[6]				"0000616100FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get, Set	
	Alarm Register 1	**M**	**M**		**1**	**0**		**0-0:97.98.0.255**				
1	logical_name			octet_string[6]				"0000616200FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get, Set	
	Alarm Filter 1	**M**	**M**		**1**	**0**		**0-0:97.98.10.255**				
1	logical_name			octet_string[6]				"0000616200FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get, Set	This filter defines, if an event is handled as an alarm when it appears
	Alarm Descriptor 1	**M**	**M**		**1**	**0**		**0-0:97.98.20.255**				
1	logical_name			octet_string[6]				"0000616214FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get, Set	Updated on Alarm register change i.e. on 0->1 transition of any of the bits in Alarm register the corresponding bit in the Alarm descriptor is set
	Alarm Monitor 1	**M**	**M**		**21**	**0**		**0-0:16.1.0.255**				
1	logical_name			octet_string[6]				"0000100100FF"	--	--	Get	
2	thresholds			array[1](double_long_unsigned)			0x08	{0}	--	--	Get	Monitored value is Alarm descriptor 1. Value 0 in threshold means if any of the bits in monitor value is newly triggered the action of the monitor will be set.
3	monitored_value			value_definition			0x10	{1, 0-0:97.98.20.255, 2}	--	--	Get	
4	actions			array[1](action_set)			0x18	{{{0-0:10.0.108.255,4} , {0-0:10.0.108.255,0}}}	--	--	Get, Set	triggers push on alarm in case of occurence, no action is triggered in case the alarm descriptor is cleared
	Alarm Register 2	**M**	**M**		**1**	**0**		**0-0:97.98.1.255**				
1	logical_name			octet_string[6]				"0000616201FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get, Set	
	Alarm Filter 2	**M**	**M**		**1**	**0**		**0-0:97.98.11.255**				
1	logical_name			octet_string[6]				"0000616200BFF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get, Set	This filter defines, if an event is handled as an alarm when it appears
	Alarm Descriptor 2	**M**	**M**		**1**	**0**		**0-0:97.98.21.255**				
1	logical_name			octet_string[6]				"0000616215FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get, Set	Updated on Alarm register change i.e. on 0->1 transition of any of the bits in Alarm register the corresponding bit in the Alarm descriptor is set
	Alarm Monitor 2	**M**	**M**		**21**	**0**		**0-0:16.1.1.255**				
1	logical_name			octet_string[6]				"0000100101FF"	--	--	Get	
2	thresholds			array[1](double_long_unsigned)			0x08	{0}	--	--	Get	Monitored value is Alarm descriptor 2. Value 0 in threshold means if any of the bits in monitor value is newly triggered the action of the monitor will be set.
3	monitored_value			value_definition			0x10	{1, 0-0:97.98.21.255, 2}	--	--	Get	
4	actions			array[1](action_set)			0x18	{{{0-0:10.0.108.255,4} , {0-0:10.0.108.255,0}}}	--	--	Get, Set	triggers push on alarm in case of occurence, no action is triggered in case the alarm descriptor is cleared
	Event Object - Standard Event Log	**M**	**M**		**1**	**0**		**0-0:96.11.0.255**				
1	logical_name			octet_string[6]				"0000600B00FF"	--	--	Get	
2	value			enum			0x08		--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	Standard Event Log	**M**	**M**		**7**	**1**		**0-0:99.98.0.255**				Standard event log containing errors and alarms
1	logical_name			octet_string[6]				"0000636200FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range
3	capture_objects			array[2](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {1,0-0:96.11.0.255,2,0}}	--	--	Get, (Set)	clock;event code
4	capture_period			double_long_unsigned			0x18	0	--	--	Get	asynchronously
5	sort_method			enum			0x20	1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28	{*,*.*.*.*.*.*.*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[100..*]	--	--	Get	
1	reset			integer			0x58		--	--	Action	
	Event Object - Fraud Detection Log	**M**	**M**		**1**	**0**		**0-0:96.11.1.255**				
1	logical_name			octet_string[6]				"0000600B01FF"	--	--	Get	
2	value			enum			0x08		--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	Fraud Detection Log	**M**	**M**		**7**	**1**		**0-0:99.98.1.255**				Event log containing all fraud detection events
1	logical_name			octet_string[6]				"0000636201FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range
3	capture_objects			array[2](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {1,0-0:96.11.1.255,2,0}}	--	--	Get, (Set)	clock;tamper event code
4	capture_period			double_long_unsigned			0x18	0	--	--	Get	asynchronously
5	sort_method			enum			0x20	1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28	{*,*.*.*.*.*.*.*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[30..*]	--	--	Get	
1	reset			integer			0x58		--	--	Action	
	Event Object - Communication Log	**M**	**M**		**1**	**0**		**0-0:96.11.5.255**				
1	logical_name			octet_string[6]				"0000600B05FF"	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
2	value			enum			0x08	--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	Communication Log	**M**	**M**		**7**	**1**	**0-0:99.98.5.255**				
1	logical_name			octet_string[6]			"0000636205FF"	--	--	Get	
2	buffer			array			0x08	--	--	Get	selective access per range
3	capture_objects			array[2](capture_object_definition)			0x10 {{8,0-0:1.0.0.255,2,0}, {1,0-0:96.11.5.255,2,0}}	--	--	Get, (Set)	clock;communication event code
4	capture_period			double_long_unsigned			0x18 0	--	--	Get	asynchronously
5	sort_method			enum			0x20 1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28 {*,*-*:*.*.*.*,*,*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30	--	--	Get	
8	profile_entries			double_long_unsigned			0x38 [100..*]	--	--	Get	
1	reset			integer			0x58	--	--	Action	
	Event Counter Object 1 - E-Meter Maintenance Total	**M**	**M**		**1**	**0**	**0-0:96.15.0.255**				
1	logical_name			octet_string[6]			"0000600F00FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 2 - E-Meter Configuration Total	**M**	**M**		**1**	**0**	**0-0:96.15.1.255**				
1	logical_name			octet_string[6]			"0000600F01FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 3 – E-Meter Billing Total	**M**	**M**		**1**	**0**	**0-0:96.15.2.255**				
1	logical_name			octet_string[6]			"0000600F02FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 4 – E-Meter Load Profile Reading Total	**M**	**M**		**1**	**0**	**0-0:96.15.3.255**				
1	logical_name			octet_string[6]			"0000600F03FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 5 – M-Bus Device 1 Billing Total	**M-M**	**M-M**		**1**	**0**	**0-0:96.15.4.255**				
1	logical_name			octet_string[6]			"0000600F04FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 6 – M-Bus Device 1 Load Profile Reading Total	**M-M**	**M-M**		**1**	**0**	**0-0:96.15.5.255**				
1	logical_name			octet_string[6]			"0000600F05FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 7 – M-Bus Device 2 Billing Total	**M-M**	**M-M**		**1**	**0**	**0-0:96.15.6.255**				
1	logical_name			octet_string[6]			"0000600F06FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 8 – M-Bus Device 2 Load Profile Reading Total	**M-M**	**M-M**		**1**	**0**	**0-0:96.15.7.255**				
1	logical_name			octet_string[6]			"0000600F07FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 9 – M-Bus Device 3 Billing Total	**M-M**	**M-M**		**1**	**0**	**0-0:96.15.8.255**				
1	logical_name			octet_string[6]			"0000600F08FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 10 – M-Bus Device 3 Load Profile Reading Total	**M-M**	**M-M**		**1**	**0**	**0-0:96.15.9.255**				
1	logical_name			octet_string[6]			"0000600F09FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 11 – M-Bus Device 4 Billing Total	**M-M**	**M-M**		**1**	**0**	**0-0:96.15.10.255**				
1	logical_name			octet_string[6]			"0000600F0AFF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Counter Object 12 – M-Bus Device 4 Load Profile Reading Total	**M-M**	**M-M**		**1**	**0**	**0-0:96.15.11.255**				
1	logical_name			octet_string[6]			"0000600F0BFF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
	Event Object - Communication Details Log	**M**	**M**		**1**	**0**	**0-0:96.11.6.255**				
1	logical_name			octet_string[6]			"0000600B06FF"	--	--	Get	
2	value			enum			0x08	--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	Communication Details Log	**M**	**M**		**7**	**1**	**0-0:99.98.6.255**				
1	logical_name			octet_string[6]			"0000636206FF"	--	--	Get	
2	buffer			array			0x08	--	--	Get	selective access per range

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
3	capture_objects			array[2](capture_object_definition)			0x10	choice({{8,0-0:1.0.0.255,2,0}, {1,0-0:96.11.6.255,2,0}, {0-0:96.15.0.255}, {0-0:96.15.1.255}, {0-0:96.15.2.255}, {0-0:96.15.3.255}, {0-0:96.15.4.255}, {0-0:96.15.5.255}, {0-0:96.15.6.255}, {0-0:96.15.7.255}, {0-0:96.15.8.255}, {0-0:96.15.9.255}, {0-0:96.15.10.255}, {0-0:96.15.11.255}} , {{8,0-0:1.0.0.255,2,0}, {1,0-0:96.11.6.255,2,0}, {0-0:96.15.0.255}, {0-0:96.15.1.255}, {0-0:96.15.2.255}, {0-0:96.15.3.255}, })	--	--	Get, (Set)	choice because of extension M
4	capture_period			double_long_unsigned			0x18	0	--	--	Get	asynchronously
5	sort_method			enum			0x20	1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28	{*.*.*.*.*.*.*.*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[100..*]	--	--	Get	
1	reset			integer			0x58		--	--	Action	
	Abstract objects - User Interface											
	Consumer Message Code - Meter Display	**M**	**M**		**1**	**0**	**0-0:96.13.1.255**					Consumer message code shown on meter's display
1	logical_name			octet_string[6]				"0000600D01FF"	--	--	Get	
2	value			octet_string[0..64]			0x08		--	Set	Get, Set	Message code to be shown on numeric display.
	Consumer Message Text - Consumer Information	**M**	**M**		**1**	**0**	**0-0:96.13.0.255**					Consumer message text sent to End consumer interface
1	logical_name			octet_string[6]				"0000600D00FF"	--	--	Get	
2	value			octet_string[0..1024]			0x08		--	Set	Get, Set	Message text sent to an external device without any further interpretation
	Push action scheduler - Consumer Information	**M-C**	**M-C**		**22**	**0**	**0-4:15.0.4.255**					
1	logical_name			octet_string[6]				"00040F0004FF"	--	--	Get	
2	executed_script			script22			0x08	{0-0:10.0.108.255,8}	--	--	Get, (Set)	
3	type			enum			0x10	5	--	--	Get, (Set)	
4	execution_time			array[0..*](execution_time_date)			0x18	{ {"FFFF0000" , "FFFFFFFFFF"}, {"FFFF0F00" , "FFFFFFFFFF"}, {"FFFF1E00" , "FFFFFFFFFF"}, {"FFFF2D00" , "FFFFFFFFFF"}, }	--	--	Get, Set	By default push every 15 seconds
	Push setup - Consumer Information	**M-C**	**M-C**		**40**	**1**	**0-6:25.9.0.255**					
1	logical_name			octet_string[6]				"0006190900FF"	--	--	Get	
2	push_object_list			array[0..*](object_definition)			0x08		--	--	Get, Set	
3	send_destination_and_method			send_destination_and_method_type			0x10	{5,"0001160000FF",0}	--	--	Get, Set	
4	communication_window			array[0..*](window_element)			0x18		--	--	Get, Set	
5	randomisation_start_interval			long_unsigned			0x20	0	--	--	Get, Set	Randomiztion time in seconds, 0 = no randomization at start of the first window defined in calling_window
6	number_of_retries			unsigned			0x28		--	--	Get, Set	Max number of repetitions of unsuccessful push
7	repetition_delay			long_unsigned			0x30		--	--	Get, Set	Delay in seconds between repetitions
8	port_reference			octet_string[0..6]			0x38		--	--	Get, Set	
9	push_client_SAP			integer			0x40		--	--	Get, Set	
10	push_protection_parameters			array			0x48	{}	--	--	Get, Set	
11	gateway_prefix			octet_string			0x50	{}				
1	push			integer			0x58		--	Action	Action	
	Security setup - Consumer Information	**M-C**	**M-C**		**64**	**0**	**0-0:43.0.1.255**					
1	logical_name			octet_string[6]				"00002B0001FF"	--	--	Get	
2	security_policy			enum			0x08	3	--	--	Get,Set	
3	security_suite			enum			0x10	0	--	--	Get,Set	0 = AES-GCM-128/AES-128
4	client_system_title			octet_string[8]			0x18		--	--	Get	
5	server_system_title			octet_string[8]			0x20		--	--	Get	
1	security_activate			enum			0x28		--	--	Action	
2	global_key_transfer			key_data			0x30		--	--	Action	
	IEC HDLC setup - Consumer Information Interface	**M-C**	**M-C**		**23**	**1**	**0-1:22.0.0.255**					Mandatory HDLC on Optical port.
1	logical_name			octet_string[6]				"0001160000FF"	--	--	Get	
2	comm_speed			enum			0x08		--	Set	Get, Set	
3	window_size_transmit			unsigned			0x10		--	Set	Get, Set	
4	window_size_receive			unsigned			0x18		--	Set	Get, Set	
5	max_info_field_lenght_transmit			long_unsigned			0x20		--	Set	Get, Set	
6	max_info_field_lenght_receive			long_unsigned			0x28		--	Set	Get, Set	
7	inter_octet_time_out			long_unsigned			0x30		--	Set	Get, Set	
8	inactivity_time_out			long_unsigned			0x38		--	Set	Get, Set	
9	device_address			long_unsigned			0x40		--	--	Get, Set	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
	Abstract objects - Firmware Upgrade											
	Image transfer	**M**	**M**		**18**	**0**		**0-0:44.0.0.255**				
1	logical_name			octet_string[6]				"00002C0000FF"	--	--	Get	
2	image_block_size			double_long_unsigned			0x08		--	--	Get	block size in bytes
3	image_transferred_block_status			bit_string			0x10		--	--	Get	each bit provides info abouot one individual image block 0 = not transferred 1 = transferred
4	image_first not_transferred_block_number			double_long_unsigned			0x18		--	--	Get	provides block number of the first missing block
5	image_transfer_enabled			boolean			0x20		--	--	Get, Set	enabled or not; detail information in image_transfer_status 0 = disabled 1 = enabled
6	image_transfer status			enum			0x28		--	--	Get	detailed status of image transfer process 0 = transfer not initiated (not active; default state) 1 = transfer initiated 2 = verification initiated 3 = verification successful 4 = verification failed 5 = activation initiated 6 = activation successful 7 = activation failed
7	image_to_activate_info			array			0x30		--	--	Get	
1	image_transfer_initiate			struct(octet_string,double_long_unsigned)			0x40		--	--	Action	initiates the transfer process; includes data in format { image_identifier image_size }
2	image_block_transfer			struct(double_long_unsigned,octet_string,)			0x48		--	Action	Action	transfers one single block; includes data in format { image_block_number image_block_value }
3	image_verify			integer			0x50		--	--	Action	verifies the integrity before activation; result can be (0) success, if the verification could be completed; (2) temporary-failure if the verification has not been completed; (250) other-reason if the verification failed.
4	image_activate			integer			0x58		--	--	Action	activates the transferred and verified image; result can be (0) success, if the verification could be completed; (2) temporary-failure if the verification has not been completed; (250) other-reason if the verification failed.
	Image transfer activation scheduler	**M**	**M**		**22**	**0**		**0-0:15.0.2.255**				
1	logical_name			octet_string[6]				"00000F0002FF"	--	--	Get	
2	executed_script			script22			0x08	{0-0:10.0.107.255,1}	--	--	Get, (Set)	activate image
3	type			enum			0x10	1	--	--	Get, (Set)	1 is mandatory (Other are optional. If only type 1 is supported Set is optional)
4	execution_time			array[0..1](execution_time_date)			0x18		--	--	Get, Set	the execution can be disabled with writing empty array or {{"FFFFFFFF", "FFFFFFFFFF"}}
	Predefined Scripts - Image activation	**M**	**M**		**9**	**0**		**0-0:10.0.107.255**				
1	logical_name			octet_string[6]				"00000A006BFF"	--	--	Get	
2	scripts			scripts			0x08		--	--	(Get)	The attribute's access rights must be declared in CTI file. Allowed possibilities - no access or Get access
1	execute			long_unsigned			0x20	1	--	--	Action	The activation of this scripts is performed by calling the execute() method to the script identifier 1 of the corresponding script object
	Active firmware identifier	**M**	**M**		**1**	**0**		**1-0:0.2.0.255**				If several FW instances are present inside the device, the B field is used for differentiation: 0=Metrologically relevant part of firmware 1=Application (metrologically non relevant part of firmware) 2=other parts (e.g. modem firmware)
1	logical_name			octet_string[6]				"0100000200FF"	--	--	Get	
2	value			octet_string			0x08		--	--	Get	
	Active firmware signature	**M**	**M**		**1**	**0**		**1-0:0.2.8.255**				
1	logical_name			octet_string[6]				"0100000208FF"	--	--	Get	
2	value			octet_string			0x08		--	--	Get	
	Active firmware identifier 1	**O**	**O**		**1**	**0**		**1-1:0.2.0.255**				
1	logical_name			octet_string[6]				"0101000200FF"	--	--	Get	
2	value			octet_string			0x08		--	--	Get	
	Active firmware signature 1	**O**	**O**		**1**	**0**		**1-1:0.2.8.255**				
1	logical_name			octet_string[6]				"0101000208FF"	--	--	Get	
2	value			octet_string			0x08		--	--	Get	
	Active firmware identifier 2	**O**	**O**		**1**	**0**		**1-2:0.2.0.255**				
1	logical_name			octet_string[6]				"0102000200FF"	--	--	Get	
2	value			octet_string			0x08		--	--	Get	
	Active firmware signature 2	**O**	**O**		**1**	**0**		**1-2:0.2.8.255**				
1	logical_name			octet_string[6]				"0102000208FF"	--	--	Get	
2	value			octet_string			0x08		--	--	Get	
	Electricity related objects - Totals											
	Active energy import (+A)	**M**	**M**		**3**	**0**		**1-0:1.8.0.255**				
1	logical_name			octet_string[6]				"0100010800FF"	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value		Public (16)	Preestablished (102)	Management (1)	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,30}	--	--	Get	
1	reset						0x28		--	--	--	
	Active energy export (–A)	**M**	**M**		**3**	**0**		**1-0:2.8.0.255**				
1	logical_name			octet_string[6]				"0100020800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,30}	--	--	Get	
1	reset						0x28		--	--	--	
	Active energy (|+A|+|-A|) Combined total	**M**	**M**		**3**	**0**		**1-0:15.8.0.255**				
1	logical_name			octet_string[6]				"01000F0800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,30}	--	--	Get	
1	reset						0x28		--	--	--	
	Active energy (|+A|-|-A|) Combined total	**M**	**M**		**3**	**0**		**1-0:16.8.0.255**				
1	logical_name			octet_string[6]				"0100100800FF"	--	--	Get	
2	value			double_long			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,30}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy QI (+Ri)	**M**	**M**		**3**	**0**		**1-0:5.8.0.255**				
1	logical_name			octet_string[6]				"0100050800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy QII (+Rc)	**M**	**M**		**3**	**0**		**1-0:6.8.0.255**				
1	logical_name			octet_string[6]				"0100060800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy QIII (-Ri)	**M**	**M**		**3**	**0**		**1-0:7.8.0.255**				
1	logical_name			octet_string[6]				"0100070800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy QIV (-Rc)	**M**	**M**		**3**	**0**		**1-0:8.8.0.255**				
1	logical_name			octet_string[6]				"0100080800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy import (+R) (QI+QII)	**M**	**M**		**3**	**0**		**1-0:3.8.0.255**				
1	logical_name			octet_string[6]				"0100030800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy export (-R) (QIII+QIV)	**M**	**M**		**3**	**0**		**1-0:4.8.0.255**				
1	logical_name			octet_string[6]				"0100040800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Apparent energy import (+VA) (QI+QIV)	**O**	**O**		**3**	**0**		**1-0:9.8.0.255**				
1	logical_name			octet_string[6]				"0100090800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,31}	--	--	Get	
1	reset						0x28		--	--	--	
	Apparent energy export (-VA) (QII+QIII)	**O**	**O**		**3**	**0**		**1-0:10.8.0.255**				
1	logical_name			octet_string[6]				"01000A0800FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,31}	--	--	Get	
1	reset						0x28		--	--	--	
	Electricity related objects - Rated registers											Should be able to perform at least 4 tariffs.
	Active energy import (+A) rate 1	**M**	**M**		**3**	**0**		**1-0:1.8.1.255**				
1	logical_name			octet_string[6]				"0100010801FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,30}	--	--	Get	
1	reset						0x28		--	--	--	
	Active energy import (+A) rate 2	**M**	**M**		**3**	**0**		**1-0:1.8.2.255**				
1	logical_name			octet_string[6]				"0100010802FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,30}	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
1	reset						0x28	--	--	--	
	Active energy import (+A) rate 3	**M**	**M**		**3**	**0**	**1-0:1.8.3.255**				
1	logical_name			octet_string[6]			"0100010803FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Active energy import (+A) rate 4	**M**	**M**		**3**	**0**	**1-0:1.8.4.255**				
1	logical_name			octet_string[6]			"0100010804FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Active energy export (–A) rate 1	**M**	**M**		**3**	**0**	**1-0:2.8.1.255**				
1	logical_name			octet_string[6]			"0100020801FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Active energy export (–A) rate 2	**M**	**M**		**3**	**0**	**1-0:2.8.2.255**				
1	logical_name			octet_string[6]			"0100020802FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Active energy export (–A) rate 3	**M**	**M**		**3**	**0**	**1-0:2.8.3.255**				
1	logical_name			octet_string[6]			"0100020803FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Active energy export (–A) rate 4	**M**	**M**		**3**	**0**	**1-0:2.8.4.255**				
1	logical_name			octet_string[6]			"0100020804FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Reactive energy (+R) rate 1	**M**	**M**		**3**	**0**	**1-0:3.8.1.255**				
1	logical_name			octet_string[6]			"0100030801FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,32}	--	--	Get	
1	reset						0x28	--	--	--	
	Reactive energy (+R) rate 2	**M**	**M**		**3**	**0**	**1-0:3.8.2.255**				
1	logical_name			octet_string[6]			"0100030802FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,32}	--	--	Get	
1	reset						0x28	--	--	--	
	Reactive energy (+R) rate 3	**M**	**M**		**3**	**0**	**1-0:3.8.3.255**				
1	logical_name			octet_string[6]			"0100030803FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,32}	--	--	Get	
1	reset						0x28	--	--	--	
	Reactive energy (+R) rate 4	**M**	**M**		**3**	**0**	**1-0:3.8.4.255**				
1	logical_name			octet_string[6]			"0100030804FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,32}	--	--	Get	
1	reset						0x28	--	--	--	
	Reactive energy (-R) rate 1	**M**	**M**		**3**	**0**	**1-0:4.8.1.255**				
1	logical_name			octet_string[6]			"0100040801FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,32}	--	--	Get	
1	reset						0x28	--	--	--	
	Reactive energy (-R) rate 2	**M**	**M**		**3**	**0**	**1-0:4.8.2.255**				
1	logical_name			octet_string[6]			"0100040802FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,32}	--	--	Get	
1	reset						0x28	--	--	--	
	Reactive energy (-R) rate 3	**M**	**M**		**3**	**0**	**1-0:4.8.3.255**				
1	logical_name			octet_string[6]			"0100040803FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,32}	--	--	Get	
1	reset						0x28	--	--	--	
	Reactive energy (-R) rate 4	**M**	**M**		**3**	**0**	**1-0:4.8.4.255**				
1	logical_name			octet_string[6]			"0100040804FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,32}	--	--	Get	
1	reset						0x28	--	--	--	
	Reactive energy (QI) rate 1	**O**	**O**		**3**	**0**	**1-0:5.8.1.255**				

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
1	logical_name			octet_string[6]				"0100050801FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QI) rate 2	**O**	**O**		**3**	**0**		**1-0:5.8.2.255**				
1	logical_name			octet_string[6]				"0100050802FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QI) rate 3	**O**	**O**		**3**	**0**		**1-0:5.8.3.255**				
1	logical_name			octet_string[6]				"0100050803FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QI) rate 4	**O**	**O**		**3**	**0**		**1-0:5.8.4.255**				
1	logical_name			octet_string[6]				"0100050804FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QII) rate 1	**O**	**O**		**3**	**0**		**1-0:6.8.1.255**				
1	logical_name			octet_string[6]				"0100060801FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QII) rate 2	**O**	**O**		**3**	**0**		**1-0:6.8.2.255**				
1	logical_name			octet_string[6]				"0100060802FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QII) rate 3	**O**	**O**		**3**	**0**		**1-0:6.8.3.255**				
1	logical_name			octet_string[6]				"0100060803FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QII) rate 4	**O**	**O**		**3**	**0**		**1-0:6.8.4.255**				
1	logical_name			octet_string[6]				"0100060804FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QIII) rate 1	**O**	**O**		**3**	**0**		**1-0:7.8.1.255**				
1	logical_name			octet_string[6]				"0100070801FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QIII) rate 2	**O**	**O**		**3**	**0**		**1-0:7.8.2.255**				
1	logical_name			octet_string[6]				"0100070802FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QIII) rate 3	**O**	**O**		**3**	**0**		**1-0:7.8.3.255**				
1	logical_name			octet_string[6]				"0100070803FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QIII) rate 4	**O**	**O**		**3**	**0**		**1-0:7.8.4.255**				
1	logical_name			octet_string[6]				"0100070804FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QIV) rate 1	**O**	**O**		**3**	**0**		**1-0:8.8.1.255**				
1	logical_name			octet_string[6]				"0100080801FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QIV) rate 2	**O**	**O**		**3**	**0**		**1-0:8.8.2.255**				
1	logical_name			octet_string[6]				"0100080802FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	
1	reset						0x28		--	--	--	
	Reactive energy (QIV) rate 3	**O**	**O**		**3**	**0**		**1-0:8.8.3.255**				
1	logical_name			octet_string[6]				"0100080803FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,32}	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
1	reset						0x28	--	--	--	
	Reactive energy (QIV) rate 4	**O**	**O**		**3**	**0**	**1-0:8.8.4.255**				
1	logical_name			octet_string[6]			"0100080804FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,32}	--	--	Get	
1	reset						0x28	--	--	--	
	Apparent energy import (+VA) rate 1	**O**	**O**		**3**	**0**	**1-0:9.8.1.255**				
1	logical_name			octet_string[6]			"0100090801FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,31}	--	--	Get	
1	reset						0x28	--	--	--	
	Apparent energy import (+VA) rate 2	**O**	**O**		**3**	**0**	**1-0:9.8.2.255**				
1	logical_name			octet_string[6]			"0100090802FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,31}	--	--	Get	
1	reset						0x28	--	--	--	
	Apparent energy import (+VA) rate 3	**O**	**O**		**3**	**0**	**1-0:9.8.3.255**				
1	logical_name			octet_string[6]			"0100090803FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,31}	--	--	Get	
1	reset						0x28	--	--	--	
	Apparent energy import (+VA) rate 4	**O**	**O**		**3**	**0**	**1-0:9.8.4.255**				
1	logical_name			octet_string[6]			"0100090804FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,31}	--	--	Get	
1	reset						0x28	--	--	--	
	Apparent energy export (-VA) rate 1	**O**	**O**		**3**	**0**	**1-0:10.8.1.255**				
1	logical_name			octet_string[6]			"01000A0801FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,31}	--	--	Get	
1	reset						0x28	--	--	--	
	Apparent energy export (-VA) rate 2	**O**	**O**		**3**	**0**	**1-0:10.8.2.255**				
1	logical_name			octet_string[6]			"01000A0802FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,31}	--	--	Get	
1	reset						0x28	--	--	--	
	Apparent energy export (-VA) rate 3	**O**	**O**		**3**	**0**	**1-0:10.8.3.255**				
1	logical_name			octet_string[6]			"01000A0803FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,31}	--	--	Get	
1	reset						0x28	--	--	--	
	Apparent energy export (-VA) rate 4	**O**	**O**		**3**	**0**	**1-0:10.8.4.255**				
1	logical_name			octet_string[6]			"01000A0804FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,31}	--	--	Get	
1	reset						0x28	--	--	--	
	Active energy Combined total (|+A|+|-A|) rate 1	**O**	**O**		**3**	**0**	**1-0:15.8.1.255**				
1	logical_name			octet_string[6]			"01000F0801FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Active energy Combined total (|+A|+|-A|) rate 2	**O**	**O**		**3**	**0**	**1-0:15.8.2.255**				
1	logical_name			octet_string[6]			"01000F0802FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Active energy Combined total (|+A|+|-A|) rate 3	**O**	**O**		**3**	**0**	**1-0:15.8.3.255**				
1	logical_name			octet_string[6]			"01000F0803FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Active energy Combined total (|+A|+|-A|) rate 4	**O**	**O**		**3**	**0**	**1-0:15.8.4.255**				
1	logical_name			octet_string[6]			"01000F0804FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,30}	--	--	Get	
1	reset						0x28	--	--	--	
	Electricity related objects - Demand										
	Demand Register 1 - Active energy import (+A)	**M**	**M**		5	0	**1-0:1.4.0.255**				
1	logical_name			octet_string[6]			"0100010400FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08	--	--	Get	
3	last_average_value			double_long_unsigned			0x10	--	--	Get	
4	scaler_unit			scal_unit_type			0x18 {0,27}	--	--	Get	
5	status			unsigned			0x20 0	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
6	capture_time			octet_string			0x28	--	--	Get	
7	start_time_current			octet_string			0x30	--	--	Get	
8	period			double_long_unsigned			0x38 900	--	--	Get, Set	Changing period of one instance will change the period of all instances automatically!
9	number_of_periods			long_unsigned			0x40 1	--	--	Get, Set	Siliding demand optiona. Changing number_of_periods of one instance will change nunmber_of_periods of all instances automatically!
1	reset						0x48	--	--	--	
2	next_period						0x50	--	--	--	
	Demand Register 2 - Active energy export (–A)	**M**	**M**		5	0	**1-0:2.4.0.255**				
1	logical_name			octet_string[6]			"0100020400FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08	--	--	Get	
3	last_average_value			double_long_unsigned			0x10	--	--	Get	
4	scaler_unit			scal_unit_type			0x18 {0,27}	--	--	Get	
5	status			unsigned			0x20 0	--	--	Get	
6	capture_time			octet_string			0x28	--	--	Get	
7	start_time_current			octet_string			0x30	--	--	Get	
8	period			double_long_unsigned			0x38 900	--	--	Get, Set	Changing period of one instance will change the period of all instances automatically!
9	number_of_periods			long_unsigned			0x40 1	--	--	Get, Set	Siliding demand optiona. Changing number_of_periods of one instance will change nunmber_of_periods of all instances automatically!
1	reset						0x48	--	--	--	
2	next_period						0x50	--	--	--	
	Demand Register 3 - Reactive energy import (+R)	**M**	**M**		5	0	**1-0:3.4.0.255**				
1	logical_name			octet_string[6]			"0100030400FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08	--	--	Get	
3	last_average_value			double_long_unsigned			0x10	--	--	Get	
4	scaler_unit			scal_unit_type			0x18 {0,29}	--	--	Get	
5	status			unsigned			0x20 0	--	--	Get	
6	capture_time			octet_string			0x28	--	--	Get	
7	start_time_current			octet_string			0x30	--	--	Get	
8	period			double_long_unsigned			0x38 900	--	--	Get, Set	Changing period of one instance will change the period of all instances automatically!
9	number_of_periods			long_unsigned			0x40 1	--	--	Get, Set	Siliding demand optiona. Changing number_of_periods of one instance will change nunmber_of_periods of all instances automatically!
1	reset						0x48	--	--	--	
2	next_period						0x50	--	--	--	
	Demand Register 4 - Reactive energy export (-R)	**M**	**M**		5	0	**1-0:4.4.0.255**				
1	logical_name			octet_string[6]			"0100040400FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08	--	--	Get	
3	last_average_value			double_long_unsigned			0x10	--	--	Get	
4	scaler_unit			scal_unit_type			0x18 {0,29}	--	--	Get	
5	status			unsigned			0x20 0	--	--	Get	
6	capture_time			octet_string			0x28	--	--	Get	
7	start_time_current			octet_string			0x30	--	--	Get	
8	period			double_long_unsigned			0x38 900	--	--	Get, Set	Changing period of one instance will change the period of all instances automatically!
9	number_of_periods			long_unsigned			0x40 1	--	--	Get, Set	Siliding demand optiona. Changing number_of_periods of one instance will change nunmber_of_periods of all instances automatically!
1	reset						0x48	--	--	--	
2	next_period						0x50	--	--	--	
	Demand Register 5 - Apparent energy import (+VA)	**O**	**O**		5	0	**1-0:9.4.0.255**				
1	logical_name			octet_string[6]			"0100090400FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08	--	--	Get	
3	last_average_value			double_long_unsigned			0x10	--	--	Get	
4	scaler_unit			scal_unit_type			0x18 {0,28}	--	--	Get	
5	status			unsigned			0x20 0	--	--	Get	
6	capture_time			octet_string			0x28	--	--	Get	
7	start_time_current			octet_string			0x30	--	--	Get	
8	period			double_long_unsigned			0x38 900	--	--	Get, Set	Changing period of one instance will change the period of all instances automatically!
9	number_of_periods			long_unsigned			0x40 1	--	--	Get, Set	Siliding demand optiona. Changing number_of_periods of one instance will change nunmber_of_periods of all instances automatically!
1	reset						0x48	--	--	--	
2	next_period						0x50	--	--	--	
	Demand Register 6 - Apparent energy export (-VA)	**O**	**O**		5	0	**1-0:10.4.0.255**				
1	logical_name			octet_string[6]			"01000A0400FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08	--	--	Get	
3	last_average_value			double_long_unsigned			0x10	--	--	Get	
4	scaler_unit			scal_unit_type			0x18 {0,28}	--	--	Get	
5	status			unsigned			0x20 0	--	--	Get	
6	capture_time			octet_string			0x28	--	--	Get	
7	start_time_current			octet_string			0x30	--	--	Get	
8	period			double_long_unsigned			0x38 900	--	--	Get, Set	Changing period of one instance will change the period of all instances automatically!
9	number_of_periods			long_unsigned			0x40 1	--	--	Get, Set	Siliding demand optiona. Changing number_of_periods of one instance will change nunmber_of_periods of all instances automatically!
1	reset						0x48	--	--	--	
2	next_period						0x50	--	--	--	
	Last Average Demand Register 1 - Active energy import (+A)	**O**	**O**		4	0	**1-0:1.5.0.255**				For display purposes only
1	logical_name			octet_string[6]			"0100010500FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
3	scaler_unit			scal_unit_type			0x10 {0,27}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Last Average Demand Register 2 - Active energy export (–A)	**O**	**O**		4	0	**1-0:2.5.0.255**				For display purposes only
1	logical_name			octet_string[6]			"0100020500FF"	--	--	Get	For the display purpose only
2	value			double_long_unsigned			0x08	--	--	Get	For the display purpose only
3	scaler_unit			scal_unit_type			0x10 {0,27}	--	--	Get	For the display purpose only
4	status			unsigned			0x18	--	--	Get	For the display purpose only
5	capture_time			octet_string			0x20	--	--	Get	For the display purpose only
1	reset			integer			0x38	--	--	--	For the display purpose only
											For the display purpose only
	Last Average Demand Register 3 - Reactive energy import (+R)	**O**	**O**		4	0	**1-0:3.5.0.255**				For display purposes only
1	logical_name			octet_string[6]			"0100030500FF"	--	--	Get	For the display purpose only
2	value			double_long_unsigned			0x08	--	--	Get	For the display purpose only
3	scaler_unit			scal_unit_type			0x10 {0,29}	--	--	Get	For the display purpose only
4	status			unsigned			0x18	--	--	Get	For the display purpose only
5	capture_time			octet_string			0x20	--	--	Get	For the display purpose only
1	reset			integer			0x38	--	--	--	For the display purpose only
											For the display purpose only
	Last Average Demand Register 4 - Reactive energy export (-R)	**O**	**O**		4	0	**1-0:4.5.0.255**				For display purposes only
1	logical_name			octet_string[6]			"0100040500FF"	--	--	Get	For the display purpose only
2	value			double_long_unsigned			0x08	--	--	Get	For the display purpose only
3	scaler_unit			scal_unit_type			0x10 {0,29}	--	--	Get	For the display purpose only
4	status			unsigned			0x18	--	--	Get	For the display purpose only
5	capture_time			octet_string			0x20	--	--	Get	For the display purpose only
1	reset			integer			0x38	--	--	--	For the display purpose only
											For the display purpose only
	Last Average Demand Register 5 - Apparent energy import (+VA)	**O**	**O**		4	0	**1-0:9.5.0.255**				For display purposes only
1	logical_name			octet_string[6]			"0100090500FF"	--	--	Get	For the display purpose only
2	value			double_long_unsigned			0x08	--	--	Get	For the display purpose only
3	scaler_unit			scal_unit_type			0x10 {0,28}	--	--	Get	For the display purpose only
4	status			unsigned			0x18	--	--	Get	For the display purpose only
5	capture_time			octet_string			0x20	--	--	Get	For the display purpose only
1	reset			integer			0x38	--	--	--	For the display purpose only
											For the display purpose only
	Last Average Demand Register 6 - Apparent energy export (-VA)	**O**	**O**		4	0	**1-0:10.5.0.255**				For display purposes only
1	logical_name			octet_string[6]			"01000A0500FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,28}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Maximum Demand Register 1 - Active energy import (+A)	**M**	**M**		4	0	**1-0:1.6.0.255**				
1	logical_name			octet_string[6]			"0100010600FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,27}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Maximum Demand Register 2 - Active energy import (+A) - rate 1	**M**	**M**		4	0	**1-0:1.6.1.255**				
1	logical_name			octet_string[6]			"0100010601FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,27}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Maximum Demand Register 3 - Active energy import (+A) - rate 2	**M**	**M**		4	0	**1-0:1.6.2.255**				
1	logical_name			octet_string[6]			"0100010602FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,27}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Maximum Demand Register 4 - Active energy import (+A) - rate 3	**M**	**M**		4	0	**1-0:1.6.3.255**				
1	logical_name			octet_string[6]			"0100010603FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,27}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Maximum Demand Register 5 - Active energy import (+A) - rate 4	**M**	**M**		4	0	**1-0:1.6.4.255**				
1	logical_name			octet_string[6]			"0100010604FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,27}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Maximum Demand Register 6 - Active energy export (–A)	**M**	**M**		4	0	**1-0:2.6.0.255**				
1	logical_name			octet_string[6]			"0100020600FF"	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value		Public (16)	Preestablished (102)	Management (1)	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,27}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 7 - Active energy export (–A) - rate 1	**M**	**M**		4	0		**1-0:2.6.1.255**				
1	logical_name			octet_string[6]				"0100020601FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,27}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 8 - Active energy export (–A) - rate 2	**M**	**M**		4	0		**1-0:2.6.2.255**				
1	logical_name			octet_string[6]				"0100020602FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,27}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 9 - Active energy export (–A) - rate 3	**M**	**M**		4	0		**1-0:2.6.3.255**				
1	logical_name			octet_string[6]				"0100020603FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,27}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 10 - Active energy export (–A) - rate 4	**M**	**M**		4	0		**1-0:2.6.4.255**				
1	logical_name			octet_string[6]				"0100020604FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,27}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 11 - Reactive energy import (+R)	**M**	**M**		4	0		**1-0:3.6.0.255**				
1	logical_name			octet_string[6]				"0100030600FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 12 - Reactive energy import (+R) - rate 1	**M**	**M**		4	0		**1-0:3.6.1.255**				
1	logical_name			octet_string[6]				"0100030601FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 13 - Reactive energy import (+R) - rate 2	**M**	**M**		4	0		**1-0:3.6.2.255**				
1	logical_name			octet_string[6]				"0100030602FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 14 - Reactive energy import (+R) - rate 3	**M**	**M**		4	0		**1-0:3.6.3.255**				
1	logical_name			octet_string[6]				"0100030603FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 15 - Reactive energy import (+R) - rate 4	**M**	**M**		4	0		**1-0:3.6.4.255**				
1	logical_name			octet_string[6]				"0100030604FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 16 - Reactive energy export (-R)	**M**	**M**		4	0		**1-0:4.6.0.255**				
1	logical_name			octet_string[6]				"0100040600FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 17 - Reactive energy export (-R) - rate 1	**M**	**M**		4	0		**1-0:4.6.1.255**				

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
1	logical_name			octet_string[6]				"0100040601FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 18 - Reactive energy export (-R) - rate 2	**M**	**M**		4	0		**1-0:4.6.2.255**				
1	logical_name			octet_string[6]				"0100040602FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 19 - Reactive energy export (-R) - rate 3	**M**	**M**		4	0		**1-0:4.6.3.255**				
1	logical_name			octet_string[6]				"0100040603FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 20 - Reactive energy export (-R) - rate 4	**M**	**M**		4	0		**1-0:4.6.4.255**				
1	logical_name			octet_string[6]				"0100040604FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	not use; return 0
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 21 - Apparent energy import (+VA)	**O**	**O**		4	0		**1-0:9.6.0.255**				
1	logical_name			octet_string[6]				"0100090600FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,28}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 22 - Apparent energy import (+VA) - rate 1	**O**	**O**		4	0		**1-0:9.6.1.255**				
1	logical_name			octet_string[6]				"0100090601FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,28}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 23 - Apparent energy import (+VA) - rate 2	**O**	**O**		4	0		**1-0:9.6.2.255**				
1	logical_name			octet_string[6]				"0100090602FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,28}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 24 - Apparent energy import (+VA) - rate 3	**O**	**O**		4	0		**1-0:9.6.3.255**				
1	logical_name			octet_string[6]				"0100090603FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,28}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 25 - Apparent energy import (+VA) - rate 4	**O**	**O**		4	0		**1-0:9.6.4.255**				
1	logical_name			octet_string[6]				"0100090604FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,28}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 26 - Apparent energy export (-VA)	**O**	**O**		4	0		**1-0:10.6.0.255**				
1	logical_name			octet_string[6]				"01000A0600FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,28}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Maximum Demand Register 27 - Apparent energy export (-VA) - rate 1	**O**	**O**		4	0		**1-0:10.6.1.255**				
1	logical_name			octet_string[6]				"01000A0601FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
3	scaler_unit			scal_unit_type			0x10 {0,28}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Maximum Demand Register 28 - Apparent energy export (-VA) - rate 2	**O**	**O**		4	0	**1-0:10.6.2.255**				
1	logical_name			octet_string[6]			"01000A0602FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,28}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Maximum Demand Register 29 - Apparent energy export (-VA) - rate 3	**O**	**O**		4	0	**1-0:10.6.3.255**				
1	logical_name			octet_string[6]			"01000A0603FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,28}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Maximum Demand Register 30 - Apparent energy export (-VA) - rate 4	**O**	**O**		4	0	**1-0:10.6.4.255**				
1	logical_name			octet_string[6]			"01000A0604FF"	--	--	Get	
2	value			double_long_unsigned			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10 {0,28}	--	--	Get	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	
1	reset			integer			0x38	--	--	--	
	Electricity related objects - Profiles										
	Profile status - Load profile with period 1	**M**	**M**		**1**	**0**	**0-0:96.10.1.255**				
1	logical_name			octet_string[6]			"0000600A01FF"	--	--	Get	
2	status			unsigned			0x08	--	--	Get	bitmask :: {PDN,reserved,CAD,reserved,DST,DNV,CIV,ERR}
	Load profile with period 1 i.e. *General Load Profile*	**M**	**M**		**7**	**1**	**1-0:99.1.0.255**				
1	logical_name			octet_string[6]			"0100630100FF"	--	--	Get	
2	buffer			array			0x08	--	--	Get	selective access per range and entry
3	capture_objects			array[2..*](capture_object_definition)			0x10 {{8,0-0:1.0.0.255,2,0},{1,0-0:96.10.1.255,2,0}, {3,1-0:1.8.0.255,2,0}, {3,1-0:2.8.0.255,2,0}, ...}	--	--	Get, Set	clock;AMR profile status; +A;-A
4	capture_period			double_long_unsigned			0x18 choice(300, 600, 900, 1800, 3600)	--	--	Get, Set	15 minutes
5	sort_method			enum			0x20 choice(1,4)	--	--	Get	unsorted (FIFO) or sorted (smallest)
6	sort_object			capture_object_definition			0x28 choice({*,*-*:*.*.*.*,*}, {8,0-0:1.0.0.255,2,0})	--	--	Get	unsorted or sorted by clock depending on choice in attr. sort_method i.e. if sort_method=sorted then sort_object=0-0:1.0.0.255
7	entries_in_use			double_long_unsigned			0x30	--	--	Get	
8	profile_entries			double_long_unsigned			0x38 [960..*]	--	--	Get, (Set)	≥10 days, optional set to restrict maximum capacity according to national laws
1	reset						0x58	--	--	Action	
	Profile status - Load profile with period 2	**M**	**M**		**1**	**0**	**0-0:96.10.2.255**				
1	logical_name			octet_string[6]			"0000600A02FF"	--	--	Get	
2	status			unsigned			0x08	--	--	Get	bitmask :: {PDN,reserved,CAD,reserved,DST,DNV,CIV,ERR}
	Load profile with period 2 i.e. *Daily Values Profile*	**M**	**M**		**7**	**1**	**1-0:99.2.0.255**				
1	logical_name			octet_string[6]			"0100630200FF"	--	--	Get	
2	buffer			array			0x08	--	--	Get	selective access per range and entry
3	capture_objects			array[2..*](capture_object_definition)			0x10 {{8,0-0:1.0.0.255,2,0},{1,0-0:96.10.2.255,2,0},{3,1-0:1.8.*.255,2,0}, ...}	--	--	Get, Set	clock;AMR profile status; choice amongst rated registers, etc.
4	capture_period			double_long_unsigned			0x18 choice(86400, 300, 600, 900, 1800, 3600)	--	--	Get, Set	daily
5	sort_method			enum			0x20 choice(1,4)	--	--	Get	unsorted (FIFO) or sorted (smallest)
6	sort_object			capture_object_definition			0x28 choice({*,*-*:*.*.*.*,*}, {8,0-0:1.0.0.255,2,0})	--	--	Get	unsorted or sorted by clock depending on choice in attr. sort_method i.e. if sort_method=sorted then sort_object=0-0:1.0.0.255
7	entries_in_use			double_long_unsigned			0x30	--	--	Get	
8	profile_entries			double_long_unsigned			0x38 [240..*]	--	--	Get, (Set)	≥10 days, optional set to restrict maximum capacity according to national laws
1	reset						0x58	--	--	Action	
	Electricity related objects - PQ, monitored values										
	Number of power failures in any phase	**M**	**M**		**1**	**0**	**0-0:96.7.21.255**				
1	logical_name			octet_string[6]			"0000600715FF"	--	--	Get	
2	value			long_unsigned			0x08	--	--	Get	
	Number of long power failures in any phase	**M**	**M**		**1**	**0**	**0-0:96.7.9.255**				
1	logical_name			octet_string[6]			"0000600709FF"	--	--	Get	
2	value			long_unsigned			0x08	--	--	Get	
	Time threshold for long power failure	**M**	**M**		**3**	**0**	**0-0:96.7.20.255**				
1	logical_name			octet_string[6]			"0000600714FF"	--	--	Get	
2	value			long_unsigned			0x08 180	--	Set	Get, Set	
3	scaler_unit			scal_unit_type			0x10 {0,7}	--	--	Get	
1	reset						0x28	--	--	--	
	Duration of last long power failure in any phase	**M**	**M**		**3**	**0**	**0-0:96.7.19.255**				

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
1	logical_name			octet_string[6]				"0000600713FF"	--	--	Get	
2	value			choice(long_unsigned, double_long_unsigned)			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Threshold for voltage sag	**M**	**M**		**3**	**0**		**1-0:12.31.0.255**				
1	logical_name			octet_string[6]				"01000C1F00FF"	--	--	Get	
2	value			long_unsigned			0x08	207	--	Set	Get, Set	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	
1	reset						0x28		--	--	--	
	Time threshold for voltage sag	**M**	**M**		**3**	**0**		**1-0:12.43.0.255**				
1	logical_name			octet_string[6]				"01000C2B00FF"	--	--	Get	
2	value			long_unsigned			0x08	30	--	Set	Get, Set	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Number of voltage sags in phase L1	**M**	**M**		**1**	**0**		**1-0:32.32.0.255**				
1	logical_name			octet_string[6]				"0100202000FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
	Number of voltage sags in phase L2	**NA**	**M**		**1**	**0**		**1-0:52.32.0.255**				
1	logical_name			octet_string[6]				"0100342000FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
	Number of voltage sags in phase L3	**NA**	**M**		**1**	**0**		**1-0:72.32.0.255**				
1	logical_name			octet_string[6]				"0100482000FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
	Duration of last voltage sag in phase L1	**M**	**M**		**3**	**0**		**1-0:32.33.0.255**				
1	logical_name			octet_string[6]				"0100202100FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Duration of last voltage sag in phase L2	**NA**	**M**		**3**	**0**		**1-0:52.33.0.255**				
1	logical_name			octet_string[6]				"0100342100FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Duration of last voltage sag in phase L3	**NA**	**M**		**3**	**0**		**1-0:72.33.0.255**				
1	logical_name			octet_string[6]				"0100482100FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Magnitude of last voltage sag in phase L1	**M**	**M**		**3**	**0**		**1-0:32.34.0.255**				
1	logical_name			octet_string[6]				"0100202200FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	
1	reset						0x28		--	--	--	
	Magnitude of last voltage sag in phase L2	**NA**	**M**		**3**	**0**		**1-0:52.34.0.255**				
1	logical_name			octet_string[6]				"0100342200FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	
1	reset						0x28		--	--	--	
	Magnitude of last voltage sag in phase L3	**NA**	**M**		**3**	**0**		**1-0:72.34.0.255**				
1	logical_name			octet_string[6]				"0100482200FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	
1	reset						0x28		--	--	--	
	Threshold for voltage swell	**M**	**M**		**3**	**0**		**1-0:12.35.0.255**				
1	logical_name			octet_string[6]				"01000C2300FF"	--	--	Get	
2	value			long_unsigned			0x08	253	--	Set	Get, Set	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	
1	reset						0x28		--	--	--	
	Time threshold for voltage swell	**M**	**M**		**3**	**0**		**1-0:12.44.0.255**				
1	logical_name			octet_string[6]				"01000C2C00FF"	--	--	Get	
2	value			long_unsigned			0x08	30	--	Set	Get, Set	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Number of voltage swells in phase L1	**M**	**M**		**1**	**0**		**1-0:32.36.0.255**				
1	logical_name			octet_string[6]				"0100202400FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
	Number of voltage swells in phase L2	**NA**	**M**		**1**	**0**		**1-0:52.36.0.255**				
1	logical_name			octet_string[6]				"0100342400FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
	Number of voltage swells in phase L3	**NA**	**M**		**1**	**0**		**1-0:72.36.0.255**				
1	logical_name			octet_string[6]				"0100482400FF"	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value		Public (16)	Preestablished (102)	Management (1)	
2	value			long_unsigned			0x08		--	--	Get	
	Duration of last voltage swell in phase L1	**M**	**M**		**3**	**0**		**1-0:32.37.0.255**				
1	logical_name			octet_string[6]				"0100202500FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Duration of last voltage swell in phase L2	**NA**	**M**		**3**	**0**		**1-0:52.37.0.255**				
1	logical_name			octet_string[6]				"0100342500FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Duration of last voltage swell in phase L3	**NA**	**M**		**3**	**0**		**1-0:72.37.0.255**				
1	logical_name			octet_string[6]				"0100482500FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Magnitude of last voltage swell in phase L1	**M**	**M**		**3**	**0**		**1-0:32.38.0.255**				
1	logical_name			octet_string[6]				"0100202600FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	
1	reset						0x28		--	--	--	
	Magnitude of last voltage swell in phase L2	**NA**	**M**		**3**	**0**		**1-0:52.38.0.255**				
1	logical_name			octet_string[6]				"0100342600FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	
1	reset						0x28		--	--	--	
	Magnitude of last voltage swell in phase L3	**NA**	**M**		**3**	**0**		**1-0:72.38.0.255**				
1	logical_name			octet_string[6]				"0100482600FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	
1	reset						0x28		--	--	--	
	Threshold for missing voltage (voltage cut)	**M**	**M**		**3**	**0**		**1-0:12.39.0.255**				
1	logical_name			octet_string[6]				"01000C2700FF"	--	--	Get	
2	value			long_unsigned			0x08	103	--	Set	Get, Set	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	
1	reset						0x28		--	--	--	
	Time threshold for voltage cut	**M**	**M**		**3**	**0**		**1-0:12.45.0.255**				
1	logical_name			octet_string[6]				"01000C2D00FF"	--	--	Get	
2	value			long_unsigned			0x08	30	--	Set	Get, Set	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Power Failure Event Log	**M**	**M**		**7**	**1**		**1-0:99.97.0.255**				
1	logical_name			octet_string[6]				"0100636100FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range
3	capture_objects			array[2](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {3,0-0:96.7.19.255,2,0}}	--	--	Get, (Set)	clock;duration of long power failures in any phase
4	capture_period			double_long_unsigned			0x18	0	--	--	Get	asynchronously
5	sort_method			enum			0x20	1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28	{*,*-*:*.*.*.*,*,*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[10..*]	--	--	Get	
1	reset			integer			0x58		--	--	Action	
	Event Object - Power Quality Log	**M**	**M**		**1**	**0**		**0-0:96.11.4.255**				
1	logical_name			octet_string[6]				"0000600B04FF"	--	--	Get	
2	value			enum			0x08		--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	Power Quality Log	**M**	**M**		**7**	**1**		**0-0:99.98.4.255**				
1	logical_name			octet_string[6]				"0000636204FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range
3	capture_objects			array[2](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {1,0-0:96.11.4.255,2,0}}	--	--	Get, (Set)	clock;event code
4	capture_period			double_long_unsigned			0x18	0	--	--	Get	asynchronously
5	sort_method			enum			0x20	1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28	{*,*-*:*.*.*.*,*,*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[100..*]	--	--	Get	
1	reset						0x58		--	--	Action	
	Profile status - Power Quality Profile	**M**	**M**		**1**	**0**		**1-0:96.10.1.255**				
1	logical_name			octet_string[6]				"0100600A01FF"	--	--	Get	
2	value			unsigned			0x08		--	--	Get	
	Power Quality Profile	**M**	**M**		**7**	**1**		**1-0:99.14.0.255**				
1	logical_name			octet_string[6]				"0100630E00FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range

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Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
3 capture_objects			array[2](capture_object_definition)			0x10	choice({ {8,0-0:1.0.0.255,2,0}, {1,1-0:96.10.1.255,2,0}, {1-0:32.24.0.255}, {1-0:52.24.0.255}, {1-0:72.24.0.255}, {1-0:31.24.0.255}, {1-0:51.24.0.255}, {1-0:71.24.0.255}, {1-0:1.4.0.255}, {1-0:2.4.0.255}, {1-0:3.4.0.255}, {1-0:4.4.0.255}}, { {8,0-0:1.0.0.255,2,0}, {1,1-0:96.10.1.255,2,0}, {1-0:32.24.0.255}, {1-0:31.24.0.255}, {1-0:1.4.0.255}, {1-0:2.4.0.255}, {1-0:3.4.0.255}, {1-0:4.4.0.255}})	--	--	Get, (Set)	choice between 3ph and 1ph
4 capture_period			double_long_unsigned			0x18	choice(60, 300,600, 900, 3600)	--	--	Get, Set	the value is equal to the averaging period of all the captured objects. Further, the averaging periods of the captured objects and the capture_period must be synchronised
5 sort_method			enum			0x20	choice(1,4)	--	--	Get	unsorted (FIFO) or sorted by clock
6 sort_object			capture_object_definition			0x28	choice({*,*-*:*.*.*.*,*,*}, {8,0-0:1.0.0.255,2,0})	--	--	Get	
7 entries_in_use			double_long_unsigned			0x30		--	--	Get	
8 profile_entries			double_long_unsigned			0x38	[960..*]	--	--	Get, (Set)	
1 reset						0x58		--	--	Action	
Instantaneous voltage L1	**M**	**M**		**3**	**0**		**1-0:32.7.0.255**				
1 logical_name			octet_string[6]				"0100200700FF"	--	--	Get	
2 value			long_unsigned			0x08		--	--	Get	
3 scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	scaler=0 or-1, unit=V resolution: 0 or 0.0 V
1 reset						0x28		--	--	--	
Average voltage L1	**M**	**M**		**3**	**0**		**1-0:32.24.0.255**				
1 logical_name			octet_string[6]				"0100201800FF"	--	--	Get	
2 value			long_unsigned			0x08		--	--	Get	10 minutes average
3 scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	scaler=0 or-1, unit=V resolution: 0 or 0.0 V
1 reset						0x28		--	--	--	
Instantaneous current L1	**M**	**M**		**3**	**0**		**1-0:31.7.0.255**				
1 logical_name			octet_string[6]				"01001F0700FF"	--	--	Get	
2 value			long_unsigned			0x08		--	--	Get	
3 scaler_unit			scal_unit_type			0x10	choice({0,33},{-1,33},{-2,33})	--	--	Get	scaler=0,-1 or-2, unit=A resolution: 0, 0.0 or 0.00 A
1 reset						0x28		--	--	--	
Average current L1	**M**	**M**		**3**	**0**		**1-0:31.24.0.255**				
1 logical_name			octet_string[6]				"01001F1800FF"	--	--	Get	
2 value			long_unsigned			0x08		--	--	Get	10 minutes average
3 scaler_unit			scal_unit_type			0x10	choice({0,33},{-1,33},{-2,33})	--	--	Get	scaler=0,-1 or-2, unit=A resolution: 0, 0.0 or 0.00 A
1 reset						0x28		--	--	--	
Sliding Average current L1 (for fuse supervision)	**M-D**	**M-D**		**5**	**0**		**1-0:31.4.0.255**				
1 logical_name			octet_string[6]				"01001F0400FF"	--	--	Get	
2 current_average_value			double_long_unsigned			0x08		--	--	Get	
3 last_average_value			double_long_unsigned			0x10		--	--	Get	
4 scaler_unit			scal_unit_type			0x18	choice({0,33},{-1,33},{-2,33})	--	--	Get	
5 status			unsigned			0x20	0	--	--	Get	
6 capture_time			octet_string			0x28		--	--	Get	
7 start_time_current			octet_string			0x30		--	--	Get	
8 period			double_long_unsigned			0x38	1	--	--	Get	
9 number_of_periods			long_unsigned			0x40	90	--	--	Get, Set	
1 reset						0x48		--	--	--	
2 next_period						0x50		--	--	--	
Instantaneous voltage L2	**NA**	**M**		**3**	**0**		**1-0:52.7.0.255**				
1 logical_name			octet_string[6]				"0100340700FF"	--	--	Get	
2 value			long_unsigned			0x08		--	--	Get	
3 scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	scaler=0 or-1, unit=V resolution: 0 or 0.0 V
1 reset						0x28		--	--	--	
Average voltage L2	**NA**	**M**		**3**	**0**		**1-0:52.24.0.255**				
1 logical_name			octet_string[6]				"0100341800FF"	--	--	Get	
2 value			long_unsigned			0x08		--	--	Get	10 minutes average
3 scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	scaler=0 or-1, unit=V resolution: 0 or 0.0 V
1 reset						0x28		--	--	--	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
	Instantaneous current L2	**NA**	**M**		**3**	**0**		**1-0:51.7.0.255**				
1	logical_name			octet_string(6)				"0100330700FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,33},{-1,33},{-2,33})	--	--	Get	scaler=0,-1 or-2, unit=A resolution: 0, 0.0 or 0.00 A
1	reset						0x28		--	--	--	
	Average current L2	**NA**	**M**		**3**	**0**		**1-0:51.24.0.255**				
1	logical_name			octet_string(6)				"0100331800FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	10 minutes
3	scaler_unit			scal_unit_type			0x10	choice({0,33},{-1,33},{-2,33})	--	--	Get	scaler=0,-1 or-2, unit=A resolution: 0, 0.0 or 0.00 A
1	reset						0x28		--	--	--	
	Sliding Average current L2 (for fuse supervision)	**NA**	**M-D**		**5**	**0**		**1-0:51.4.0.255**				
1	logical_name			octet_string(6)				"0100330400FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08		--	--	Get	
3	last_average_value			double_long_unsigned			0x10		--	--	Get	
4	scaler_unit			scal_unit_type			0x18	choice({0,33},{-1,33},{-2,33})	--	--	Get	
5	status			unsigned			0x20	0	--	--	Get	
6	capture_time			octet_string			0x28		--	--	Get	
7	start_time_current			octet_string			0x30		--	--	Get	
8	period			double_long_unsigned			0x38	1	--	--	Get	
9	number_of_periods			long_unsigned			0x40	90	--	--	Get, Set	
1	reset						0x48		--	--	--	
2	next_period						0x50		--	--	--	
	Instantaneous voltage L3	**NA**	**M**		**3**	**0**		**1-0:72.7.0.255**				
1	logical_name			octet_string(6)				"0100480700FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	scaler=0 or-1, unit=V resolution: 0 or 0.0 V
1	reset						0x28		--	--	--	
	Average voltage L3	**NA**	**M**		**3**	**0**		**1-0:72.24.0.255**				
1	logical_name			octet_string(6)				"0100481800FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	10 minutes average voltage
3	scaler_unit			scal_unit_type			0x10	choice({0,35},{-1,35})	--	--	Get	scaler=0 or-1, unit=V resolution: 0 or 0.0 V
1	reset						0x28		--	--	--	
	Instantaneous current L3	**NA**	**M**		**3**	**0**		**1-0:71.7.0.255**				
1	logical_name			octet_string(6)				"0100470700FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,33},{-1,33},{-2,33})	--	--	Get	scaler=0,-1 or-2, unit=A resolution: 0, 0.0 or 0.00 A
1	reset						0x28		--	--	--	
	Average current L3	**NA**	**M**		**3**	**0**		**1-0:71.24.0.255**				
1	logical_name			octet_string(6)				"0100471800FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	10 minutes
3	scaler_unit			scal_unit_type			0x10	choice({0,33},{-1,33},{-2,33})	--	--	Get	scaler=0,-1 or-2, unit=A resolution: 0, 0.0 or 0.00 A
1	reset						0x28		--	--	--	
	Sliding Average current L3 (for fuse supervision)	**NA**	**M-D**		**5**	**0**		**1-0:71.4.0.255**				
1	logical_name			octet_string(6)				"0100470400FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08		--	--	Get	
3	last_average_value			double_long_unsigned			0x10		--	--	Get	
4	scaler_unit			scal_unit_type			0x18	choice({0,33},{-1,33},{-2,33})	--	--	Get	
5	status			unsigned			0x20	0	--	--	Get	
6	capture_time			octet_string			0x28		--	--	Get	
7	start_time_current			octet_string			0x30		--	--	Get	
8	period			double_long_unsigned			0x38	1	--	--	Get	
9	number_of_periods			long_unsigned			0x40	90	--	--	Get, Set	
1	reset						0x48		--	--	--	
2	next_period						0x50		--	--	--	
	Instantaneous current (sum over all phases)	**M**	**M**		**3**	**0**		**1-0:90.7.0.255**				
1	logical_name			octet_string(6)				"01005A0700FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	total current (sum of all phases)
3	scaler_unit			scal_unit_type			0x10	choice({0,33},{-1,33},{-2,33})	--	--	Get	scaler=0,-1 or-2, unit=A resolution: 0, 0.0 or 0.00 A
1	reset						0x28		--	--	--	
	Instantaneous net frequency; any phase	**O**	**O**		**3**	**0**		**1-0:14.7.0.255**				
1	logical_name			octet_string(6)				"01000E0700FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	choice({0,44},{-1,44},{-2,44})	--	--	Get	scaler=0,-1 or-2, unit=Hz resolution: 0, 0.0 or 0.00 Hz
1	reset						0x28		--	--	--	
	Instantaneous active power (|+A|+|-A|)	**M**	**M**		**3**	**0**		**1-0:15.7.0.255**				
1	logical_name			octet_string(6)				"01000F0700FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	sum of all phases: |QI+QIV|+|QII+QIII|
3	scaler_unit			scal_unit_type			0x10	{0,27}	--	--	Get	
1	reset						0x28		--	--	--	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
	Instantaneous active import power (+A)	**M**	**M**		**3**	**0**		**1-0:1.7.0.255**				
1	logical_name			octet_string[6]				"0100010700FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,27}	--	--	Get	
1	reset						0x28		--	--	--	
	Instantaneous active export power (-A)	**M**	**M**		**3**	**0**		**1-0:2.7.0.255**				
1	logical_name			octet_string[6]				"0100020700FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,27}	--	--	Get	
1	reset						0x28		--	--	--	
	Instantaneous reactive import power (+R)	**M**	**M**		**3**	**0**		**1-0:3.7.0.255**				
1	logical_name			octet_string[6]				"0100030700FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
1	reset						0x28		--	--	--	
	Instantaneous reactive export power (-R)	**M**	**M**		**3**	**0**		**1-0:4.7.0.255**				
1	logical_name			octet_string[6]				"0100040700FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,29}	--	--	Get	
1	reset						0x28		--	--	--	
	Instantaneous apparent import power (+VA)	**O**	**O**		**3**	**0**		**1-0:9.7.0.255**				
1	logical_name			octet_string[6]				"0100090700FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,28}	--	--	Get	
1	reset						0x28		--	--	--	
	Instantaneous apparent export power (-VA)	**O**	**O**		**3**	**0**		**1-0:10.7.0.255**				
1	logical_name			octet_string[6]				"01000A0700FF"	--	--	Get	
2	value			double_long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{0,28}	--	--	Get	
1	reset						0x28		--	--	--	
	Average Import Power (+A)	**M-D**	**M-D**		**5**	**0**		**1-0:1.24.0.255**				Sliding demand with its own period
1	logical_name			octet_string[6]				"0100011800FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08		--	--	Get	
3	last_average_value			double_long_unsigned			0x10		--	--	Get	
4	scaler_unit			scal_unit_type			0x18	{0,27}	--	--	Get	
5	status			unsigned			0x20	0	--	--	Get	
6	capture_time			octet_string			0x28		--	--	Get	
7	start_time_current			octet_string			0x30		--	--	Get	
8	period			double_long_unsigned			0x38	60	--	--	Get	
9	number_of_periods			long_unsigned			0x40	15	--	--	Get, Set	
1	reset						0x48		--	--	--	
2	next_period						0x50		--	--	--	
	Average Net Power (|+A|-|-A|)	**O**	**O**		**5**	**0**		**1-0:16.24.0.255**				Sliding demand with its own period
1	logical_name			octet_string[6]				"0100101800FF"	--	--	Get	
2	current_average_value			double_long			0x08		--	--	Get	
3	last_average_value			double_long			0x10		--	--	Get	
4	scaler_unit			scal_unit_type			0x18	{0,27}	--	--	Get	
5	status			unsigned			0x20	0	--	--	Get	
6	capture_time			octet_string			0x28		--	--	Get	
7	start_time_current			octet_string			0x30		--	--	Get	
8	period			double_long_unsigned			0x38	60	--	--	Get	
9	number_of_periods			long_unsigned			0x40	15	--	--	Get, Set	
1	reset						0x48		--	--	--	
2	next_period						0x50		--	--	--	
	Average Total Power (|+A|+|-A|)	**M-D**	**M-D**		**5**	**0**		**1-0:15.24.0.255**				Sliding demand with its own period
1	logical_name			octet_string[6]				"01000F1800FF"	--	--	Get	
2	current_average_value			double_long_unsigned			0x08		--	--	Get	
3	last_average_value			double_long_unsigned			0x10		--	--	Get	
4	scaler_unit			scal_unit_type			0x18	{0,27}	--	--	Get	
5	status			unsigned			0x20	0	--	--	Get	
6	capture_time			octet_string			0x28		--	--	Get	
7	start_time_current			octet_string			0x30		--	--	Get	
8	period			double_long_unsigned			0x38	60	--	--	Get	
9	number_of_periods			long_unsigned			0x40	15	--	--	Get, Set	
1	reset						0x48		--	--	--	
2	next_period						0x50		--	--	--	
	Instantaneous Power factor (+A/+VA)	**O**	**O**		**3**	**0**		**1-0:13.7.0.255**				
1	logical_name			octet_string[6]				"01000D0700FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{-3,255}	--	--	Get	
1	reset						0x28		--	--	--	
	Last Average Power factor (+A/+VA)	**O**	**O**		4	0		**1-0:13.5.0.255**				For display purposes only
1	logical_name			octet_string[6]				"01000D0500FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{-3,255}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Minimum Power factor (+A/+VA)	**O**	**O**		4	0	**1-0:13.3.0.255**					For display purposes only
1	logical_name			octet_string[6]				"01000D0300FF"	--	--	Get	
2	value			long_unsigned			0x08		--	--	Get	
3	scaler_unit			scal_unit_type			0x10	{-3,255}	--	--	Get	
4	status			unsigned			0x18		--	--	Get	
5	capture_time			octet_string			0x20		--	--	Get	
1	reset			integer			0x38		--	--	--	
	Measurement Period 3 for Instantaneous values	**O**	**O**		**3**	**0**	**1-0:0.8.2.255**					Used with simple average scheme (see IC3, D=24). If object not used period is considered as implicit!
1	logical_name			octet_string[6]				"0100000802FF"	--	--	Get	
2	value			double_long_unsigned			0x08	600	--	Set	Get, Set	
3	scaler_unit			scal_unit_type			0x10	{0,7}	--	--	Get	
1	reset						0x28		--	--	--	
	Local communication - M-Bus											
	M-Bus master port setup 1	**M-M**	**M-M**		**74**	**0**	**0-0:24.6.0.255**					Wired M-Bus port setup
1	logical_name			octet_string[6]				"0000180600FF"	--	--	Get	
2	comm_speed			enum			0x08	3	--	--	Get	3== 2400 bps fix
	M-Bus master port setup 2	**O**	**O**		**74**	**0**	**0-1:24.6.0.255**					Wireless M-Bus port setup
1	logical_name			octet_string[6]				"0001180600FF"	--	--	Get	
2	comm_speed			enum			0x08		--	--	Get	(0) 300 bps, (1) 600 bps, (2) 1200 bps, (3) 2400 bps, (4) 4800 bps, (5) 9600 bps, (6) 19200 bps...
	M-Bus client channel 1	**M-M**	**M-M**		**72**	**1**	**0-1:24.1.0.255**					
1	logical_name			octet_string[6]				"0001180100FF"	--	--	Get	
2	mbus_port_reference			octet_string[6]			0x08	choice(0-0:24.6.0.255, 0-1:24.6.0.255)	--	--	Get, Set	
3	capture_definition			array			0x10		--	--	Get, Set	
4	capture_period			double_long_unsigned			0x18		--	Set	Get, Set	
5	primary_address			unsigned			0x20		--	--	Get, Set	
6	identification_number			double_long_unsigned			0x28		--	--	Get	
7	manufacturer_id			long_unsigned			0x30		--	--	Get	
8	version			unsigned			0x38		--	--	Get	
9	device_type			unsigned			0x40		--	--	Get	
10	access_number			unsigned			0x48		--	--	Get	
11	status			unsigned			0x50		--	--	Get	
12	alarm			unsigned			0x58		--	--	Get	
13	configuration			long-unsigned			0x60		--	--	Get	
14	encryption_key_status			enum			0x68		--	--	Get	
1	slave_install			unsigned			0x70		--	--	Action	
2	slave_deinstall			unsigned			0x78		--	--	Action	
3	capture			integer			0x80		--	--	(Action)	
4	reset_alarm			integer			0x88		--	--	(Action)	
5	synchronize_clock			integer			0x90		--	--	(Action)	
6	data_send			array			0x98		--	--	(Action)	
7	set_encryption_key			octet_string			0xA0		--	--	Action	
8	transfer_key			octet_string			0xA8		--	--	Action	
	M-Bus client channel 2	**M-M**	**M-M**		**72**	**1**	**0-2:24.1.0.255**					
1	logical_name			octet_string[6]				"0002180100FF"	--	--	Get	
2	mbus_port_reference			octet_string[6]			0x08	choice(0-0:24.6.0.255, 0-1:24.6.0.255)	--	--	Get, Set	
3	capture_definition			array			0x10		--	--	Get, Set	
4	capture_period			double_long_unsigned			0x18		--	Set	Get, Set	
5	primary_address			unsigned			0x20		--	--	Get, Set	
6	identification_number			double_long_unsigned			0x28		--	--	Get	
7	manufacturer_id			long_unsigned			0x30		--	--	Get	
8	version			unsigned			0x38		--	--	Get	
9	device_type			unsigned			0x40		--	--	Get	
10	access_number			unsigned			0x48		--	--	Get	
11	status			unsigned			0x50		--	--	Get	
12	alarm			unsigned			0x58		--	--	Get	
13	configuration			long-unsigned			0x60		--	--	Get	
14	encryption_key_status			enum			0x68		--	--	Get	
1	slave_install			unsigned			0x70		--	--	Action	
2	slave_deinstall			unsigned			0x78		--	--	Action	
3	capture			integer			0x80		--	--	(Action)	
4	reset_alarm			integer			0x88		--	--	(Action)	
5	synchronize_clock			integer			0x90		--	--	(Action)	
6	data_send			array			0x98		--	--	(Action)	
7	set_encryption_key			octet_string			0xA0		--	--	Action	
8	transfer_key			octet_string			0xA8		--	--	Action	
	M-Bus client channel 3	**M-M**	**M-M**		**72**	**0**	**0-3:24.1.0.255**					
1	logical_name			octet_string[6]				"0003180100FF"	--	--	Get	
2	mbus_port_reference			octet_string[6]			0x08	choice(0-0:24.6.0.255, 0-1:24.6.0.255)	--	--	Get, Set	
3	capture_definition			array			0x10		--	--	Get, Set	
4	capture_period			double_long_unsigned			0x18		--	Set	Get, Set	
5	primary_address			unsigned			0x20		--	--	Get, Set	
6	identification_number			double_long_unsigned			0x28		--	--	Get	
7	manufacturer_id			long_unsigned			0x30		--	--	Get	
8	version			unsigned			0x38		--	--	Get	
9	device_type			unsigned			0x40		--	--	Get	
10	access_number			unsigned			0x48		--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
11	status			unsigned			0x50	--	--	Get	
12	alarm			unsigned			0x58	--	--	Get	
13	configuration			long-unsigned			0x60	--	--	Get	
14	encryption_key_status			enum			0x68	--	--	Get	
1	slave_install			unsigned			0x70	--	--	Action	
2	slave_deinstall			unsigned			0x78	--	--	Action	
3	capture			integer			0x80	--	--	(Action)	
4	reset_alarm			integer			0x88	--	--	(Action)	
5	synchronize_clock			integer			0x90	--	--	(Action)	
6	data_send			array			0x98	--	--	(Action)	
7	set_encryption_key			octet_string			0xA0	--	--	Action	
8	transfer_key			octet_string			0xA8	--	--	Action	
	M-Bus client channel 4	**M-M**	**M-M**		**72**	**0**	**0-4:24.1.0.255**				
1	logical_name			octet_string[6]			"0004180100FF"	--	--	Get	
2	mbus_port_reference			octet_string[6]			0x08 choice(0-0:24.6.0.255, 0-1:24.6.0.255)	--	--	Get, Set	
3	capture_definition			array			0x10	--	--	Get, Set	
4	capture_period			double_long_unsigned			0x18	--	Set	Get, Set	
5	primary_address			unsigned			0x20	--	--	Get, Set	
6	identification_number			double_long_unsigned			0x28	--	--	Get	
7	manufacturer_id			long_unsigned			0x30	--	--	Get	
8	version			unsigned			0x38	--	--	Get	
9	device_type			unsigned			0x40	--	--	Get	
10	access_number			unsigned			0x48	--	--	Get	
11	status			unsigned			0x50	--	--	Get	
12	alarm			unsigned			0x58	--	--	Get	
13	configuration			long-unsigned			0x60	--	--	Get	
14	encryption_key_status			enum			0x68	--	--	Get	
1	slave_install			unsigned			0x70	--	--	Action	
2	slave_deinstall			unsigned			0x78	--	--	Action	
3	capture			integer			0x80	--	--	(Action)	
4	reset_alarm			integer			0x88	--	--	(Action)	
5	synchronize_clock			integer			0x90	--	--	(Action)	
6	data_send			array			0x98	--	--	(Action)	
7	set_encryption_key			octet_string			0xA0	--	--	Action	
8	transfer_key			octet_string			0xA8	--	--	Action	
	M-Bus Value channel 1, instance 1	**M-M**	**M-M**		**4**	**0**	**0-1:24.2.1.255**				
1	logical_name			octet_string[6]			"0001180201FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 1, instance 2	**M-M**	**M-M**		**4**	**0**	**0-1:24.2.2.255**				
1	logical_name			octet_string[6]			"0001180202FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 1, instance 3	**M-M**	**M-M**		**4**	**0**	**0-1:24.2.3.255**				
1	logical_name			octet_string[6]			"0001180203FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 1, instance 4	**M-M**	**M-M**		**4**	**0**	**0-1:24.2.4.255**				
1	logical_name			octet_string[6]			"0001180204FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 1, instance 5	**M-M**	**M-M**		**4**	**0**	**0-1:24.2.5.255**				
1	logical_name			octet_string[6]			"0001180205FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 1, instance 6	**M-M**	**M-M**		**4**	**0**	**0-1:24.2.6.255**				
1	logical_name			octet_string[6]			"0001180206FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 1, instance 7	**M-M**	**M-M**		**4**	**0**	**0-1:24.2.7.255**				
1	logical_name			octet_string[6]			"0001180207FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 1, instance 8	**M-M**	**M-M**		**4**	**0**	**0-1:24.2.8.255**				
1	logical_name			octet_string[6]			"0001180208FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 2, instance 1	**M-M**	**M-M**		**4**	**0**	**0-2:24.2.1.255**				
1	logical_name			octet_string[6]			"0002180201FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 2, instance 2	**M-M**	**M-M**		**4**	**0**	**0-2:24.2.2.255**				
1	logical_name			octet_string[6]			"0002180202FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 2, instance 3	**M-M**	**M-M**		**4**	**0**	**0-2:24.2.3.255**				
1	logical_name			octet_string[6]			"0002180203FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 2, instance 4	**M-M**	**M-M**		**4**	**0**	**0-2:24.2.4.255**				
1	logical_name			octet_string[6]			"0002180204FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 2, instance 5	**M-M**	**M-M**		**4**	**0**	**0-2:24.2.5.255**				
1	logical_name			octet_string[6]			"0002180205FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 2, instance 6	**M-M**	**M-M**		**4**	**0**	**0-2:24.2.6.255**				
1	logical_name			octet_string[6]			"0002180206FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 2, instance 7	**M-M**	**M-M**		**4**	**0**	**0-2:24.2.7.255**				
1	logical_name			octet_string[6]			"0002180207FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3	scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4	status			unsigned			0x18	--	--	Get	
5	capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1	reset						0x38	--	--	(Action)	
	M-Bus Value channel 2, instance 8	**M-M**	**M-M**		**4**	**0**	**0-2:24.2.8.255**				
1	logical_name			octet_string[6]			"0002180208FF"	--	--	Get	

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Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	
M-Bus Value channel 3, instance 1	**M-M**	**M-M**		**4**	**0**	**0-3:24.2.1.255**				
1 logical_name			octet_string[6]			"0003180201FF"	--	--	Get	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	
M-Bus Value channel 3, instance 2	**M-M**	**M-M**		**4**	**0**	**0-3:24.2.2.255**				
1 logical_name			octet_string[6]			"0003180202FF"	--	--	Get	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	
M-Bus Value channel 3, instance 3	**M-M**	**M-M**		**4**	**0**	**0-3:24.2.3.255**				
1 logical_name			octet_string[6]			"0003180203FF"	--	--	Get	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	
M-Bus Value channel 3, instance 4	**M-M**	**M-M**		**4**	**0**	**0-3:24.2.4.255**				
1 logical_name			octet_string[6]			"0003180204FF"	--	--	Get	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	
M-Bus Value channel 3, instance 5	**M-M**	**M-M**		**4**	**0**	**0-3:24.2.5.255**				
1 logical_name			octet_string[6]			"0003180205FF"	--	--	Get	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	
M-Bus Value channel 3, instance 6	**M-M**	**M-M**		**4**	**0**	**0-3:24.2.6.255**				
1 logical_name			octet_string[6]			"0003180206FF"	--	--	Get	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	
M-Bus Value channel 3, instance 7	**M-M**	**M-M**		**4**	**0**	**0-3:24.2.7.255**				
1 logical_name			octet_string[6]			"0003180207FF"	--	--	Get	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	
M-Bus Value channel 3, instance 8	**M-M**	**M-M**		**4**	**0**	**0-3:24.2.8.255**				
1 logical_name			octet_string[6]			"0003180208FF"	--	--	Get	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	
M-Bus Value channel 4, instance 1	**M-M**	**M-M**		**4**	**0**	**0-4:24.2.1.255**				
1 logical_name			octet_string[6]			"0004180201FF"	--	--	Get	
2 value			choice(double_long,double_long_unsigned,octet_string)			0x08	--	--	Get	
3 scaler_unit			scal_unit_type			0x10	--	--	Get, (Set)	
4 status			unsigned			0x18	--	--	Get	
5 capture_time			octet_string			0x20	--	--	Get	time of last successful readout; format same as clock.time
1 reset						0x38	--	--	(Action)	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
	M-Bus Value channel 4, instance 2	**M-M**	**M-M**		**4**	**0**	**0-4:24.2.2.255**				
1	logical_name			octet_string[6]			"0004180202FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)		0x08		--	--	Get	
3	scaler_unit			scal_unit_type		0x10		--	--	Get, (Set)	
4	status			unsigned		0x18		--	--	Get	
5	capture_time			octet_string		0x20		--	--	Get	time of last successful readout; format same as clock.time
1	reset					0x38		--	--	(Action)	
	M-Bus Value channel 4, instance 3	**M-M**	**M-M**		**4**	**0**	**0-4:24.2.3.255**				
1	logical_name			octet_string[6]			"0004180203FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)		0x08		--	--	Get	
3	scaler_unit			scal_unit_type		0x10		--	--	Get, (Set)	
4	status			unsigned		0x18		--	--	Get	
5	capture_time			octet_string		0x20		--	--	Get	time of last successful readout; format same as clock.time
1	reset					0x38		--	--	(Action)	
	M-Bus Value channel 4, instance 4	**M-M**	**M-M**		**4**	**0**	**0-4:24.2.4.255**				
1	logical_name			octet_string[6]			"0004180204FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)		0x08		--	--	Get	
3	scaler_unit			scal_unit_type		0x10		--	--	Get, (Set)	
4	status			unsigned		0x18		--	--	Get	
5	capture_time			octet_string		0x20		--	--	Get	time of last successful readout; format same as clock.time
1	reset					0x38		--	--	(Action)	
	M-Bus Value channel 4, instance 5	**M-M**	**M-M**		**4**	**0**	**0-4:24.2.5.255**				
1	logical_name			octet_string[6]			"0004180205FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)		0x08		--	--	Get	
3	scaler_unit			scal_unit_type		0x10		--	--	Get, (Set)	
4	status			unsigned		0x18		--	--	Get	
5	capture_time			octet_string		0x20		--	--	Get	time of last successful readout; format same as clock.time
1	reset					0x38		--	--	(Action)	
	M-Bus Value channel 4, instance 6	**M-M**	**M-M**		**4**	**0**	**0-4:24.2.6.255**				
1	logical_name			octet_string[6]			"0004180206FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)		0x08		--	--	Get	
3	scaler_unit			scal_unit_type		0x10		--	--	Get, (Set)	
4	status			unsigned		0x18		--	--	Get	
5	capture_time			octet_string		0x20		--	--	Get	time of last successful readout; format same as clock.time
1	reset					0x38		--	--	(Action)	
	M-Bus Value channel 4, instance 7	**M-M**	**M-M**		**4**	**0**	**0-4:24.2.7.255**				
1	logical_name			octet_string[6]			"0004180207FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)		0x08		--	--	Get	
3	scaler_unit			scal_unit_type		0x10		--	--	Get, (Set)	
4	status			unsigned		0x18		--	--	Get	
5	capture_time			octet_string		0x20		--	--	Get	time of last successful readout; format same as clock.time
1	reset					0x38		--	--	(Action)	
	M-Bus Value channel 4, instance 8	**M-M**	**M-M**		**4**	**0**	**0-4:24.2.8.255**				
1	logical_name			octet_string[6]			"0004180208FF"	--	--	Get	
2	value			choice(double_long,double_long_unsigned,octet_string)		0x08		--	--	Get	
3	scaler_unit			scal_unit_type		0x10		--	--	Get, (Set)	
4	status			unsigned		0x18		--	--	Get	
5	capture_time			octet_string		0x20		--	--	Get	time of last successful readout; format same as clock.time
1	reset					0x38		--	--	(Action)	
	M-Bus Device ID 1 channel 1	**M-M**	**M-M**		**1**	**0**	**0-1:96.1.0.255**				
1	logical_name			octet_string[6]			"0001600100FF"	--	--	Get	
2	value			octet_string[0..48]		0x08		--	--	Get, Set	
	M-Bus Device ID 1 channel 2	**M-M**	**M-M**		**1**	**0**	**0-2:96.1.0.255**				
1	logical_name			octet_string[6]			"0002600100FF"	--	--	Get	
2	value			octet_string[0..48]		0x08		--	--	Get, Set	
	M-Bus Device ID 1 channel 3	**M-M**	**M-M**		**1**	**0**	**0-3:96.1.0.255**				
1	logical_name			octet_string[6]			"0003600100FF"	--	--	Get	
2	value			octet_string[0..48]		0x08		--	--	Get, Set	
	M-Bus Device ID 1 channel 4	**M-M**	**M-M**		**1**	**0**	**0-4:96.1.0.255**				
1	logical_name			octet_string[6]			"0004600100FF"	--	--	Get	
2	value			octet_string[0..48]		0x08		--	--	Get, Set	
	M-Bus Device ID 2 channel 1	**M-M**	**M-M**		**1**	**0**	**0-1:96.1.1.255**				
1	logical_name			octet_string[6]			"0001600101FF"	--	--	Get	
2	value			octet_string[0..48]		0x08		--	--	Get, Set	
	M-Bus Device ID 2 channel 2	**M-M**	**M-M**		**1**	**0**	**0-2:96.1.1.255**				
1	logical_name			octet_string[6]			"0002600101FF"	--	--	Get	
2	value			octet_string[0..48]		0x08		--	--	Get, Set	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
	M-Bus Device ID 2 channel 3	**M-M**	**M-M**		**1**	**0**		**0-3:96.1.1.255**				
1	logical_name			octet_string[6]				"0003600101FF"	--	--	Get	
2	value			octet_string[0..48]			0x08		--	--	Get, Set	
	M-Bus Device ID 2 channel 4	**M-M**	**M-M**		**1**	**0**		**0-4:96.1.1.255**				
1	logical_name			octet_string[6]				"0004600101FF"	--	--	Get	
2	value			octet_string[0..48]			0x08		--	--	Get, Set	
	Profile status for M-Bus Master Load profile 1	**M-M**	**M-M**		**1**	**0**		**0-1:96.10.3.255**				
1	logical_name			octet_string[6]				"0001600A03FF"	--	--	Get	
2	status			unsigned			0x08		--	--	Get	
	Profile status for M-Bus Master Load profile 2	**M-M**	**M-M**		**1**	**0**		**0-2:96.10.3.255**				
1	logical_name			octet_string[6]				"0002600A03FF"	--	--	Get	
2	status			unsigned			0x08		--	--	Get	
	Profile status for M-Bus Master Load profile 3	**M-M**	**M-M**		**1**	**0**		**0-3:96.10.3.255**				
1	logical_name			octet_string[6]				"0003600A03FF"	--	--	Get	
2	status			unsigned			0x08		--	--	Get	
	Profile status for M-Bus Master Load profile 4	**M-M**	**M-M**		**1**	**0**		**0-4:96.10.3.255**				
1	logical_name			octet_string[6]				"0004600A03FF"	--	--	Get	
2	status			unsigned			0x08		--	--	Get	
	M-Bus Master Load profile for channel 1	**M-M**	**M-M**		**7**	**1**		**0-1:24.3.0.255**				Hourly interval readings of M-Bus devices
1	logical_name			octet_string[6]				"0001180300FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range and entry
3	capture_objects			array[2..*](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {1, 0-1:96.10.3.255,2,0}, {4, 0-1:24.2.[1..4].255,2,0}, {4, 0-1:24.2.[1..4].255,2,0}, {4, 0-1:24.2.[1..4].255,2,0}, {4, 0-1:24.2.[1..4].255,2,0}}	--	--	Get, Set	clock;AMR status; M-Bus Master value object;
4	capture_period			double_long_unsigned			0x18	3600	--	--	Get, Set	every hour
5	sort_method			enum			0x20	choice(1,4)	--	--	Get	unsorted (FIFO) or sorted (smallest)
6	sort_object			capture_object_definition			0x28	choice({*,*-*:*.*.*.*.*.*}, {8,0-0:1.0.0.255,2,0})	--	--	Get	unsorted or sorted by clock depending on choice in attr. sort_method i.e. if sort_method=sorted then sort_object=0-0:1.0.0.255
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[240..*]	--	--	Get	≥10 days, optional set to restrict maximum capacity according to national laws
1	reset						0x58		--	--	Action	
	M-Bus Master Load profile for channel 2	**M-M**	**M-M**		**7**	**1**		**0-2:24.3.0.255**				Hourly interval readings of M-Bus devices
1	logical_name			octet_string[6]				"0002180300FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range and entry
3	capture_objects			array[2..*](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {1, 0-2:96.10.3.255,2,0}, {4, 0-2:24.2.[1..4].255,2,0}, {4, 0-2:24.2.[1..4].255,2,0}, {4, 0-2:24.2.[1..4].255,2,0}, {4, 0-2:24.2.[1..4].255,2,0}}	--	--	Get, Set	clock;AMR status; M-Bus Master value object;
4	capture_period			double_long_unsigned			0x18	3600	--	--	Get, Set	every hour
5	sort_method			enum			0x20	choice(1,4)	--	--	Get	unsorted (FIFO) or sorted (smallest)
6	sort_object			capture_object_definition			0x28	choice({*,*-*:*.*.*.*.*.*}, {8,0-0:1.0.0.255,2,0})	--	--	Get	unsorted or sorted by clock depending on choice in attr. sort_method i.e. if sort_method=sorted then sort_object=0-0:1.0.0.255
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[240..*]	--	--	Get	≥10 days, optional set to restrict maximum capacity according to national laws
1	reset						0x58		--	--	Action	
	M-Bus Master Load profile for channel 3	**M-M**	**M-M**		**7**	**1**		**0-3:24.3.0.255**				Hourly interval readings of M-Bus devices
1	logical_name			octet_string[6]				"0003180300FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range and entry
3	capture_objects			array[2..*](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {1, 0-3:96.10.3.255,2,0}, {4, 0-3:24.2.[1..4].255,2,0}, {4, 0-3:24.2.[1..4].255,2,0}, {4, 0-3:24.2.[1..4].255,2,0}, {4, 0-3:24.2.[1..4].255,2,0}}	--	--	Get, Set	clock;AMR status; M-Bus Master value object;
4	capture_period			double_long_unsigned			0x18	3600	--	--	Get, Set	every hour
5	sort_method			enum			0x20	choice(1,4)	--	--	Get	unsorted (FIFO) or sorted (smallest)
6	sort_object			capture_object_definition			0x28	choice({*,*-*:*.*.*.*.*.*}, {8,0-0:1.0.0.255,2,0})	--	--	Get	unsorted or sorted by clock depending on choice in attr. sort_method i.e. if sort_method=sorted then sort_object=0-0:1.0.0.255
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[240..*]	--	--	Get	≥10 days, optional set to restrict maximum capacity according to national laws
1	reset						0x58		--	--	Action	
	M-Bus Master Load profile for channel 4	**M-M**	**M-M**		**7**	**1**		**0-4:24.3.0.255**				Hourly interval readings of M-Bus devices
1	logical_name			octet_string[6]				"0004180300FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range and entry

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
3	capture_objects			array[2..*](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {1, 0-4:96.10.3.255,2,0}, {4, 0-4:24.2.[1..4].255,2,0}, {4, 0-4:24.2.[1..4].255,2,0}, {4, 0-4:24.2.[1..4].255,2,0}, {4, 0-4:24.2.[1..4].255,2,0}}	--	--	Get, Set	clock;AMR status; M-Bus Master value object;
4	capture_period			double_long_unsigned			0x18	3600	--	--	Get, Set	every hour
5	sort_method			enum			0x20	choice(1,4)	--	--	Get	unsorted (FIFO) or sorted (smallest)
6	sort_object			capture_object_definition			0x28	choice({*,*-*:*.*.*.*,*,*}, {8,0-0:1.0.0.255,2,0})	--	--	Get	unsorted or sorted by clock depending on choice in attr. sort_method i.e. if sort_method=sorted then sort_object=0-0:1.0.0.255
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[240..*]	--	--	Get	≥10 days, optional set to restrict maximum capacity according to national laws
1	reset						0x58		--	--	Action	
	M-Bus Master Disconnect control object 1	**M-M**	**M-M**		**70**	**0**		**0-1:24.4.0.255**				Controls the opening and closing of an M-Bus disconnector (e.g. gas valve)
1	logical_name			octet_string[6]				"0001180400FF"	--	--	Get	
2	output_state			boolean			0x08		--	--	Get	
3	control_state			enum			0x10		--	--	Get	
4	control_mode			enum			0x18		--	Set	Get, Set	
1	remote_disconnect			integer			0x20		--	--	Action	
2	remote_connect			integer			0x28		--	--	Action	
	M-Bus Master Disconnect control object 2	**M-M**	**M-M**		**70**	**0**		**0-2:24.4.0.255**				Controls the opening and closing of an M-Bus disconnector (e.g. gas valve)
1	logical_name			octet_string[6]				"0002180400FF"	--	--	Get	
2	output_state			boolean			0x08		--	--	Get	
3	control_state			enum			0x10		--	--	Get	
4	control_mode			enum			0x18		--	Set	Get, Set	
1	remote_disconnect			integer			0x20		--	--	Action	
2	remote_connect			integer			0x28		--	--	Action	
	M-Bus Master Disconnect control object 3	**M-M**	**M-M**		**70**	**0**		**0-3:24.4.0.255**				Controls the opening and closing of an M-Bus disconnector (e.g. gas valve)
1	logical_name			octet_string[6]				"0003180400FF"	--	--	Get	
2	output_state			boolean			0x08		--	--	Get	
3	control_state			enum			0x10		--	--	Get	
4	control_mode			enum			0x18		--	Set	Get, Set	
1	remote_disconnect			integer			0x20		--	--	Action	
2	remote_connect			integer			0x28		--	--	Action	
	M-Bus Master Disconnect control object 4	**M-M**	**M-M**		**70**	**0**		**0-4:24.4.0.255**				Controls the opening and closing of an M-Bus disconnector (e.g. gas valve)
1	logical_name			octet_string[6]				"0004180400FF"	--	--	Get	
2	output_state			boolean			0x08		--	--	Get	
3	control_state			enum			0x10		--	--	Get	
4	control_mode			enum			0x18		--	Set	Get, Set	
1	remote_disconnect			integer			0x20		--	--	Action	
2	remote_connect			integer			0x28		--	--	Action	
	M-Bus Disconnect control scheduler	**M-M**	**M-M**		**22**	**0**		**0-1:15.0.1.255**				
1	logical_name			octet_string[6]				"00010F0001FF"	--	--	Get	
2	executed_script			script22			0x08	{0-1:10.0.106.255, [1..16]}	--	--	Get, Set	
3	type			enum			0x10	1	--	--	Get, (Set)	1 is mandatory (Other are optional. If only type 1 is supported Set is optional)
4	execution_time			array[0..1](execution_time_date)			0x18	{{"FFFFFFFF", "FFFFFFFFFF"}}	--	--	Get, Set	dedicated timepoint for connection or disconnection
	M-Bus Disconnect Control Activity Calendar	**M-M**	**M-M**		**20**	**0**		**0-1:13.0.1.255**				
1	logical_name			octet_string[6]				"00010D0001FF"	--	--	Get	
2	calendar_name_active			octet_string[0..8]			0x08		--	--	Get	
3	season_profile_active			array[0..*]			0x10		--	--	Get	
4	week_profile_table_active			array[0..*]			0x18		--	--	Get	
5	day_profile_table_active			array[0..*]			0x20		--	--	Get	
6	calendar_name_passive			octet_string[0..8]			0x28		--	Set	Get, Set	
7	season_profile_passive			array[0..*]			0x30		--	Set	Get, Set	
8	week_profile_table_passive			array[0..*]			0x38		--	Set	Get, Set	
9	day_profile_table_passive			array[0..*]			0x40		--	Set	Get, Set	
10	activate_passive_calendar_time			octet_string[12]			0x48		--	Set	Get, Set	
1	activate_passive_calendar			integer			0x50		--	(Action)	(Action)	
	M-Bus Disconnect Control Special Days Table	**M-M**	**M-M**		**11**	**0**		**0-1:11.0.1.255**				
1	logical_name			octet_string[6]				"00010B0001FF"	--	--	Get	
2	entries			array[0..30]			0x08		--	Set	Get, Set	
	M-Bus Disconnector script table	**M-M**	**M-M**		**9**	**0**		**0-1:10.0.106.255**				
1	logical_name			octet_string[6]				"00010A006AFF"	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	

2	scripts			array[16](script9)			0x08 {
							{1,{{1,70, 0-1:24.4.0.255,3,2}}},
							{2,{{1,70, 0-1:24.4.0.255,3,1}}},
							{3,{{1,70, 0-2:24.4.0.255,3,2}}},
							{4,{{1,70, 0-2:24.4.0.255,3,1}}},
							{5,{{1,70, 0-3:24.4.0.255,3,2}}},
							{6,{{1,70, 0-3:24.4.0.255,3,1}}},
							{7,{{1,70, 0-4:24.4.0.255,3,2}}},
							{8,{{1,70, 0-4:24.4.0.255,3,1}}},
							{9,{{2,70, 0-1:24.4.0.255,1,0}}},
							{10,{{2,70, 0-1:24.4.0.255,2,0}}},
							{11,{{2,70, 0-2:24.4.0.255,1,0}}},
							{12,{{2,70, 0-2:24.4.0.255,2,0}}},
							{13,{{2,70, 0-3:24.4.0.255,1,0}}},
							{14,{{2,70, 0-3:24.4.0.255,2,0}}},
							{15,{{2,70, 0-4:24.4.0.255,1,0}}},
							{16,{{2,70, 0-4:24.4.0.255,2,0}}}
							}
								--	--	Get	local and remote connect and disconnect methods of the M-Bus disconnector objects, see E disconnector script table for further details

1	execute			long_unsigned			0x20	--	--	--	Executes the script specified in parameter data. If data matches one of the script_identifiers in the script table, then the corresponding action_specification is executed. No direct method call possible via remote communication
	Event Objects - M-Bus Master Control logs 1	**M-M**	**M-M**		**1**	**0**	**0-1:96.11.4.255**				
1	logical_name			octet_string[6]			"0001600B04FF"	--	--	Get	
2	value			enum			0x08	--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	Event Objects - M-Bus Master Control logs 2	**M-M**	**M-M**		**1**	**0**	**0-2:96.11.4.255**				
1	logical_name			octet_string[6]			"0002600B04FF"	--	--	Get	
2	value			enum			0x08	--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	Event Objects - M-Bus Master Control logs 3	**M-M**	**M-M**		**1**	**0**	**0-3:96.11.4.255**				
1	logical_name			octet_string[6]			"0003600B04FF"	--	--	Get	
2	value			enum			0x08	--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	Event Objects - M-Bus Master Control logs 4	**M-M**	**M-M**		**1**	**0**	**0-4:96.11.4.255**				
1	logical_name			octet_string[6]			"0004600B04FF"	--	--	Get	
2	value			enum			0x08	--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	M-Bus Master Control log object 1	**M-M**	**M-M**		**7**	**1**	**0-1:24.5.0.255**				Changes of the states related to the disconnect control are recorded (open, close)
1	logical_name			octet_string[6]			"0001180500FF"	--	--	Get	
2	buffer			array			0x08	--	--	Get	selective access per range
3	capture_objects			array[2](capture_object_definition)			0x10 {{8,0-0:1.0.0.255,2,0}, {1, 0-1:96.11.4.255, 2, 0}}	--	--	Get, (Set)	clock;control event code
4	capture_period			double_long_unsigned			0x18 0	--	--	Get	asynchronously
5	sort_method			enum			0x20 1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28 {*,*-*:*.*.*.*,*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30	--	--	Get	
8	profile_entries			double_long_unsigned			0x38 [10..*]	--	--	Get	
1	reset						0x58	--	--	Action	
	M-Bus Master Control log object 2	**M-M**	**M-M**		**7**	**1**	**0-2:24.5.0.255**				Changes of the states related to the disconnect control are recorded (open, close)
1	logical_name			octet_string[6]			"0002180500FF"	--	--	Get	
2	buffer			array			0x08	--	--	Get	selective access per range
3	capture_objects			array[2](capture_object_definition)			0x10 {{8,0-0:1.0.0.255,2,0}, {1, 0-2:96.11.4.255, 2, 0}}	--	--	Get, (Set)	clock;control event code
4	capture_period			double_long_unsigned			0x18 0	--	--	Get	asynchronously
5	sort_method			enum			0x20 1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28 {*,*-*:*.*.*.*,*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30	--	--	Get	
8	profile_entries			double_long_unsigned			0x38 [10..*]	--	--	Get	
1	reset						0x58	--	--	Action	
	M-Bus Master Control log object 3	**M-M**	**M-M**		**7**	**1**	**0-3:24.5.0.255**				Changes of the states related to the disconnect control are recorded (open, close)
1	logical_name			octet_string[6]			"0003180500FF"	--	--	Get	
2	buffer			array			0x08	--	--	Get	selective access per range
3	capture_objects			array[2](capture_object_definition)			0x10 {{8,0-0:1.0.0.255,2,0}, {1, 0-3:96.11.4.255, 2, 0}}	--	--	Get, (Set)	clock;control event code
4	capture_period			double_long_unsigned			0x18 0	--	--	Get	asynchronously
5	sort_method			enum			0x20 1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28 {*,*-*:*.*.*.*,*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30	--	--	Get	
8	profile_entries			double_long_unsigned			0x38 [10..*]	--	--	Get	
1	reset						0x58	--	--	Action	
	M-Bus Master Control log object 4	**M-M**	**M-M**		**7**	**1**	**0-4:24.5.0.255**				Changes of the states related to the disconnect control are recorded (open, close)
1	logical_name			octet_string[6]			"0004180500FF"	--	--	Get	
2	buffer			array			0x08	--	--	Get	selective access per range

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
3	capture_objects			array[2](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {1, 0-4:96.11.4.255, 2, 0}}	--	--	Get, (Set)	clock;control event code
4	capture_period			double_long_unsigned			0x18	0	--	--	Get	asynchronously
5	sort_method			enum			0x20	1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28	{*.*.*.*.*.*.*.*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[10..*]	--	--	Get	
1	reset						0x58		--	--	Action	
	Event Object - M-Bus Event Log	**M-M**	**M-M**		**1**	**0**		**0-0:96.11.3.255**				
1	logical_name			octet_string[6]				"0000600B03FF"	--	--	Get	
2	value			enum			0x08		--	--	Get	event number (0 to 255); at first power up or if no events were yet generated the value should return 255
	M-Bus Event Log	**M-M**	**M-M**		**7**	**1**		**0-0:99.98.3.255**				Event log containing errors and alarms related to all M-Bus devices
1	logical_name			octet_string[6]				"0000636203FF"	--	--	Get	
2	buffer			array			0x08		--	--	Get	selective access per range
3	capture_objects			array[2](capture_object_definition)			0x10	{{8,0-0:1.0.0.255,2,0}, {1, 0-0:96.11.3.255, 2, 0}}	--	--	Get, (Set)	clock;event code
4	capture_period			double_long_unsigned			0x18	0	--	--	Get	asynchronously
5	sort_method			enum			0x20	1	--	--	Get	unsorted (FIFO)
6	sort_object			capture_object_definition			0x28	{*.*.*.*.*.*.*.*}	--	--	Get	unsorted
7	entries_in_use			double_long_unsigned			0x30		--	--	Get	
8	profile_entries			double_long_unsigned			0x38	[10..*]	--	--	Get	
1	reset						0x58		--	--	Action	
	M-Bus function control	**M-M**	**M-M**		**122**	**0**		**0-1:44.1.0.255**				
1	logical_name			octet_string[6]				"00012C0100FF"	--	--	Get	
2	activation_status			array					--	--	Get	
3	function_list			array[4](functional_block)				{ {"\LPCAP_M1",{7,0-1:24.3.0.255}}, {"\LPCAP_M2",{7,0-2:24.3.0.255}}, {"\LPCAP_M3",{7,0-3:24.3.0.255}}, {"\LPCAP_M4",{7,0-4:24.3.0.255}} }	--	--	Get	
1	set_function_status							array	--	--	Action	
	M-Bus function control scheduler	**M-M**	**M-M**		**22**	**0**		**0-1:15.0.7.255**				
1	logical_name			octet_string[6]				"00010F0007FF"	--	--	Get	
2	executed_script			script22			0x08	{0-1:10.0.111.255, [1..10]}	--	--	Get, Set	
3	type			enum			0x10	1	--	--	Get, (Set)	1 is mandatory (others are optional. If only type 1 is supported Set is optional)
4	execution_time			array[0..1](execution_time_date)			0x18		--	--	Get, Set	dedicated timepoint for enabling and disabling functions; the execution can be disabled with writing empty array or {{"FFFFFFFF", "FFFFFFFFF"}}
	M-Bus function control script table	**M-M**	**M-M**		**9**	**0**		**0-1:10.0.111.255**				
1	logical_name			octet_string[6]				"00010A006FFF"	--	--	Get	
2	scripts			array[10](script9)			0x08	{ {1,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M1",1},{"\LPCAP_M2",1},{"\LPCAP_M3",1},{"\LPCAP_M4",1}}}}}, //enable M-Bus profile 1..4 {2,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M1",0},{"\LPCAP_M2",0},{"\LPCAP_M3",0},{"\LPCAP_M4",0}}}}}, //disable M-Bus profile 1..4 {3,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M1",1}}}}}, //enable M-Bus profile 1 {4,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M1",0}}}}}, //disable M-Bus profile 1 {5,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M2",1}}}}}, //enable M-Bus profile 2 {6,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M2",0}}}}}, //disable M-Bus profile 2 {7,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M3",1}}}}}, //enable M-Bus profile 3 {8,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M3",0}}}}}, //disable M-Bus profile 3 {9,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M4",1}}}}}, //enable M-Bus profile 4 {10,{{2,122,0-1:44.1.0.255,1,{{"\LPCAP_M4",0}}}}} //disable M-Bus profile 4 }	--	--	Get	scripts to enable and disable the capturing of load profile 1 and 2 (opt-in, opt-out)
1	execute			long_unsigned			0x20		--	--	--	Executes the script specified in parameter data. If data matches one of the script_identifiers in the script table, then the corresponding action_specification is executed. No direct method call possible via remote communication
	Local communication - IEC/HDLC Optical port											
	IEC local port setup - IEC Optical port	**O**	**O**		**19**	**1**		**0-0:20.0.0.255**				Used in case IEC1107 is supported on the Optical port s well
1	logical_name			octet_string[6]				"0000140000FF"	--	--	Get	
2	default_mode			enum			0x08		--	Set	Get, Set	
3	default_baud			enum			0x10		--	Set	Get, Set	
4	prop_baud			enum			0x18		--	Set	Get, Set	
5	response_time			enum			0x20		--	Set	Get, Set	
6	device_addr			octet_string			0x28		--	Set	Get, Set	
7	pass_p1			octet_string			0x30		--	Set	Set	
8	pass_p2			octet_string			0x38		--	Set	Set	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
9	pass_p3			octet_string			0x40	--	Set	Set	
	IEC HDLC setup - HDLC Optical port	**M**	**M**		**23**	**1**	**0-0:22.0.0.255**				Mandatory HDLC on Optical port.
1	logical_name			octet_string[6]			"0000160000FF"	--	--	Get	
2	comm_speed			enum			0x08 5	--	Set	Get, Set	
3	window_size_transmit			unsigned			0x10 1	--	Set	Get, Set	
4	window_size_receive			unsigned			0x18 1	--	Set	Get, Set	
5	max_info_field_lenght_transmit			long_unsigned			0x20 [32..2030]	--	Set	Get, Set	
6	max_info_field_lenght_receive			long_unsigned			0x28 [32..2030]	--	Set	Get, Set	
7	inter_octet_time_out			long_unsigned			0x30 [20..6000]	--	Set	Get, Set	
8	inactivity_time_out			long_unsigned			0x38	--	Set	Get, Set	
9	device_address			long_unsigned			0x40	--	--	Get, Set	
	Abstract objects - TCP/IP profile setup										
	TCP-UDP setup	**M**	**M**		**41**	**0**	**0-0:25.0.0.255**				
1	logical_name			octet_string[6]			"0000190000FF"	--	--	Get	
2	TCP-UDP_port			long_unsigned			0x08 choice(4059, 61616)	--	--	Get, Set	
3	IP_reference			octet_string[6]			0x10 Choice(0-0:25.1.0.255,0-0:25.7.0.255)	--	--	Get, (Set)	optional set to allow switching from IPv4 to IPv6
4	MSS			long_unsigned			0x18 1280	--	--	Get, Set	
5	nb_of_sim_conn			unsigned			0x20 1	--	--	Get, Set	
6	inactivity_time_out			long_unsigned			0x28 180	--	--	Get, Set	
	IPv4 setup	**M-I4**	**M-I4**		**42**	**0**	**0-0:25.1.0.255**				
1	logical_name			octet_string[6]			"0000190100FF"	--	--	Get	
2	DL_reference			octet_string[6]			0x08 Choice(0-0:25.3.0.255, 0-0:25.2.0.255)	--	--	Get, Set	
3	IP_address			double_long_unsigned			0x10	--	--	Get, Set	
4	multicast_IP_address			array[0..*]			0x18	--	--	Get, Set	
5	IP_options			array[0..*]			0x20	--	--	Get, Set	
6	subnet_mask			double_long_unsigned			0x28	--	--	Get, Set	
7	gateway_IP_address			double_long_unsigned			0x30	--	--	Get, Set	
8	use_DHCP_flag			boolean			0x38	--	--	Get, Set	
9	primary_DNS_address			double_long_unsigned			0x40	--	--	Get, Set	
10	secondary_DNS_address			double_long_unsigned			0x48	--	--	Get, Set	
1	add_mc_IP_address			double_long_unsigned			0x60	--	--	--	
2	delete_mc_IP_address			double_long_unsigned			0x68	--	--	--	
3	get_nbof_mc_IP_addresses			unsigned			0x70	--	--	--	
	IPv6 setup	**M-I6**	**M-I6**		**48**	**0**	**0-0:25.7.0.255**				
1	logical_name			octet_string[6]			"0000190700FF"	--	--	Get	
2	DL_reference			octet_string[6]			0x08 Choice(0-0:25.3.0.255, 0-0:25.2.0.255)	--	--	Get, Set	
3	address_config_mode			enum			0x10 0	--	--	Get, Set	
4	unicast_IPv6_addresses			array[0..*](IPv6_address)			0x18	--	--	Get, Set	
5	multicast_IPv6_addresses			array[0..*](IPv6_address)			0x20	--	--	Get, Set	
6	gateway_IPv6_address			array[0..*](IPv6_address)			0x28	--	--	Get, Set	
7	primary_DNS_address			IPv6_address			0x30	--	--	Get, Set	
8	secondary_DNS_address			IPv6_address			0x38	--	--	Get, Set	
9	traffic_class			unsigned			0x40	--	--	Get, Set	
10	neighbor_discovery_setup			array[0..*](neighbour_discovery_setup_element)			0x48	--	--	Get, Set	
1	add_IPv6_address			IPv6_address_entry			0x60	--	--	Action	
2	remove_IPv6_address			IPv6_address_entry			0x68	--	--	Action	
	PPP setup	**M-G**	**M-G**		**44**	**0**	**0-0:25.3.0.255**				
1	logical_name			octet_string[6]			"0000190300FF"	--	--	Get	
2	PHY_refernce			octet_string[6]			0x08 0-0:25.4.0.255	--	--	Get, Set	
3	LCP_options			array[1..8](LCP_options_element)			0x10 { {3, 4, 0xC023},... }	--	--	Get, Set	{ LCP-option-type := Auth-Prot, LCP-Option-Length := 2, LCP-option-data := PAP}
4	IPCP_options			array[1..5](IPCP_options_element)			0x18 { {2, 4, 0x0000},... }	--	--	Get, Set	{ IPCP-option-type := IP-comp-prot, IPCP-option-Length := 2, IPCP-option-data := No-IP-Compresion-is-used}
5	PPP_authentication			PPP_authentication_type			0x20	--	--	Get, Set	{ choice(NULL, { user-name, PAP-password}) }
	Abstract objects - Push setup										
	Push setup - Interval_1	**M**	**M**		**40**	**1**	**0-1:25.9.0.255**				
1	logical_name			octet_string[6]			"0001190900FF"	--	--	Get	
2	push_object_list			array[0..*](capture_object_definition)			0x08	--	--	Get, Set	
3	send_destination_and_method			send_destination_and_method_type			0x10 {[0..4],*,0}	--	--	Get, Set	
4	communication_window			array[0..*](window_element)			0x18	--	--	Get, Set	
5	randomisation_start_interval			long_unsigned			0x20 0	--	--	Get, Set	Randomiztion time in seconds, 0 = no randomization at start of the first window defined in calling_window
6	number_of_retries			unsigned			0x28	--	--	Get, Set	Max number of repetitions of unsuccessful push
7	repetition_delay			long_unsigned			0x30	--	--	Get, Set	Delay in seconds between repetitions
8	port_reference			octet_string[0..6]			0x38	--	--	Get, Set	
9	push_client_SAP			integer			0x40	--	--	Get, Set	
10	push_protection_parameters			array			0x48 {}	--	--	Get, Set	
11	gateway_prefix			octet_string			0x50 {}				
1	push			integer			0x58	--	Action	Action	
	Push setup - Interval_2	**M**	**M**		**40**	**1**	**0-2:25.9.0.255**				
1	logical_name			octet_string[6]			"0002190900FF"	--	--	Get	
2	push_object_list			array[0..*](capture_object_definition)			0x08	--	--	Get, Set	
3	send_destination_and_method			send_destination_and_method_type			0x10 {[0..4],*,0}	--	--	Get, Set	
4	communication_window			array[0..*](window_element)			0x18	--	--	Get, Set	
5	randomisation_start_interval			long_unsigned			0x20 0	--	--	Get, Set	Randomiztion time in seconds, 0 = no randomization at start of the first window defined in calling_window
6	number_of_retries			unsigned			0x28	--	--	Get, Set	Max number of repetitions of unsuccessful push
7	repetition_delay			long_unsigned			0x30	--	--	Get, Set	Delay in seconds between repetitions

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value		Public (16)	Preestablished (102)	Management (1)	
8	port_reference			octet_string[0..6]			0x38		--	--	Get, Set	
9	push_client_SAP			integer			0x40		--	--	Get, Set	
10	push_protection_parameters			array			0x48	{}	--	--	Get, Set	
11	gateway_prefix			octet_string			0x50	{}				
1	push			integer			0x58		--	Action	Action	
	Push setup - Interval_3	**M**	**M**		**40**	**1**		**0-3:25.9.0.255**				
1	logical_name			octet_string[6]				"0003190900FF"	--	--	Get	
2	push_object_list			array[0..*](capture_object_definition)			0x08		--	--	Get, Set	
3	send_destination_and_method			send_destination_and_method_type			0x10	{[0..4],*,0}	--	--	Get, Set	
4	communication_window			array[0..*](window_element)			0x18		--	--	Get, Set	
5	randomisation_start_interval			long_unsigned			0x20	0	--	--	Get, Set	Randomiztion time in seconds, 0 = no randomization at start of the first window defined in calling_window
6	number_of_retries			unsigned			0x28		--	--	Get, Set	Max number of repetitions of unsuccessful push
7	repetition_delay			long_unsigned			0x30		--	--	Get, Set	Delay in seconds between repetitions
8	port_reference			octet_string[0..6]			0x38		--	--	Get, Set	
9	push_client_SAP			integer			0x40		--	--	Get, Set	
10	push_protection_parameters			array			0x48	{}	--	--	Get, Set	
11	gateway_prefix			octet_string			0x50	{}				
1	push			integer			0x58		--	Action	Action	
	Push setup - On Alarm	**M**	**M**		**40**	**1**		**0-4:25.9.0.255**				
1	logical_name			octet_string[6]				"0004190900FF"	--	--	Get	
2	push_object_list			array[0..*](capture_object_definition)			0x08		--	--	Get, Set	
3	send_destination_and_method			send_destination_and_method_type			0x10	{[0..4],*,0}	--	--	Get, Set	
4	communication_window			array[0..*](window_element)			0x18		--	--	Get, Set	
5	randomisation_start_interval			long_unsigned			0x20	0	--	--	Get, Set	Randomiztion time in seconds, 0 = no randomization at start of the first window defined in calling_window
6	number_of_retries			unsigned			0x28		--	--	Get, Set	Max number of repetitions of unsuccessful push
7	repetition_delay			long_unsigned			0x30		--	--	Get, Set	Delay in seconds between repetitions
8	port_reference			octet_string[0..6]			0x38		--	--	Get, Set	
9	push_client_SAP			integer			0x40		--	--	Get, Set	
10	push_protection_parameters			array			0x48	{}	--	--	Get, Set	
11	gateway_prefix			octet_string			0x50	{}				
1	push			integer			0x58		--	Action	Action	
	Push setup - On Connectivity	**M**	**M**		**40**	**1**		**0-0:25.9.0.255**				
1	logical_name			octet_string[6]				"0000190900FF"	--	--	Get	
2	push_object_list			array[0..*](capture_object_definition)			0x08		--	--	Get, Set	
3	send_destination_and_method			send_destination_and_method_type			0x10	{[0..4],*,0}	--	--	Get, Set	
4	communication_window			array[0..*](window_element)			0x18		--	--	Get, Set	
5	randomisation_start_interval			long_unsigned			0x20	0	--	--	Get, Set	Randomiztion time in seconds, 0 = no randomization at start of the first window defined in calling_window
6	number_of_retries			unsigned			0x28		--	--	Get, Set	Max number of repetitions of unsuccessful push
7	repetition_delay			long_unsigned			0x30		--	--	Get, Set	Delay in seconds between repetitions
8	port_reference			octet_string[0..6]			0x38		--	--	Get, Set	
9	push_client_SAP			integer			0x40		--	--	Get, Set	
10	push_protection_parameters			array			0x48	{}	--	--	Get, Set	
11	gateway_prefix			octet_string			0x50	{}				
1	push			integer			0x58		--	Action	Action	
	Push setup - On Installation	**M**	**M**		**40**	**1**		**0-7:25.9.0.255**				
1	logical_name			octet_string[6]				"0007190900FF"	--	--	Get	
2	push_object_list			array[0..*](capture_object_definition)			0x08		--	--	Get, Set	
3	send_destination_and_method			send_destination_and_method_type			0x10	{[0..4],*,0}	--	--	Get, Set	
4	communication_window			array[0..*](window_element)			0x18		--	--	Get, Set	
5	randomisation_start_interval			long_unsigned			0x20	0	--	--	Get, Set	Randomiztion time in seconds, 0 = no randomization at start of the first window defined in calling_window
6	number_of_retries			unsigned			0x28		--	--	Get, Set	Max number of repetitions of unsuccessful push
7	repetition_delay			long_unsigned			0x30		--	--	Get, Set	Delay in seconds between repetitions
8	port_reference			octet_string[0..6]			0x38		--	--	Get, Set	
9	push_client_SAP			integer			0x40		--	--	Get, Set	
10	push_protection_parameters			array			0x48	{}	--	--	Get, Set	
11	gateway_prefix			octet_string			0x50	{}				
1	push			integer			0x58		--	Action	Action	
	Push setup - On Power down	**O**	**O**		**40**	**1**		**0-5:25.9.0.255**				
1	logical_name			octet_string[6]				"0005190900FF"	--	--	Get	
2	push_object_list			array[0..*](capture_object_definition)			0x08		--	--	Get, Set	
3	send_destination_and_method			send_destination_and_method_type			0x10	{[0..4],*,0}	--	--	Get, Set	
4	communication_window			array[0..*](window_element)			0x18		--	--	Get, Set	
5	randomisation_start_interval			long_unsigned			0x20	0	--	--	Get, Set	Randomiztion time in seconds, 0 = no randomization at start of the first window defined in calling_window
6	number_of_retries			unsigned			0x28		--	--	Get, Set	Max number of repetitions of unsuccessful push
7	repetition_delay			long_unsigned			0x30		--	--	Get, Set	Delay in seconds between repetitions
8	port_reference			octet_string[0..6]			0x38		--	--	Get, Set	
9	push_client_SAP			integer			0x40		--	--	Get, Set	
10	push_protection_parameters			array			0x48	{}	--	--	Get, Set	
11	gateway_prefix			octet_string			0x50	{}				
1	push			integer			0x58		--	Action	Action	
	Push script table	**M**	**M**		**9**	**0**		**0-0:10.0.108.255**				
1	logical_name			octet_string[6]				"00000A006CFF"	--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.	OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
2	scripts			array[6..*](script9)			0x08	choice({ {1,{{2,40,0-1:25.9.0.255,1,0}}}, //Push - Interval_1 {2,{{2,40,0-2:25.9.0.255,1,0}}}, //Push - Interval_2 {3,{{2,40,0-3:25.9.0.255,1,0}}}, //Push - Interval_3 {4,{{2,40,0-4:25.9.0.255,1,0}}}, //Push - On Alarm {5,{{2,40,0-0:25.9.0.255,1,0}}}, //Push - On Connectivity {6,{{2,40,0-7:25.9.0.255,1,0}}}, //Push - On Installation }, { {1,{{2,40,0-1:25.9.0.255,1,0}}}, //Push - Interval_1 {2,{{2,40,0-2:25.9.0.255,1,0}}}, //Push - Interval_2 {3,{{2,40,0-3:25.9.0.255,1,0}}}, //Push - Interval_3 {4,{{2,40,0-4:25.9.0.255,1,0}}}, //Push - On Alarm {5,{{2,40,0-0:25.9.0.255,1,0}}}, //Push - On Connectivity {6,{{2,40,0-7:25.9.0.255,1,0}}}, //Push - On Installation {8,{{2,40,0-6:25.9.0.255,1,0}}} //Push - Consumer Information }}	--	--	Get, (Set)	The first 6 scripts are mandatory, script 7 and 8 only need to be present if the optional elements (push setup on power down and CIP) are present as well. Please note that the assignment of the script_identifier is fixed.
1	execute			long_unsigned			0x20		--	--	Action	
	Push action scheduler - Interval_1	**M**	**M**		**22**	**0**		**0-1:15.0.4.255**				
1	logical_name			octet_string[6]				"00010F0004FF"	--	--	Get	
2	executed_script			script22			0x08	{0-0:10.0.108.255,1}	--	--	Get, (Set)	
3	type			enum			0x10	5	--	--	Get, (Set)	
4	execution_time			array[0..*](execution_time_date)			0x18		--	--	Get, Set	
	Push action scheduler - Interval_2	**M**	**M**		**22**	**0**		**0-2:15.0.4.255**				
1	logical_name			octet_string[6]				"00020F0004FF"	--	--	Get	
2	executed_script			script22			0x08	{0-0:10.0.108.255,2}	--	--	Get, (Set)	
3	type			enum			0x10	5	--	--	Get, (Set)	
4	execution_time			array[0..*](execution_time_date)			0x18		--	--	Get, Set	
	Push action scheduler - Interval_3	**M**	**M**		**22**	**0**		**0-3:15.0.4.255**				
1	logical_name			octet_string[6]				"00030F0004FF"	--	--	Get	
2	executed_script			script22			0x08	{0-0:10.0.108.255,3}	--	--	Get, (Set)	
3	type			enum			0x10	5	--	--	Get, (Set)	
4	execution_time			array[0..*](execution_time_date)			0x18		--	--	Get, Set	
	Remote communication - Abstract Objects for Network Mgmt											
	Auto connect	**M**	**M**		**29**	**2**		**0-0:2.1.0.255**				
1	logical_name			octet_string[6]				"0000020100FF"	--	--	Get	
2	mode			enum			0x08	101	--	--	Get, Set	
3	repetitions			unsigned			0x10	0	--	--	Get, Set	
4	repetition_delay			long_unsigned			0x18	0	--	--	Get, Set	
5	calling_window			array[0..*](window_element)			0x20		--	--	Get, Set	
6	destination_list			array[0..*]			0x28		--	--	Get, (Set)	
1	connect			integer			0x30		--	--	Action	
	MAC address setup	**M-E|M-G3**	**M-E|M-G3**		**43**	**0**		**0-0:25.2.0.255**				
1	logical_name			octet_string[6]				"0000190200FF"	--	--	Get	
2	MAC_address			octet_string			0x08		--	--	Get	
	Remote communication - GSM/GPRS Setup Objects											
	GPRS modem setup	**M-G**	**M-G**		**45**	**0**		**0-0:25.4.0.255**				
1	logical_name			octet_string[6]				"0000190400FF"	--	--	Get	
2	APN			octet_string[0..64]			0x08		--	--	Get, Set	An APN with a size of 64 bytes at least is supported, i.e. [0.64] is mandatoty but can be longer
3	PIN_code			long_unsigned			0x10		--	--	Get, Set	
4	quality_of_service			quality_of_service_type			0x18		--	--	Get, Set	
	Modem configuration	**M-G**	**M-G**		**27**	**1**		**0-0:2.0.0.255**				
1	logical_name			octet_string[6]				"0000020000FF"	--	--	Get	
2	comm_speed			enum			0x08		--	--	Get, Set	
3	initialization_string			array			0x10		--	--	Get, Set	
4	modem_profile			array			0x18		--	--	Get, Set	
	Auto answer	**M-G**	**M-G**		**28**	**2**		**0-0:2.2.0.255**				
1	logical_name			octet_string[6]				"0000020200FF"	--	--	Get	
2	mode			enum			0x08		--	--	Get, Set	
3	listening_window			array[0..*]			0x10		--	--	Get, Set	Array[0] equals always listening
4	status			enum			0x18		--	--	Get, Set	
5	number_of_calls			unsigned			0x20		--	--	Get, Set	
6	number_of_rings			nr_rings_type			0x28		--	--	Get, Set	
7	list_of_allowed_callers			array[0..*](list_of_allowed_callers_element)			0x30		--	--	Get, Set	phone number is an ASCII coded octet-string of phone number length
	GSM Diagonstic	**M-G**	**M-G**		**47**	**0**		**0-0:25.6.0.255**				
1	logical_name			octet_string[6]				"0000190600FF"	--	--	Get	
2	operator			visible_string					--	--	Get	
3	status			enum					--	--	Get	
4	cs_attachment			enum					--	--	Get	
5	ps_status			enum					--	--	Get	
6	cell_info			struct(double_long_unsigned,long_unsigned,unsigned,unsigned,long_unsigned,long_unsigned,double_long_unsigned)					--	--	Get	
7	adjacent_cells			array					--	--	Get	

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	Object / Attribute Name	104 1ph IP	105 3ph IP	Attribute Type	Class	Ver.		OBIS code / Default Value	Public (16)	Preestablished (102)	Management (1)	
8	capture_time			date-time					--	--	Get	
	Remote communication - G3-PLC Setup Objects											
	G3-PLC MAC layer counters	**M-G3**	**M-G3**		**90**	**1**		**0-0:29.0.0.255**				
1	logical_name			octet_string[6]				"00001D0000FF"	--	--	Get	
2	mac_Tx_data_packet_count			double_long_unsigned			0x08		--	--	Get	
3	mac_Rx_data_packet_count			double_long_unsigned			0x10		--	--	Get	
4	mac_Tx_cmd_packet_count			double_long_unsigned			0x18		--	--	Get	
5	mac_Rx_cmd_packet_count			double_long_unsigned			0x20		--	--	Get	
6	mac_CSMA_fail_count			double_long_unsigned			0x28		--	--	Get	
7	mac_CSMA_no_ACK_count			double_long_unsigned			0x30		--	--	Get	
8	mac_bad_CRC_count			double_long_unsigned			0x38		--	--	Get	
9	mac_Tx_data_broadcast_count			double_long_unsigned			0x40		--	--	Get	
10	mac_Rx_data_broadcast_count			double_long_unsigned			0x48		--	--	Get	
1	reset			integer			0x50		--	--	Action	
	G3-PLC MAC setup	**M-G3**	**M-G3**		**91**	**1**		**0-0:29.1.0.255**				
1	logical_name			octet_string[6]				"00001D0100FF"	--	--	Get	
2	mac_short_address			long_unsigned			0x08		--	--	Get	
3	mac_RC_coord			long_unsigned			0x10		--	--	Get	
4	mac_PAN_id			long_unsigned			0x18		--	--	Get	
5	mac_key_table			array			0x20		--	--	Set	
6	mac_frame_counter			double_long_unsigned			0x28		--	--	Get	
7	mac_tone_mask			bit_string			0x30	"\0000000000000000000000000000000000000000011111111111111111111111111111111111111"	--	--	Get, Set	
8	mac_TMR_TTL			unsigned			0x38	2	--	Set	Get, Set	
9	mac_max_frame_retries			unsigned			0x40	5	--	Set	Get, Set	
10	mac_neighbour_table_entry_TTL			unsigned			0x48	255	--	Set	Get, Set	
11	mac_neighbour_table			array			0x50		--	--	Get	
12	mac_high_priority_window_size			unsigned			0x58	7	--	Set	Get, Set	
13	mac_CSMA_fairness_limit			unsigned			0x60	25	--	Set	Get, Set	
14	mac_beacon_randomization_window_length			unsigned			0x68	12	--	Set	Get, Set	
15	mac_A			unsigned			0x70	8	--	Set	Get, Set	
16	mac_K			unsigned			0x78	5	--	Set	Get, Set	
17	mac_min_CW_attempts			unsigned			0x80	10	--	Set	Get, Set	
18	mac_cenelec_legacy_mode			unsigned			0x88	1	--	--	Get	
19	mac_FCC_legacy_mode			unsigned			0x90	1	--	--	Get	
20	mac_max_BE			unsigned			0x98	8	--	Set	Get, Set	
21	mac_max_CSMA_backoffs			unsigned			0xA0	50	--	Set	Get, Set	
22	mac_min_BE			unsigned			0xA8	3	--	Set	Get, Set	
1	mac_get_neighbour_table_entry			long_unsigned			0xB0		--	--	Action	
	G3-PLC MAC 6LoWPAN adaptation layer setup	**M-G3**	**M-G3**		**92**	**1**		**0-0:29.2.0.255**				
1	logical_name			octet_string[6]				"00001D0200FF"	--	--	Get	
2	adp_max_hops			unsigned			0x08		--	--	Get, Set	
3	adp_weak_LQI_value			unsigned			0x10		--	--	Get, Set	
4	adp_security_level			unsigned			0x18	5	--	--	Get, Set	*keys must be defined: Shall we define defaults or add it to CTI file ?*
5	adp_prefix_table			array			0x20		--	--	Get	
6	adp_routing_configuration			array			0x28		--	--	Get, Set	
7	adp_broadcast_log_table_entry_TTL			long_unsigned			0x30	2	--	--	Get, Set	
8	adp_routing_table			array			0x38		--	--	Get	
9	adp_context_information_table			array			0x40		--	--	Get	
10	adp_blacklist_table			array			0x48		--	--	Get	
11	adp_broadcast_log_table			array			0x50		--	--	Get	
12	adp_group_table			array			0x58		--	--	Get	
13	adp_max_join_wait_time			long_unsigned			0x60	20	--	--	Get, Set	
14	adp_path_discovery_time			unsigned			0x68	40	--	--	Get, Set	
15	adp_active_key_index			unsigned			0x70		--	--	Get	
16	adp_metric_type			unsigned			0x78	0x0F	--	--	Get, Set	
17	adp_coord_short_address			long_unsigned			0x80	0x0000	--	--	Get, Set	
18	adp_disable_default_routing			boolean			0x88	0	--	--	Get, Set	
19	adp_device_type			enum			0x90	0	--	--	Get, Set	

3.3 Event definitions and assignments

Number	Name	Description	Standard Event Log	Fraud Detection Log	Disconnector Control Log	M-Bus Event Log	M-Bus Control Log 1	M-Bus Control Log 2	M-Bus Control Log 3	M-Bus Control Log 4	Power Quality Event Log
1	Power Down	Indicates a complete power down of the device. Please note that this is related to the device and not necessarily to the network.	x								
2	Power Up	Indicates that the device is powered again after a complete power down.	x								
3	Daylight saving time enabled or disabled	Indicates the regular change from and to daylight saving time. The time stamp shows the time before the change. This event is not set in case of manual clock changes and in case of power failures.	x								
4	Clock adjusted (old date/time)	Indicates that the clock has been adjusted. The date/time that is stored in the event log is the old date/time before adjusting the clock.	x								
5	Clock adjusted (new date/time)	Indicates that the clock has been adjusted. The date/time that is stored in the event log is the new date/time after adjusting the clock.	x								
6	Clock invalid	Indicates that clock may be invalid, i.e. if the power reserve of the clock has exhausted. It is set at power up.	x								
7	Replace Battery	Indicates that the battery must be exchanged due to the expected end of life time.	x								
8	Battery voltage low	Indicates that the current battery voltage is low.	x								
9	TOU activated	Indicates that the passive TOU has been activated.	x								
10	Error register cleared	Indicates that the error register was cleared.	x								
11	Alarm register cleared	Indicates that the alarm register was cleared.	x								
12	Program memory error	Indicates a physical or a logical error in the program memory.	x								
13	RAM error	Indicates a physical or a logical error in the RAM.	x								
14	NV memory error	Indicates a physical or a logical error in the non volatile memory	x								
15	Watchdog error	Indicates a watch dog reset or a hardware reset of the microcontroller.	x								
16	Measurement system error	Indicates a logical or physical error in the measurement system	x								
17	Firmware ready for activation	Indicates that the new firmware has been successfully downloaded and verified, i.e. it is ready for activation	x								
18	Firmware activated	Indicates that a new firmware has been activated	x								
19	Passive TOU programmed	The passive structures of TOU or a new activation date/time were programmed	x								
20	External alert detected	Indicates a signal detected on the meter's input terminal	x								
21	End of non-periodic billing interval	Indicates the end of a non-periodic billing interval	x								
22	IF_LO_2W enabled	2 way communication on local port enabled	x								
23	IF_LO_2W disabled	2 way communication on local port disabled	x								
24	*reserved for future use*										
25	*reserved for future use*										
26	Communication start on remote interface I3 or I3.1	Indicates that the communication was started on the remote interface I_3 or $I_{3.1}$									
27	Communication end on remote interface I3 or I3.1	Indicates that the communication has ended on the remote interface I_3 or $I_{3.1}$									
28	Communication start on local interface I_{E-M}	Indicates that the communication was started on the local interface I_{E-M}									
29	Communication end on local interface I_{E-M}	Indicates that the communication has ended on the local interface I_{E-M}									

Number	Name	Description	Standard Event Log	Fraud Detection Log	Disconnector Control Log	M-Bus Event Log	M-Bus Control Log 1	M-Bus Control Log 2	M-Bus Control Log 3	M-Bus Control Log 4	Power Quality Event Log
30	M-Bus ch1 Disconnector activity calendar activated	Indicates that the passive Disconnector activity calendar has been activated					x				
31	M-Bus ch1 Disconnector passive activity calendar programmed	Indicates that the passive Disconnector activity calendar has been programmed					x				
32	M-Bus ch2 Disconnector activity calendar activated	Indicates that the passive Disconnector activity calendar has been activated						x			
33	M-Bus ch2 Disconnector passive activity calendar programmed	Indicates that the passive Disconnector activity calendar has been programmed						x			
34	M-Bus ch3 Disconnector activity calendar activated	Indicates that the passive Disconnector activity calendar has been activated							x		
35	M-Bus ch3 Disconnector passive activity calendar programmed	Indicates that the passive Disconnector activity calendar has been programmed							x		
36	M-Bus ch4 Disconnector activity calendar activated	Indicates that the passive Disconnector activity calendar has been activated								x	
37	M-Bus ch4 Disconnector passive activity calendar programmed	Indicates that the passive Disconnector activity calendar has been programmed								x	
38	M-Bus FW ready for activation	Indicates that on M-Bus channel x the FW has been successfully downloaded and verified; i.e. it is ready for activation.				x					
39	M-Bus FW activated	Indicates that on an M-Bus channel x the FW has been activated				x					
40	Terminal cover removed	Indicates that the terminal cover has been removed.		x							
41	Terminal cover closed	Indicates that the terminal cover has been closed.		x							
42	Strong DC field detected	Indicates that a strong magnetic DC field has been detected.		x							
43	No strong DC field anymore	Indicates that the strong magnetic DC field has disappeared.		x							
44	Meter cover removed	Indicates that the meter cover has been removed.		x							
45	Meter cover closed	Indicates that the meter cover has been closed.		x							
46	Association authentication failure (n time failed authentication)	Indicates that a user tried to gain LLS access with wrong password (intrusion detection) or HLS access challenge processing failed n-times		x							
47	One or more parameters changed	Indicates the change of at least one parameter	x								
48	Global key(s) changed	One or more global keys changed	x								
49	Decryption or authentication failure (n time failure)	Decryption with currently valid key (global or dedicated) failed to generate a valid APDU or authentication tag		x							
50	Replay attack	Receive frame counter value less or equal to the last successfully received frame counter in the received APDU Event signalizes as well the situation when the DC has lost the frame counter synchronization.		x							
51	FW verification failed	Indicates the transferred firmware verification failed i.e. cannot be activated.	x								
52	Unexpected consumption	Indicates consumption is detected at least on one phase when the disconnector has been disconnected	x								
53	LPCAP_M1 enabled	Capturing of M-Bus profile 1 has been enabled				x					
54	LPCAP_M1 disabled	Capturing of M-Bus profile 1 has been disabled				x					
55	LPCAP_M2 enabled	Capturing of M-Bus profile 2 has been enabled				x					
56	LPCAP_M2 disabled	Capturing of M-Bus profile 2 has been disabled				x					
57	LPCAP_M3 enabled	Capturing of M-Bus profile 3 has been enabled				x					
58	LPCAP_M3 disabled	Capturing of M-Bus profile 3 has been disabled				x					
59	Disconnector ready for manual reconnection	Indicates that the disconnector has been set into the Ready_for_reconnection state and can be manually reconnected			x						

Number	Name	Description	Standard Event Log	Fraud Detection Log	Disconnector Control Log	M-Bus Event Log	M-Bus Control Log 1	M-Bus Control Log 2	M-Bus Control Log 3	M-Bus Control Log 4	Power Quality Event Log
60	Manual disconnection	Indicates that the disconnector has been manually disconnected.			x						
61	Manual connection	Indicates that the disconnector has been manually connected.			x						
62	Remote disconnection	Indicates that the disconnector has been remotely disconnected.			x						
63	Remote connection	Indicates that the disconnector has been remotely connected.			x						
64	Local disconnection	Indicates that the disconnector has been locally disconnected (i.e. via the limiter or current supervision monitors).			x						
65	Limiter threshold exceeded	Indicates that the limiter threshold has been exceeded.			x						
66	Limiter threshold ok	Indicates that the monitored value of the limiter dropped below the threshold.			x						
67	Limiter threshold changed	Indicates that the limiter threshold has been changed			x						
68	Disconnect/Reconnect failure	Indicates that the a failure of disconnection or reconnection has happened (control state does not match output state)			x						
69	Local reconnection	Indicates that the disconnector has been locally re-connected (i.e. via the limiter or current supervision monitors).			x						
70	Supervision monitor 1 threshold exceeded	Indicates that the supervision monitor threshold has been exceeded.			x						
71	Supervision monitor 1 threshold ok	Indicates that the monitored value dropped below the threshold.			x						
72	Supervision monitor 2 threshold exceeded	Indicates that the supervision monitor threshold has been exceeded.			x						
73	Supervision monitor 2 threshold ok	Indicates that the monitored value dropped below the threshold.			x						
74	Supervision monitor 3 threshold exceeded	Indicates that the supervision monitor threshold has been exceeded.			x						
75	Supervision monitor 3 threshold ok	Indicates that the monitored value dropped below the threshold.			x						
76	Undervoltage L1	Indicates undervoltage on at least L1 phase was detected.									x
77	Undervoltage L2	Indicates undervoltage on at least L2 phase was detected.									x
78	Undervoltage L3	Indicates undervoltage on at least L3 phase was detected.									x
79	Overvoltage L1	Indicates overvoltage on at least L1 phase was detected.									x
80	Overvoltage L2	Indicates overvoltage on at least L2 phase was detected.									x
81	Overvoltage L3	Indicates overvoltage on at least L3 phase was detected.									x
82	Missing voltage L1	Indicates that the voltage on at least L1 phase has fallen below the Umin threshold for longer than the time delay.									x
83	Missing voltage L2	Indicates that the voltage on at least L2 phase has fallen below the Umin threshold for longer than the time delay.									x
84	Missing voltage L3	Indicates that the voltage on at least L3 phase has fallen below the Umin threshold for longer than the time delay.									x
85	Voltage L1 normal	Indicates that the mains voltage is in normal limits again, e.g. after overvoltage.									x
86	Voltage L2 normal	Indicates that the mains voltage is in normal limits again, e.g. after overvoltage.									x
87	Voltage L3 normal	Indicates that the mains voltage is in normal limits again, e.g. after overvoltage.									x
88	Phase sequence reversal	Indicates wrong mains connection. Usually indicates fraud or wrong installation. For poly phase connection only!	x								

Number	Name	Description	Standard Event Log	Fraud Detection Log	Disconnector Control Log	M-Bus Event Log	M-Bus Control Log 1	M-Bus Control Log 2	M-Bus Control Log 3	M-Bus Control Log 4	Power Quality Event Log
89	Missing neutral	Indicates that the neutral connection from the supplier to the meter is interrupted (but the neutral connection to the load prevails). The phase voltages measured by the meter may differ from their nominal values	x								
90	Phase Asymmetry	Indicates phase asymmetry due to large unbalance of loads connected									x
91	Current Reversal	Indicates unexpected energy export (for devices which are configured for energy import measurement only)		x							
92	Bad Voltage Quality L1	Indicates that during each period of one week 95 % of the 10 min mean r.m.s, values of the supply voltage are within the range of Un+/- 10 % and all 10 min mean r.m.s. values of the supply voltage shall be within the range of Un + 10%/- 15%. (acc. EN50160:2010, section 4.2.2)									x
93	Bad Voltage Quality L2	Indicates that during each period of one week 95 % of the 10 min mean r.m.s, values of the supply voltage are within the range of Un+/- 10 % and all 10 min mean r.m.s. values of the supply voltage shall be within the range of Un + 10%/- 15%. (acc. EN50160:2010, section 4.2.2)									x
94	Bad Voltage Quality L3	Indicates that during each period of one week 95 % of the 10 min mean r.m.s, values of the supply voltage are within the range of Un+/- 10 % and all 10 min mean r.m.s. values of the supply voltage shall be within the range of Un + 10%/- 15%. (acc. EN50160:2010, section 4.2.2)									x
95	Disconnector activity calendar activated	Indicates that the passive Disconnector activity calendar has been *activated*			x						
96	Disconnector passive activity calendar programmed	Indicates that the passive Disconnector activity calendar has been *programmed*			x						
97	Load Mgmt activity calendar activated	Indicates that the passive load management activity calendar has been *activated*	x								
98	Load Mgmt passive activity calendar programmed	Indicates that the passive load management activity calendar has been *programmed*	x								
99	M-Bus FW verification failed	M-Bus FW verification failed				x					
100	Communication error M-Bus channel 1	Indicates a communication problem when reading the meter connected to channel 1 of the M-Bus				x					
101	Communication ok M-Bus channel 1	Indicates that the communication with the M-Bus meter connected to channel 1 of the M-Bus is ok again.				x					
102	Replace Battery M-Bus channel 1	Indicates that the battery must be exchanged due to the expected end of life time.				x					
103	Fraud attempt M-Bus channel 1	Indicates that a fraud attempt has been registered.				x					
104	Clock adjusted M-Bus channel 1	Indicates that the clock has been adjusted.				x					
105	New M-Bus device installed channel 1	Indicated the meter (M-Bus master) has registered a M-Bus device connected to channel 1 with a new serial number				x					
106	Permanent Error M-Bus channel 1	Severe error reported by M-Bus device				x					
107	LPCAP_M4 enabled	Capturing of M-Bus profile 4 has been enabled				x					
108	LPCAP_M4 disabled	Capturing of M-Bus profile 4 has been disabled				x					
109	LPCAP_1 enabled	Capturing of load profile 1 has been enabled	x								
110	Communication error M-bus channel 2	Indicates a communication problem when reading the meter connected to channel 2 of the M-Bus				x					
111	Communication ok M-bus channel 2	Indicates that the communication with the M-Bus meter connected to channel 2 of the M-Bus is ok again.				x					

Number	Name	Description	Standard Event Log	Fraud Detection Log	Disconnector Control Log	M-Bus Event Log	M-Bus Control Log 1	M-Bus Control Log 2	M-Bus Control Log 3	M-Bus Control Log 4	Power Quality Event Log
112	Replace Battery M-Bus channel 2	Indicates that the battery must be exchanged due to the expected end of life time.				x					
113	Fraud attempt M-Bus channel 2	Indicates that a fraud attempt has been registered in the M-Bus device.				x					
114	Clock adjusted M-Bus channel 2	Indicates that the clock has been adjusted.				x					
115	New M-Bus device installed channel 2	Indicated the meter (M-Bus master) has registered a M-Bus device connected to channel 2 with a new serial number				x					
116	Permanent Error M-Bus channel 2	Severe error reported by M-Bus device (Bit 3 in MBUS status EN13757)				x					
117	LPCAP_1 disabled	Capturing of load profile 1 has been disabled	x								
118	LPCAP_2 enabled	Capturing of load profile 2 has been enabled	x								
119	LPCAP_2 disabled	Capturing of load profile 2 has been disabled	x								
120	Communication error M-bus channel 3	Indicates a communication problem when reading the meter connected to channel 3 of the M-Bus				x					
121	Communication ok M-bus channel 3	Indicates that the communication with the M-Bus meter connected to channel 3 of the M-Bus is ok again.				x					
122	Replace Battery M-Bus channel 3	Indicates that the battery must be exchanged due to the expected end of life time.				x					
123	Fraud attempt M-Bus channel 3	Indicates that a fraud attempt has been registered.				x					
124	Clock adjusted M-Bus channel 3	Indicates that the clock has been adjusted.				x					
125	New M-Bus device installed channel 3	Indicated the meter (M-Bus master) has registered a M-Bus device connected to channel 3 with a new serial number				x					
126	Permanent Error M-Bus channel 3	Severe error reported by M-Bus device (Bit 3 in MBUS status EN13757)				x					
127	*reserved for future use*										
128	*reserved for future use*	Indicates that on M-Bus channel x the FW verification failed									
129	*reserved for future use*										
130	Communication error M-bus channel 4	Indicates a communication problem when reading the meter connected to channel 4 of the M-Bus				x					
131	Communication ok M-bus channel 4	Indicates that the communication with the M-Bus meter connected to channel 4 of the M-Bus is ok again.				x					
132	Replace Battery M-Bus channel 4	Indicates that the battery must be exchanged due to the expected end of life time.				x					
133	Fraud attempt M-Bus channel 4	Indicates that a fraud attempt has been registered.				x					
134	Clock adjusted M-Bus channel 4	Indicates that the clock has been adjusted.				x					
135	New M-Bus device installed channel 4	Indicated the meter (M-Bus master) has registered a M-Bus device connected to channel 4 with a new serial number				x					
136	Permanent Error M-Bus channel 4	Severe error reported by M-Bus device (Bit 3 in MBUS status EN13757)				x					
137	*reserved for future use*										
138	*reserved for future use*										
139	*reserved for future use*										
140	No connection timeout	There has been no remote communication on application layer for a predefined period of time; i.e. meter could not be reached remotely.									
141	Modem Initialization failure	Modem's response to initialization AT command(s) is invalid or ERROR or no response received									
142	SIM Card failure	SIM card is not inserted or is not recognized. The detection of the SIM card status is supported by the corresponding AT commands as listed in 3GPP TS 27.007									
143	SIM Card ok	SIM card has been correctly detected									

Number	Name	Description	Standard Event Log	Fraud Detection Log	Disconnector Control Log	M-Bus Event Log	M-Bus Control Log 1	M-Bus Control Log 2	M-Bus Control Log 3	M-Bus Control Log 4	Power Quality Event Log
144	GSM registration failure	Modem's registration on GSM network was not successful									
145	GPRS registration failure	Modem's registration on GPRS network was not successful									
146	PDP context established	PDP context is established									
147	PDP context destroyed	PDP context is destroyed									
148	PDP context failure	No Valid PDP context(s) retrieved									
149	Modem SW reset	Modem restarted by SW reset									
150	Modem HW reset	Modem restarted by HW reset (this event is not issued after a general power resume)									
151	GSM outgoing connection	Modem is successfully connected, initiated by an outgoing call.									
152	GSM incoming connection	Modem is successfully connected, initiated by an incoming call									
153	GSM hang-up	Modem is disconnected									
154	Diagnostic failure	Modem's response to diagnostic AT command(s) ("+CPIN?", "+CSQ", "+CREG?", "+CGREG?", "+COPS?", "+CGACT?", "+CPMS?") is invalid or ERROR or no response received.									
155	User initialization failure	Modem's initialization AT command(s) – specified in attribute 3 of the modem configuration object - is invalid. Error message or no response from the modem.									
156	Signal quality low	Signal strength too low, not known, or not detectable									
157	Auto Answer Number of calls exceeded	Number of calls has exceeded (in mode(1) or mode(2)) the values given in the attribute number_of_calls.									
158	Local communication attempt	Indicates a successful communication on any local port has been initiated.									
159	*reserved for future use*										
160	Manual disconnection M-Bus channel 1	Indicates that the disconnector has been manually disconnected.					x				
161	Manual connection M-Bus channel 1	Indicates that the disconnector has been manually connected.					x				
162	Remote disconnection M-Bus channel 1	Indicates that the disconnector has been remotely disconnected.					x				
163	Remote connection M-Bus channel 1	Indicates that the disconnector has been remotely connected.					x				
164	Valve alarm M-Bus channel 1	Indicates that a valve alarm has been registered.					x				
165	Local disconnection M-Bus channel 1	Indicates that the disconnector has been locally disconnected.					x				
166	Local connection M-Bus channel 1	Indicates that the disconnector has been locally connected.					x				
167	*reserved for future use*										
168	*reserved for future use*										
169	*reserved for future use*										
170	Manual disconnection M-Bus channel 2	Indicates that the disconnector has been manually disconnected.						x			
171	Manual connection M-Bus channel 2	Indicates that the disconnector has been manually connected.						x			
172	Remote disconnection M-Bus channel 2	Indicates that the disconnector has been remotely disconnected.						x			
173	Remote connection M-Bus channel 2	Indicates that the disconnector has been remotely connected.						x			
174	Valve alarm M-Bus channel 2	Indicates that a valve alarm has been registered.						x			
175	Local disconnection M-Bus channel 2	Indicates that the disconnector has been locally disconnected.						x			

Number	Name	Description	Standard Event Log	Fraud Detection Log	Disconnector Control Log	M-Bus Event Log	M-Bus Control Log 1	M-Bus Control Log 2	M-Bus Control Log 3	M-Bus Control Log 4	Power Quality Event Log
176	Local connection M-Bus channel 2	Indicates that the disconnector has been locally connected.						x			
177	*reserved for future use*										
178	*reserved for future use*										
179	*reserved for future use*										
180	Manual disconnection M-Bus channel 3	Indicates that the disconnector has been manually disconnected.							x		
181	Manual connection M-Bus channel 3	Indicates that the disconnector has been manually connected.							x		
182	Remote disconnection M-Bus channel 3	Indicates that the disconnector has been remotely disconnected.							x		
183	Remote connection M-Bus channel 3	Indicates that the disconnector has been remotely connected.							x		
184	Valve alarm M-Bus channel 3	Indicates that a valve alarm has been registered.							x		
185	Local disconnection M-Bus channel 3	Indicates that the disconnector has been locally disconnected.							x		
186	Local connection M-Bus channel 3	Indicates that the disconnector has been locally connected.							x		
187	*reserved for future use*										
188	*reserved for future use*										
189	*reserved for future use*										
190	Manual disconnection M-Bus channel 4	Indicates that the disconnector has been manually disconnected.								x	
191	Manual connection M-Bus channel 4	Indicates that the disconnector has been manually connected.								x	
192	Remote disconnection M-Bus channel 4	Indicates that the disconnector has been remotely disconnected.								x	
193	Remote connection M-Bus channel 4	Indicates that the disconnector has been remotely connected.								x	
194	Valve alarm M-Bus channel 4	Indicates that a valve alarm has been registered.								x	
195	Local disconnection M-Bus channel 4	Indicates that the disconnector has been locally disconnected.								x	
196	Local connection M-Bus channel 4	Indicates that the disconnector has been locally connected.								x	
197	*reserved for future use*										
198	*reserved for future use*										
199	*reserved for future use*										
200	*manufacturer specific*										
201	*manufacturer specific*										
202	*manufacturer specific*										
203	*manufacturer specific*										
204	*manufacturer specific*										
205	*manufacturer specific*										
206	*manufacturer specific*										
207	*manufacturer specific*										
208	*manufacturer specific*										
209	*manufacturer specific*										
210	*manufacturer specific*										
211	*manufacturer specific*										
212	*manufacturer specific*										
213	*manufacturer specific*										
214	*manufacturer specific*										
215	*manufacturer specific*										
216	*manufacturer specific*										
217	*manufacturer specific*										

Number	Name	Description	Standard Event Log	Fraud Detection Log	Disconnector Control Log	M-Bus Event Log	M-Bus Control Log 1	M-Bus Control Log 2	M-Bus Control Log 3	M-Bus Control Log 4	Power Quality Event Log
218	*manufacturer specific*										
219	*manufacturer specific*										
220	*manufacturer specific*										
221	*manufacturer specific*										
222	*manufacturer specific*										
223	*manufacturer specific*										
224	*manufacturer specific*										
225	*manufacturer specific*										
226	*manufacturer specific*										
227	*manufacturer specific*										
228	*manufacturer specific*										
229	*manufacturer specific*										
230	*manufacturer specific*										
231	*manufacturer specific*										
232	*manufacturer specific*										
233	*manufacturer specific*										
234	*manufacturer specific*										
235	*manufacturer specific*										
236	*manufacturer specific*										
237	*manufacturer specific*										
238	*manufacturer specific*										
239	*manufacturer specific*										
240	*manufacturer specific*										
241	*manufacturer specific*										
242	*manufacturer specific*										
243	*manufacturer specific*										
244	*manufacturer specific*										
245	*manufacturer specific*										
246	*manufacturer specific*										
247	*manufacturer specific*										
248	*manufacturer specific*										
249	*manufacturer specific*										
250	*manufacturer specific*										
251	*manufacturer specific*										
252	*manufacturer specific*										
253	*manufacturer specific*										
254	Load profile cleared	Any of the profiles cleared. NOTE: If it appears in Standard Event Log then any of the E-load profiles was cleared. If the event appears in the M-Bus Event log then one of the M-Bus load profiles was cleared	x			x					
255	Event log cleared	Indicates that the event log was cleared. This is always the first entry in an event log. It is only stored in the affected event log.	x	x	x	x	x	x	x	x	x

ALARM REGISTER 1 [0-0:97.98.0.255]			ALAI
Bit	Alarm Name	Triggering Event	Bit
0	Clock invalid	6	0
1	Battery replace	7	1
2	Reserved for future use	-	2
3	Reserved for future use	-	3
4	Reserved for future use	-	4
5	Reserved for future use	-	5
6	Reserved for future use	-	6
7	Reserved for future use	-	7
8	Program memory error	12	8
9	RAM error	13	9
10	NV memory error	14	10
11	Measurement system error	16	11
12	Watchdog error	15	12
13	Fraud attempt	40, 42, 44, 46, 49, 50[1]	13
14	Reserved for future use	-	14
15	Reserved for future use	-	15
16	M-Bus communication error ch1	100	16
17	M-Bus communication error ch2	110	17
18	M-Bus communication error ch3	120	18
19	M-Bus communication error ch4	130	19
20	M-Bus fraud attempt ch1	103	20
21	M-Bus fraud attempt ch2	113	21
22	M-Bus fraud attempt ch3	123	22
23	M-Bus fraud attempt ch4	133	23
24	Reserved for future use	-	24
25	Reserved for future use	-	25
26	Reserved for future use	-	26
27	Reserved for future use	-	27
28	Reserved for future use	-	28
29	Reserved for future use	-	29
30	Reserved for future use	-	30
31	Reserved for future use	-	31

[1] Any of the listed events shall trigger the alarm.

RM REGISTER 2 [0-0:97.98.1.255]	
Alarm Name	**Triggering Event**
Total Power Failure	1
Power Resume	2
Voltage Phase Failure L1	82
Voltage Phase Failure L2	83
Voltage Phase Failure L3	84
Voltage Phase Resume L1	85
Voltage Phase Resume L2	86
Voltage Phase Resume L3	87
Missing Neutral	89
Phase Asymmetry	90
Current Reversal	91
Wrong Phase Sequence	88
Unexpected Consumption	52
Key Exchanged	48
Bad Voltage Quality L1	92
Bad Voltage Quality L2	93
Bad Voltage Quality L3	94
External Alert	20
Local communication attempt	53
New M-Bus Device Installed Ch1	105
New M-Bus Device Installed Ch2	115
New M-Bus Device Installed Ch3	125
New M-Bus Device Installed Ch4	135
Reserved for future use	-
Reserved for future use	-
Reserved for future use	-
Reserved for future use	-
M-Bus valve alarm Ch1	164
M-Bus valve alarm Ch2	174
M-Bus valve alarm Ch3	184
M-Bus valve alarm Ch4	194
Disconnect/Reconnect Failure	68

Extension D

Objects concerning disconnector.

#	Object / Attribute Name	Supported Objects		IC
		104 1ph IP	105 3ph IP	Class
	Disconnect control scheduler	**M-D**	**M-D**	**22**
	Disconnector script table	**M-D**	**M-D**	**9**
	Disconnect control	**M-D**	**M-D**	**70**
	Disconnect Control Activity Calendar	**M-D**	**M-D**	**20**
	Disconnect Control Special Days Table	**M-D**	**M-D**	**11**
	Event Object - Disconnector Control log	**M-D**	**M-D**	**1**
	Disconnector Control Log	**M-D**	**M-D**	**7**
	Limiter	**M-D**	**M-D**	**71**
	Supervision monitor 1 - Fuse supervision L1	**M-D**	**M-D**	**21**
	Supervision monitor 2 - Fuse supervision L2	**NA**	**M-D**	**21**
	Supervision monitor 3 - Fuse supervision L3	**NA**	**M-D**	**21**
	Sliding Average current L1 (for fuse supervision)	**M-D**	**M-D**	**5**
	Sliding Average current L2 (for fuse supervision)	**NA**	**M-D**	**5**
	Sliding Average current L3 (for fuse supervision)	**NA**	**M-D**	**5**
	Average Instantaneous Import Power (+A)	**M-D**	**M-D**	**5**
	Average Instantaneous Net Power (|+A|-|-A|)	**O**	**O**	**5**
	Average Instantaneous Total Power (|+A|+|-A|)	**M-D**	**M-D**	**5**

M-D=mandatory for extension D, O=optional, NA=not applicable

OBIS code	
Ver.	OBIS code
0	**0-0:15.0.1.255**
0	**0-0:10.0.106.255**
0	**0-0:96.3.10.255**
0	**0-0:13.0.1.255**
0	**0-0:11.0.1.255**
0	**0-0:96.11.2.255**
1	**0-0:99.98.2.255**
0	**0-0:17.0.0.255**
0	**1-0:31.4.0.255**
0	**1-0:51.4.0.255**
0	**1-0:71.4.0.255**
0	**1-0:31.4.0.255**
0	**1-0:51.4.0.255**
0	**1-0:71.4.0.255**
0	**1-0:1.24.0.255**
0	**1-0:16.24.0.255**
0	**1-0:15.24.0.255**

Extension M

Objects concerning Multi Utility

#	Object / Attribute Name
	M-Bus master port setup 1
	M-Bus master port setup 2
	M-Bus client channel 1
	M-Bus client channel 2
	M-Bus client channel 3
	M-Bus client channel 4
	M-Bus Value channel 1, instance 1
	M-Bus Value channel 1, instance 2
	M-Bus Value channel 1, instance 3
	M-Bus Value channel 1, instance 4
	M-Bus Value channel 1, instance 5
	M-Bus Value channel 1, instance 6
	M-Bus Value channel 1, instance 7
	M-Bus Value channel 1, instance 8
	M-Bus Value channel 2, instance 1
	M-Bus Value channel 2, instance 2
	M-Bus Value channel 2, instance 3
	M-Bus Value channel 2, instance 4
	M-Bus Value channel 2, instance 5
	M-Bus Value channel 2, instance 6
	M-Bus Value channel 2, instance 7
	M-Bus Value channel 2, instance 8
	M-Bus Value channel 3, instance 1
	M-Bus Value channel 3, instance 2
	M-Bus Value channel 3, instance 3
	M-Bus Value channel 3, instance 4
	M-Bus Value channel 3, instance 5
	M-Bus Value channel 3, instance 6
	M-Bus Value channel 3, instance 7
	M-Bus Value channel 3, instance 8
	M-Bus Value channel 4, instance 1
	M-Bus Value channel 4, instance 2
	M-Bus Value channel 4, instance 3
	M-Bus Value channel 4, instance 4
	M-Bus Value channel 4, instance 5
	M-Bus Value channel 4, instance 6
	M-Bus Value channel 4, instance 7
	M-Bus Value channel 4, instance 8
	M-Bus Device ID 1 channel 1
	M-Bus Device ID 1 channel 2
	M-Bus Device ID 1 channel 3
	M-Bus Device ID 1 channel 4
	M-Bus Device ID 2 channel 1
	M-Bus Device ID 2 channel 2
	M-Bus Device ID 2 channel 3
	M-Bus Device ID 2 channel 4
	Profile status for M-Bus Master Load profile 1
	Profile status for M-Bus Master Load profile 2

Profile status for M-Bus Master Load profile 3
Profile status for M-Bus Master Load profile 4
M-Bus Master Load profile for channel 1
M-Bus Master Load profile for channel 2
M-Bus Master Load profile for channel 3
M-Bus Master Load profile for channel 4
M-Bus Master Disconnect control object 1
M-Bus Master Disconnect control object 2
M-Bus Master Disconnect control object 3
M-Bus Master Disconnect control object 4
M-Bus Disconnect control scheduler
M-Bus Disconnector script table
M-Bus Disconnect Control Activity Calendar
M-Bus Disconnect Control Special Days Table
Event Objects - M-Bus Master Control logs 1
Event Objects - M-Bus Master Control logs 2
Event Objects - M-Bus Master Control logs 3
Event Objects - M-Bus Master Control logs 4
M-Bus Master Control log object 1
M-Bus Master Control log object 2
M-Bus Master Control log object 3
M-Bus Master Control log object 4
Event Object - M-Bus Event Log
M-Bus Event Log
Event Counter Object 5 – M-Bus Device 1 Billing Total
Event Counter Object 6 – M-Bus Device 1 Load Profile Reading Total
Event Counter Object 7 – M-Bus Device 2 Billing Total
Event Counter Object 8 – M-Bus Device 2 Load Profile Reading Total
Event Counter Object 9 – M-Bus Device 3 Billing Total
Event Counter Object 10 – M-Bus Device 3 Load Profile Reading Total
Event Counter Object 11 – M-Bus Device 4 Billing Total
Event Counter Object 12 – M-Bus Device 4 Load Profile Reading Total
M-Bus function control
M-Bus function control scheduler
M-Bus function control script table

M-M=mandatory for extension M, O=optional, NA=not applicable

Supported		IC		OBIS code
104 1ph IP	105 3ph IP	Class	Ver.	OBIS code
M-M	M-M	74	0	0-0:24.6.0.255
O	O	74	0	0-1:24.6.0.255
M-M	M-M	72	1	0-1:24.1.0.255
M-M	M-M	72	1	0-2:24.1.0.255
M-M	M-M	72	1	0-3:24.1.0.255
M-M	M-M	72	1	0-4:24.1.0.255
M-M	M-M	4	0	0-1:24.2.1.255
M-M	M-M	4	0	0-1:24.2.2.255
M-M	M-M	4	0	0-1:24.2.3.255
M-M	M-M	4	0	0-1:24.2.4.255
M-M	M-M	4	0	0-1:24.2.5.256
M-M	M-M	4	0	0-1:24.2.6.256
M-M	M-M	4	0	0-1:24.2.7.256
M-M	M-M	4	0	0-1:24.2.8.256
M-M	M-M	4	0	0-2:24.2.1.255
M-M	M-M	4	0	0-2:24.2.2.255
M-M	M-M	4	0	0-2:24.2.3.255
M-M	M-M	4	0	0-2:24.2.4.255
M-M	M-M	4	0	0-2:24.2.5.255
M-M	M-M	4	0	0-2:24.2.6.255
M-M	M-M	4	0	0-2:24.2.7.255
M-M	M-M	4	0	0-2:24.2.8.255
M-M	M-M	4	0	0-3:24.2.1.255
M-M	M-M	4	0	0-3:24.2.2.255
M-M	M-M	4	0	0-3:24.2.3.255
M-M	M-M	4	0	0-3:24.2.4.255
M-M	M-M	4	0	0-3:24.2.5.255
M-M	M-M	4	0	0-3:24.2.6.255
M-M	M-M	4	0	0-3:24.2.7.255
M-M	M-M	4	0	0-3:24.2.8.255
M-M	M-M	4	0	0-4:24.2.1.255
M-M	M-M	4	0	0-4:24.2.2.255
M-M	M-M	4	0	0-4:24.2.3.255
M-M	M-M	4	0	0-4:24.2.4.255
M-M	M-M	4	0	0-4:24.2.5.255
M-M	M-M	4	0	0-4:24.2.6.255
M-M	M-M	4	0	0-4:24.2.7.255
M-M	M-M	4	0	0-4:24.2.8.255
M-M	M-M	1	0	0-1:96.1.0.255
M-M	M-M	1	0	0-2:96.1.0.255
M-M	M-M	1	0	0-3:96.1.0.255
M-M	M-M	1	0	0-4:96.1.0.255
M-M	M-M	1	0	0-1:96.1.1.255
M-M	M-M	1	0	0-2:96.1.1.255
M-M	M-M	1	0	0-3:96.1.1.255
M-M	M-M	1	0	0-4:96.1.1.255
M-M	M-M	1	0	0-1:96.10.3.255
M-M	M-M	1	0	0-2:96.10.3.255

M-M	M-M	1	0	0-3:96.10.3.255
M-M	M-M	1	0	0-4:96.10.3.255
M-M	M-M	7	1	0-1:24.3.0.255
M-M	M-M	7	1	0-2:24.3.0.255
M-M	M-M	7	1	0-3:24.3.0.255
M-M	M-M	7	1	0-4:24.3.0.255
M-M	M-M	70	0	0-1:24.4.0.255
M-M	M-M	70	0	0-2:24.4.0.255
M-M	M-M	70	0	0-3:24.4.0.255
M-M	M-M	70	0	0-4:24.4.0.255
M-M	M-M	22	0	0-1:15.0.1.255
M-M	M-M	9	0	0-1:10.0.106.255
M-M	M-M	20	0	0-1:13.0.1.255
M-M	M-M	11	0	0-1:11.0.1.255
M-M	M-M	1	0	0-1:96.11.4.255
M-M	M-M	1	0	0-2:96.11.4.255
M-M	M-M	1	0	0-3:96.11.4.255
M-M	M-M	1	0	0-4:96.11.4.255
M-M	M-M	7	1	0-1:24.5.0.255
M-M	M-M	7	1	0-2:24.5.0.255
M-M	M-M	7	1	0-3:24.5.0.255
M-M	M-M	7	1	0-4:24.5.0.255
M-M	M-M	1	1	0-0:96.11.3.255
M-M	M-M	7	1	0-0:99.98.3.255
M-M	M-M	1	0	0-0:96.15.4.255
M-M	M-M	1	0	0-0:96.15.5.255
M-M	M-M	1	0	0-0:96.15.6.255
M-M	M-M	1	0	0-0:96.15.7.255
M-M	M-M	1	0	0-0:96.15.8.255
M-M	M-M	1	0	0-0:96.15.9.255
M-M	M-M	1	0	0-0:96.15.10.255
M-M	M-M	1	0	0-0:96.15.11.255
M-M	M-M	122	0	0-1:44.1.0.255
M-M	M-M	22	0	0-1:15.0.7.255
M-M	M-M	9	0	0-1:10.0.111.255

Extension L

Objects concerning Load management with relays

#	Object / Attribute Name	Supported		IC	
		104 1ph IP	105 3ph IP	Class	Ver.
	Load Mgmt script table	**M-L**	**M-L**	**9**	**0**
	Load Mgmt - Relay control 1	**M-L**	**M-L**	**70**	**0**
	Load Mgmt - Relay control 2	**O**	**O**	**70**	**0**
	Load Mgmt - Relay control N	**O**	**O**	**70**	**0**
	Load Mgmt Activity Calendar	**M-L**	**M-L**	**20**	**0**
	Load Mgmt Special Days Table	**M-L**	**M-L**	**11**	**0**

M-L=mandatory for extension L, O=optional, NA=not applicable

OBIS code OBIS code
0-0:10.0.103.255
0-1:96.3.10.255
0-2:96.3.10.255
0-n:96.3.10.255
0-0:13.0.2.255
0-0:11.0.2.255